Wireless InSite®

3D Wireless Communication and Propagation Prediction Software

USER'S GUIDE



REMCOM®

Electromagnetic Simulation Software

Wireless InSite® User's Guide

Version 3.4.4

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Chapter 1

Introduction

This guide provides an overview and a set of tutorials that will teach you...

- → How to interact with the Wireless InSite® GUI to create a complete project
- → How to request various types of output and run the calculation models to produce them
- → How to view and analyze the results of a model simulation

These tutorials provide step-by-step instructions for creating a project, running calculations, and viewing results. They also include examples of using Wireless InSite for prediction of radio propagation for indoor models, indoor-outdoor models, and terrain models.

- ✓ We highly recommend reading the Overview chapter in the Reference Manual for the Wireless InSite GUI before working through this tutorial.

1.1 Wireless InSite Summary

Wireless InSite 3.4.4 is an electromagnetic simulation tool for predicting the effects of buildings and terrain on the propagation of electromagnetic waves. It predicts how the locations of the transmitters and receivers within an urban area affect signal strength. Wireless InSite models the physical characteristics of irregular terrain and urban building features, performs the electromagnetic calculations, and then evaluates the signal propagation characteristics.

The virtual building and terrain environment is either constructed using Wireless InSite's editing tools or imported from a number of popular file formats, such as `DXF`, `shapefile`, `DTED` and `USGS`.

Transmitter and receiver locations can be specified using Wireless InSite's site-defining tools, or imported from an external data file. Separate calculations for portions of the overall area may be specified by defining study areas.

The calculations are made by shooting rays from the transmitters and propagating them through the defined geometry. These rays interact with geometrical features and make their way to receiver locations.

Ray interactions include *reflections* from feature faces, *diffractions* around feature edges, and *transmissions* through features faces. Wireless InSite's ray-based solvers use the Uniform Theory of Diffraction (UTD) to evaluate a ray path's electric field. UTD provides accurate results when the scenario geometry is large compared to the wavelength of the propagating wave. For typical applications, the UTD-based models provide accurate predictions from approximately 100 MHz to approximately 30 GHz. The X3D propagation model includes atmospheric absorption, extending the validity of its propagation calculations up to millimeter wave frequencies.

At each receiver location, contributions from arriving ray paths are combined and evaluated to determine predicted quantities such as electric and magnetic field strength, received power, interference measures, path loss, delay spread, direction of arrival, impulse response, electric field vs. time, electric field vs. frequency, and power delay profile.

Wireless InSite presents results in a number of ways. It provides visual representation of some results, such as transmitter coverage areas and power distributions, placing these visually within the modeled environment. Wireless InSite is also capable of playing movies of time-domain E-field and H-field evolution. For other types of data, Wireless InSite provides an advanced plotting system. Overlays of data allow quick comparison to imported measurements, or even previous Wireless InSite calculations. All output files produced by Wireless InSite are in a readable ASCII format.

1.2 How to Read this Guide

After installing Wireless InSite, we recommend that the user read the overview chapter in the reference manual and work through the tutorials in this guide. Following this, new users of Wireless InSite should understand the organization of the software and be able to quickly set up projects, run calculations, and view the propagation predictions. The user may then proceed to the Reference Guide to look up further details of the software and its capabilities.

The units used for data values in the tutorials are as listed below. It is recommended that you return your preferences to their default settings in order to make the tutorials easier to follow. If the units are not restored to their default settings, then values will appear in a different format from what is described in this chapter.

- angles in degrees
- distances in meters
- frequency in Megahertz
- power in dBm
- path loss in dB
- time in seconds

When viewing the RESULTS tree, it is also assumed that you are using the default sorting order. If you have changed this it is recommended that you reset this as well. To restore the default settings:

1. Select *Edit→Preferences* in the MAIN WINDOW.
2. Under the *Output* tab, set the fields sort order:
    - *Output filter*

- *Data type*
- *Transmitter set*
- *Receiver set*

The tutorials directory is specified in this guide using a placeholder name because this location can be changed as a part of the installation process and may vary depending on the version of the operating system you are using.

`tutorials_location\` The tutorial files are installed into C:\Program Files\Remcom\Wireless InSite 3.4.4 .x \tutorials. Since this folder can have restricted access, you may not be able to save to it. When you open a project in this file location, you will be prompted to save it to a location where you have write permissions.

Chapter 2

Simple Irregular Terrain

In this tutorial, we use the VERTICAL PLANE model to simulate radio wave propagation over irregular terrain. We compare the result of our calculation with path gain measurements performed by the Institute for Telecommunication Sciences (ITS) in Colorado. This data was extracted from the ITS website and stored in `*.plt` (Plot) files, which are readable by Wireless InSite.

- The ITS path gain measurements are available by request from [1].

Loading a USGS DEM file

To start the example, open Wireless InSite and begin a new project.

The *Terrain* data for this project will be a US Geological Survey Digital Elevation Model (or USGS DEM). The area we we will work with is Greeley, Colorado.

1. Navigate to *File→Import→Geometry→GDAL Terrain* to open the IMPORT TERRAIN window.

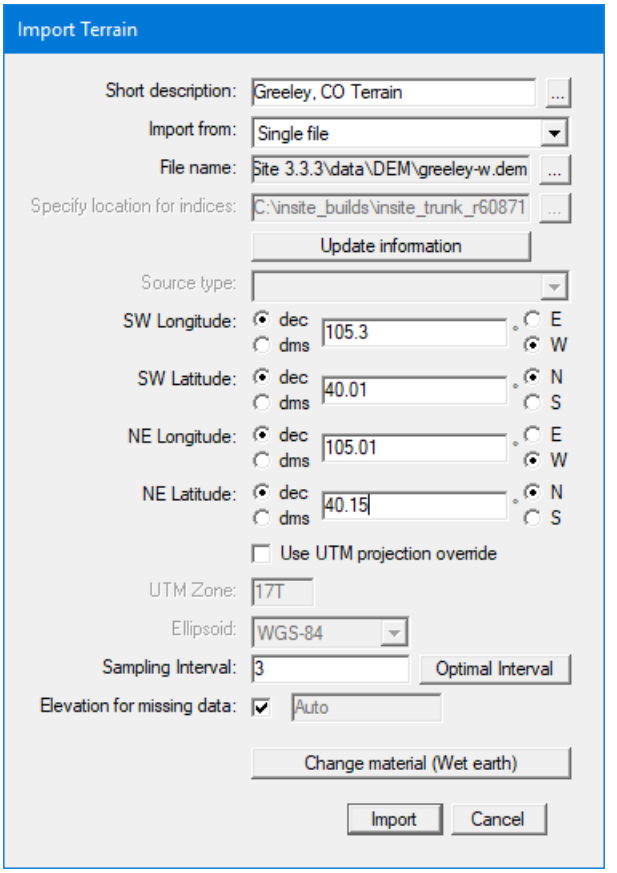


Figure 2.1: IMPORT TERRAIN window with coordinates

2. Enter “Greeley, CO Terrain” in the *Short description* field.
3. Select “Single file” from the *Import from* drop-down menu.
4. Select the terrain file that will be imported.
   (a) Click the “…” button next to the *File name* field.
   (b) In the lower right of the OPEN FILE window, change the file type to *DEM (*.dem)*.
   (c) Navigate to the `install_location\data\DEM` directory.
   (d) Select `greeley-w.dem` and click OPEN.
5. Define the coordinates of the section of the terrain to be imported.
   (a) *SW Longitude*: “105.3”
   (b) *SW Latitude*: “40.01”
   (c) *NE Longitude*: “105.01”
   (d) *NE Latitude*: “40.15”

▶ See page 3 for information about the location of the data directory on your machine.

6. Click IMPORT, and the importer will create a faceted *Terrain* from the DEM data.

After importing the terrain, it is possible to view it in the GEOMETRY VIEW in 3D SOLID BODY mode. Due to the large extent of the *Terrain*, it is difficult to perceive the elevation variation of the terrain. The GEOMETRY VIEWś *Z-scaling* and *Color by height* capabilities can draw out the contours of the terrain by stretching it in the Z-direction and coloring the terrain facets according to their elevation.

To turn on *Z-scaling* and *Color by height*:

1. Click *View*→*Z Scaling*→*On* in the GEOMETRY VIEW.
2. Type “3” in the edit box.
3. Click OK.
4. In the GEOMETRY VIEW, choose *Select*→*Geometry*.
5. Click on the *Terrain* in the GEOMETRY VIEW.
6. After the *Terrain* has been selected, right-click to access the terrain’s context menu and choose *Color by height*.

When in 3D SOLID BODY mode, the *Terrain* should look like Figure 2.2.

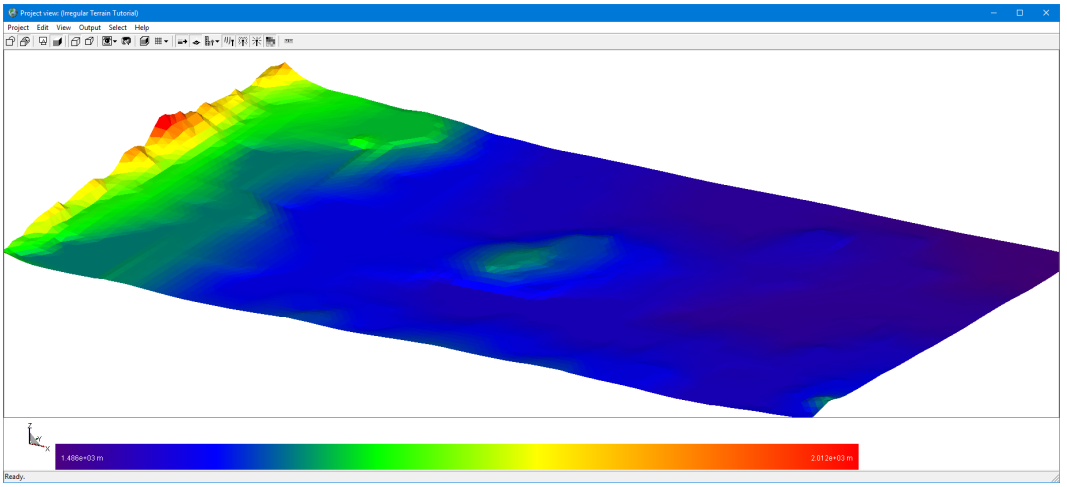


Figure 2.2: Imported terrain in 3D Solid Body Mode with Color by Height

Use the context menu to turn off *Color by height* in preparation for the subsequent steps of the tutorial. Choose *Select*→*Clear* to clear the selection of the terrain.

Loading a USGS DRG file

Wireless InSite supports loading GeoTiff files and overlaying these files on *Terrain*. The GeoTiff files do not affect the calculation; they are only used to help place other features such as TRANSMITTERS and

RECEIVERS in the project. Digital Raster Graphic (DRG) files are a specific type of GeoTiff created from scanned topographic maps which have been georeferenced, and are widely available on the Internet.

1. To open the DRG file, click *File→Open→Image*. An OPEN Window will appear.
2. Navigate to the `install_location\data\DRG` directory and open `Colorado.Plains.USGS.DRG.100K.tiff`. The IMAGE PROPERTIES Window, as seen in Figure 2.3, will prompt you for what part of the image you want to import.

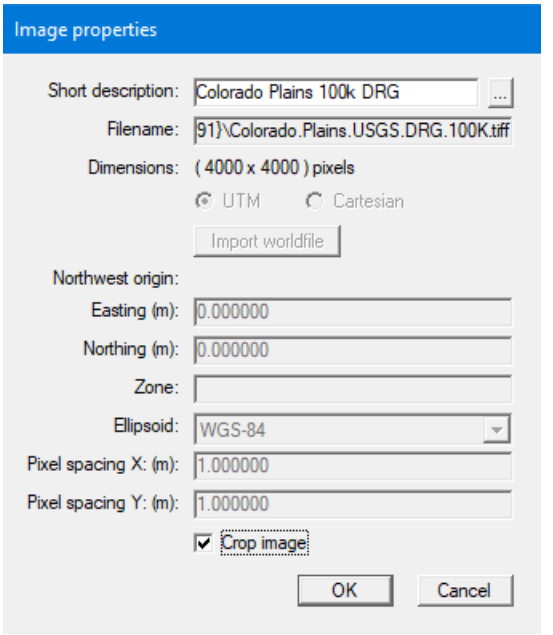


Figure 2.3: Open Image Window

3. Enter “Colorado Plains 100k DRG” for a *Short description*, or accept the default short description.
4. Choose *Crop image*, and click OK. The CROP IMAGE Window appears, as seen in Figure 2.4.
5. Choose FIT TO TERRAIN and click OK. The image imports.

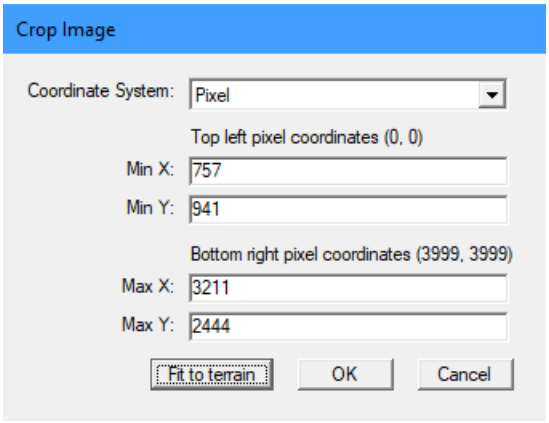


Figure 2.4: Crop Image Window

After the DRG imports, it will cover the *Terrain*. The image will appear green because the image colors are blended with the terrain colors.

To change the color of the terrain:

1. Click the MATERIALS tab in the MAIN WINDOW.
2. Double-click on the material listed. The MATERIAL PROPERTIES Window appears.
3. At the bottom of the left-hand column of the MATERIAL PROPERTIES Window is the color of the material. Click the COLOR button.
4. In the COLOR Window, choose a light gray color. Click OK.
5. Click OK in the MATERIAL PROPERTIES Window to accept the color changes.
6. After choosing a light gray background, the *Terrain* should look like Figure 2.5.

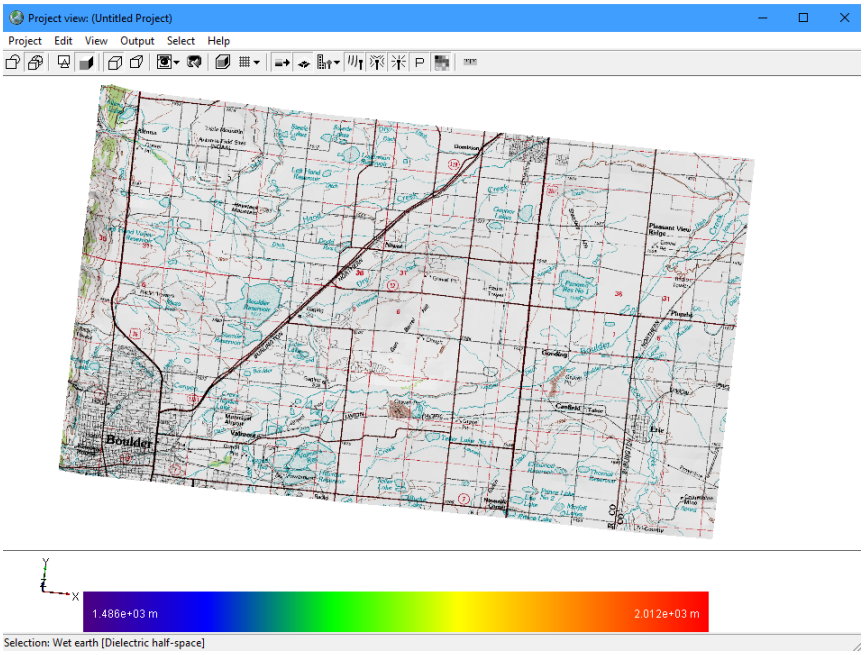


Figure 2.5: Terrain with DRG Overlay

Loading a USGS DOQ File

Another type of GeoTiff file is the USGS Digital Orthophoto Quadrangle (DOQ). USGS DOQ files are aerial photographs that have been corrected for camera orientation and terrain distortion to make them cartographically correct. Opening a USGS DOQ is the same as importing the USGS DRG because they are both GeoTiff files.

1. To open the DOQ file, click *File→Open→Image*. In the OPEN FILE window, navigate to the `install_location\data\DOQ` directory and open `Colorado.Plains.USGS.DOQ.tiff`.
2. Enter "Colorado Plains DOQ" for a *Short description*.
3. When opening the DOQ, do not click the CROP IMAGE button in the first dialog window. The opened DOQ should look like Figure 2.6.

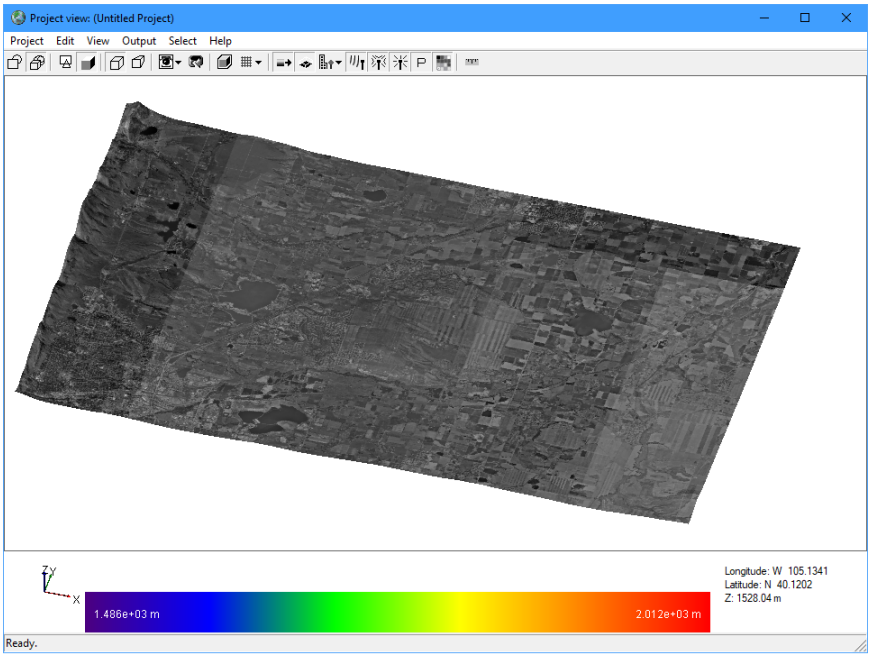


Figure 2.6: Terrain with DOQ overlay

✓ Notice that only one image can be displayed at a certain location at the same time. To switch between images, go to the IMAGES tab in the MAIN WINDOW and select the image you want to view. Make sure it is active and the other images loaded in the project are inactive.

Creating a New Waveform

WAVEFORMS in Wireless InSite describe the time and frequency dependence of the signal radiated from the transmitting ANTENNA.

1. To create a new waveform, click *File→New→Waveform* and choose *Raised cosine*.
2. In the *Short description* box, enter “230 MHz”.
3. In the *Carrier frequency* box, enter “230”.
4. In the *Pulse width* box, enter “4.0e-8”.
5. Ensure the *Phase* and *Roll-off* values are set to “0”.
6. Click OK.

Creating a New Antenna

Now that there is a waveform in the project, an antenna can be created by following these steps:

1. Select *File→New→Antenna*. The CREATE NEW ANTENNA Window appears.

2. Select *Isotropic* from the *Create a new antenna of type* drop-down list.
3. Click OK. The ISOTROPIC ANTENNA PROPERTIES Window will appear.
4. In the *Short description* field, enter “Horizontal Isotropic”.
5. Choose “230 MHz” as the *Waveform*.
6. Select “Horizontal” in the *Polarization* drop-down menu and click OK.

Placing Transmitters

With a waveform and antenna definition, we can now place a TRANSMITTER into the project.

The transmitter height is 6.6 m above the *Terrain*. To add a transmitter, create a new transmitter point:

1. Set *File*→*New*→*Transmitter Set*→*Points*. The GEOMETRY VIEW will switch to 2D.
2. To view latitude/longitude coordinates, select *Lat/Long*→*decimal degrees* from the context menu of the COORDINATE VIEWING Window, as seen in Figure 2.7.

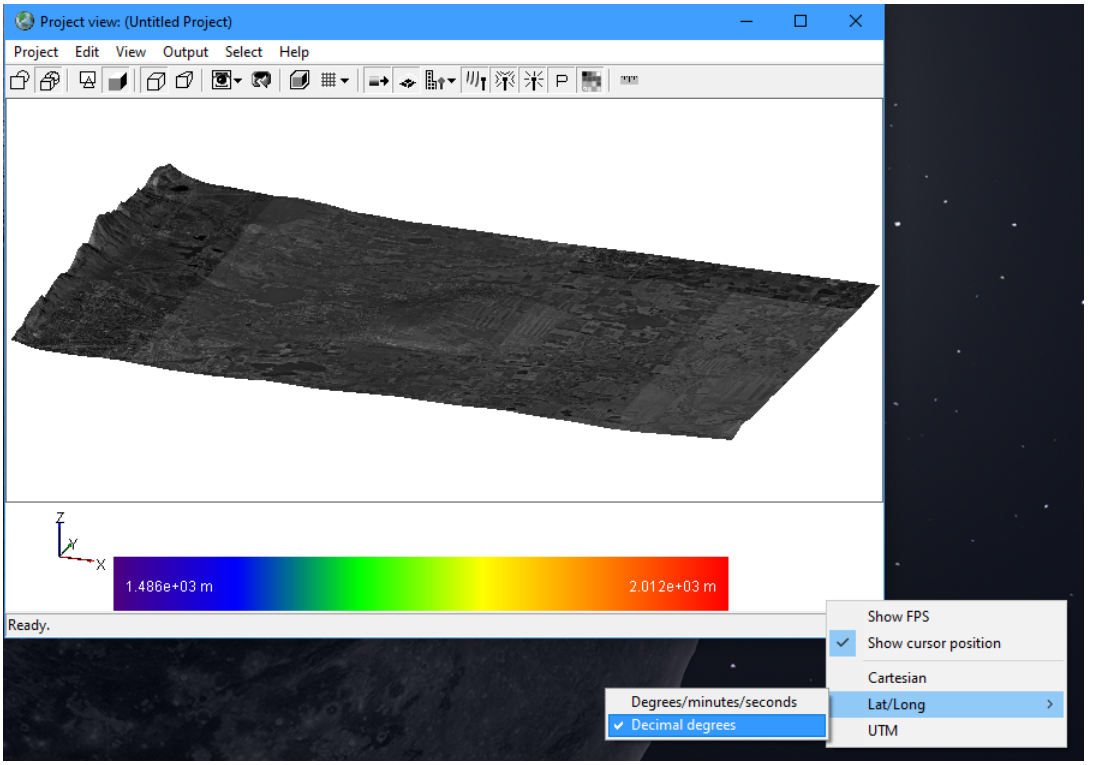


Figure 2.7: Selecting latitude/longitude coordinates for the viewing window

3. The dialog window that appears gives instructions for placing a TRANSMITTER points set. After reading the message, click OK to continue.

4. Move the cursor over the terrain, near 105.2345 W, 40.0653 N. Left click to place a point, and right-click to finish placing points. The TX/RX PROPERTIES Window will appear, as seen in Figure 2.8.
5. Enter “R1-010-T2-O” into the *Short description* edit box.
6. Change the *Origin longitude* and *Origin latitude* to the values above.
7. Click on the TRANSMITTER PROPERTIES button to access the TRANSMITTER PROPERTIES window.
8. Make sure the antenna is set to *Horizontal Isotropic*.
9. Make sure the waveform is set to “Antenna’s waveform” or “230 MHz”.
10. Make sure the input power is set to 0 dBm.
11. Click OK to exit the TRANSMITTER PROPERTIES window.

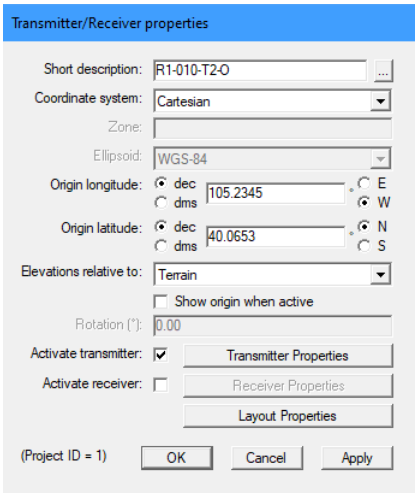


Figure 2.8: The Tx/Rx Properties Window

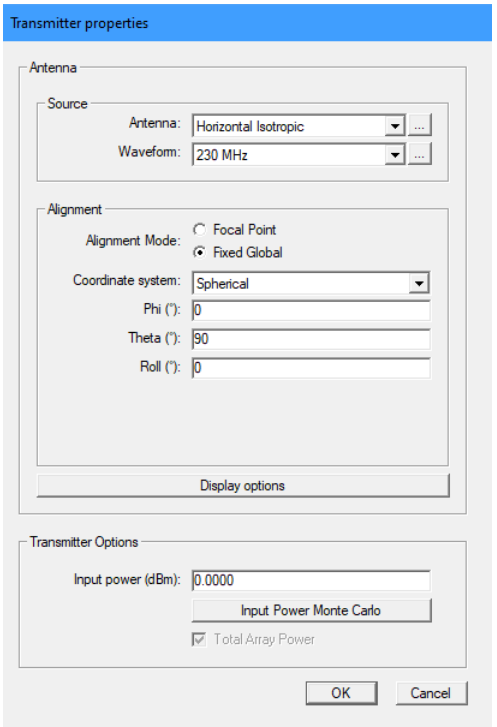


Figure 2.9: The Transmitter Properties Window

✓ The size of the cube representing a transmitter, receiver, or transceiver can be adjusted clicking the LAYOUT PROPERTIES button and changing the *Rendered size* in the TX/RX LAYOUT PROPERTIES Window, as seen in Figure 2.10. Depending on the size of the terrain in the project, it may be necessary to adjust this value to make the TRANSMITTER easily visible in the GEOMETRY VIEW.

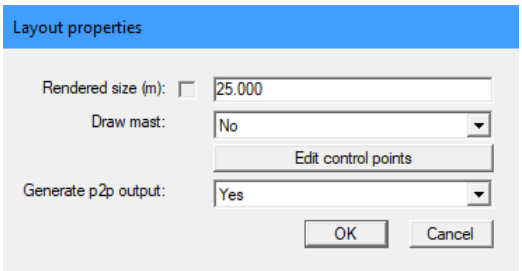


Figure 2.10: The Tx/Rx Layout Properties window for a Point set

12. Remember to set the transmitter's height to 6.6 m.

    (a) Click on the LAYOUT PROPERTIES button.

    (b) Click the EDIT CONTROL POINTS button in the TX/RX LAYOUT PROPERTIES Window to access the VIEW/EDIT VERTICES Window.

(c) Double-click the X-Y-Z entry in the VIEW/EDIT VERTICES dialog box to get the EDIT VERTEX dialog box.

(d) Change the Z value to "6.6" m.

13. Close all open windows by clicking their respective OK buttons.

Creating A Route of Receivers

It is now time place the RECEIVERS. A vertical Receiver *Route* is going to be created to simulate measuring path gain at different heights.

To create a vertical Receiver *Route*:

1. Click *File*→*New*→*Receiver Set*→*Route*. The GEOMETRY VIEW will switch to 2D.
2. To create the vertical *Route*, move the mouse cursor to a point close to 105.1255W, 40.0939N.
3. Click twice, and then right-click. This places two receivers at the same location, and we will manually define their heights through the TX/RX LAYOUT PROPERTIES Window.

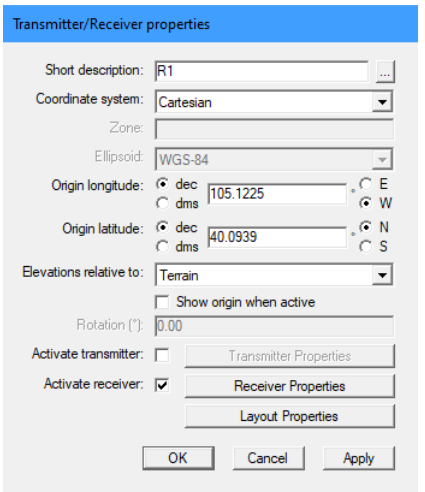


Figure 2.11: The Route's Tx/Rx Properties Window

4. In the *Short description* field enter "R1".
5. Change the *Origin longitude* and *Origin latitude* to the values shown in Figure 2.11.
6. Click on the LAYOUT PROPERTIES button.
7. In the *Spacing* field enter "0.50 m".
8. To change the height of the control points of the *Route*, click on the EDIT CONTROL POINTS button. The VIEW/EDIT VERTICES Window will open and have two points listed inside of it.
9. Double click on the first point. The EDIT VERTEX Window will open.

   (a) Enter "0" m for X, "0" m for Y, and "0.1" m for Z.

(b) Click OK.

(c) Double-click on the second vertex and enter “0” m for X, “0” m for Y, and “15.1” m for Z.

(d) Click OK.

10. Click OK to close the VIEW/EDIT VERTICES Window.

✓ The rendered size of the receivers can be changed in the TX/RX LAYOUT PROPERTIES, if desired.

11. Click OK to exit the TX/RX LAYOUT PROPERTIES window

12. Click OK to complete creation of the vertical Receiver *Route*.

Creating a Study Area

In order to run calculations, each project must have a STUDY AREA defined. Figure 2.12 shows the complete study area properties.

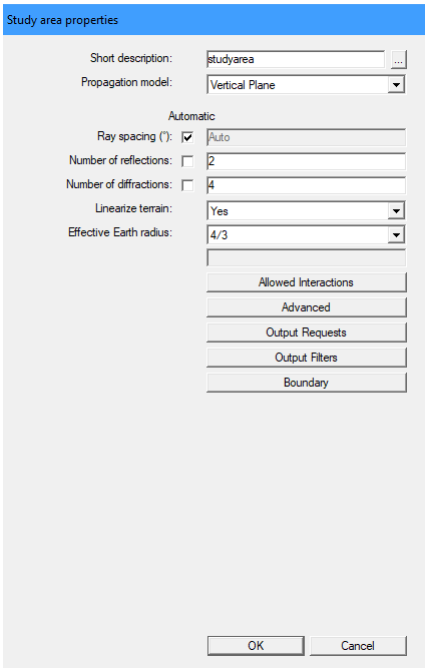


Figure 2.12: The Study Area Properties Window

To create the study area:

1. Click *File*→*New*→*Study Area*.
2. Choose *Fit to geometry*.
3. Click BEGIN. The STUDY AREA PROPERTIES Window opens.

4. Enter parameters for the calculation model.
   (a) The *Propagation model* should be set to “Vertical Plane”.
   (b) Deselect the *Automatic* checkbox on *Number of reflections* and set this to “2”.
   (c) Deselect the *Automatic* checkbox on *Number of diffractions* and set this to “4”.
   (d) Set *Effective Earth radius* to “4/3”.
5. Click OK.

Now we will view the original, non-linearized *Terrain* profile.

1. Select *Project View→Select→Transmitter/Receiver Pair*, and click on both the transmitter *Point* and receiver *Route*.

   If descriptions are shown in the GEOMETRY VIEW, they may interfere with the selection process. To deactivate them select *Project View→View→Descriptions*.

2. Select *Graph terrain profile* from the context menu.
3. Disable the field *Draw propagation paths*, but leave the other options checked.
4. Click OK. (The propagation paths may be drawn after running the calculation.) A Wireless InSite graph will be generated with the original profile, the transmitter and the receiver. An example of the terrain profile is shown in Figure 2.13.

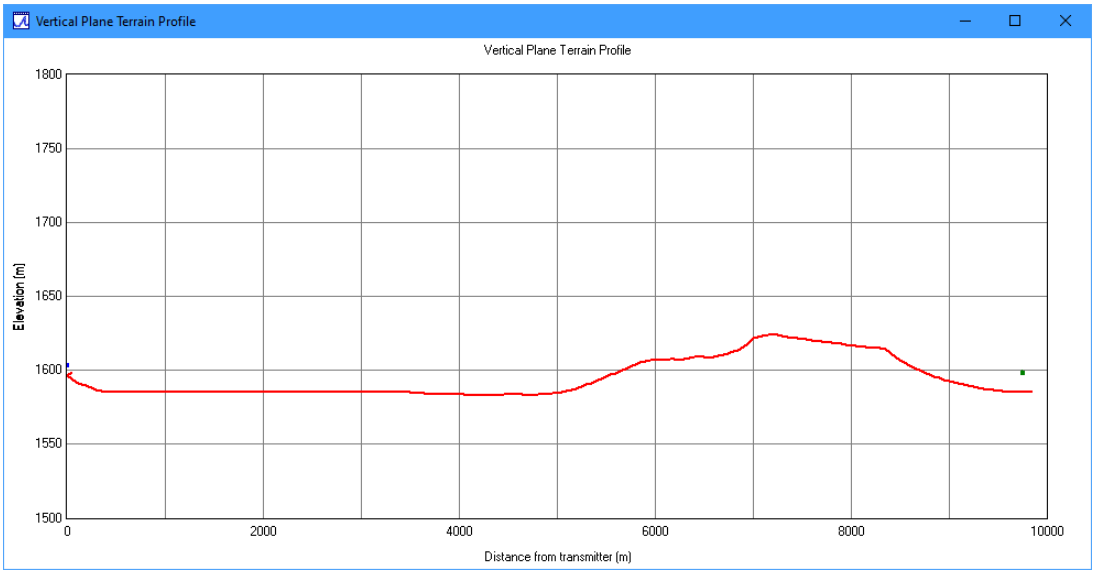


Figure 2.13: A terrain profile graph

Running the Calculation

Now the project is ready to run. To start, click RUN→*New*. Wireless InSite will ask you to save the project and terrain files before running. Save them in a new folder. Wireless InSite will then run the calculation.

Viewing Propagation Paths

Propagation paths can be viewed once the calculation has run. To view them, follow these steps:

1. In the GEOMETRY VIEW, select *Select→Transmitter-Receiver Pair*.
2. Click on the receiver and a transmitter to select them.
3. Choose *Graph terrain profile* from the context menu.
4. Leave all options checked.
5. Click OK to display the graph.
6. Select *Adjust scale→Y axis* from the graph context menu to set a manual minimum and manual maximum. The minimum Y value should be around 1580 m, and the maximum Y value should be about 1625 m. This provides maximum separation between paths for easy viewing. The graph is seen in Figure 2.14.

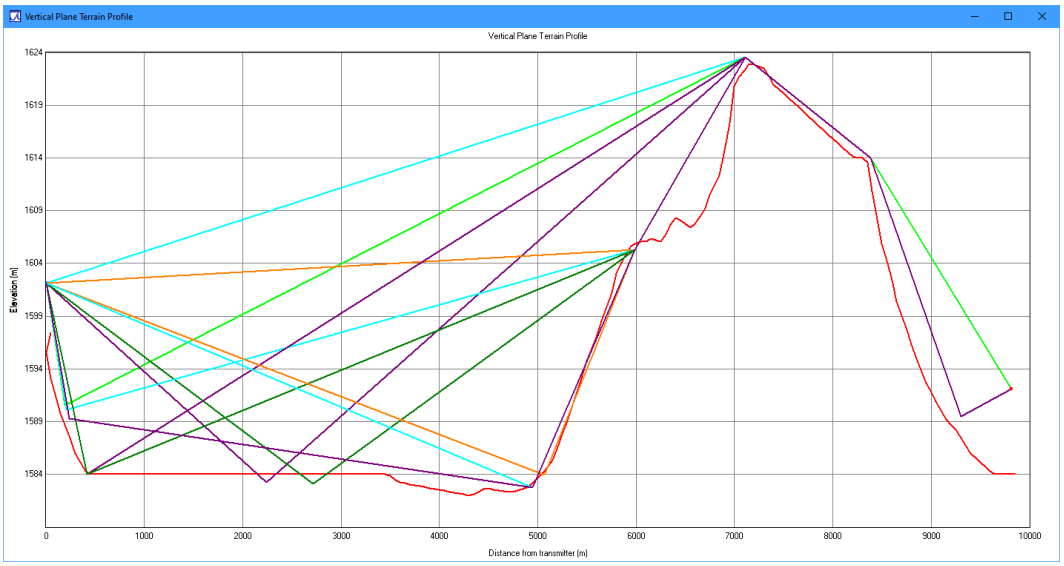


Figure 2.14: Terrain profile graph with propagation paths visible

Comparison to Measured Path Gain

Once the calculation is finished, requested output can be viewed in the GEOMETRY VIEW or plotted. To plot the path gain from the TRANSMITTER R1-010-T2-O to RECEIVER route R1:

1. Click on RESULTS in the MAIN WINDOW.
2. Expand *Area: studyarea→Point to multipoint→Path gain*. The path gain for each transmitter point to receiver set pairing is listed under *Path gain*.
3. Right click on *R1-010-T2-O→R1* and select *Plot* from its context menu. The CHOOSE PLOT Window will open.

4. Under *Select axis*, choose “Distance”.
5. Click OK.
6. A graph is created that includes a plot of the calculated path gain between the receiver set and the R1-010-T2-O transmitter.
7. Right-click on the graph and choose *Properties*.
8. Set the *Title* to ”R1-010-T2-O Path gain vs. Height”.
9. Set the *X label* to ”Height”.
10. Click OK.
11. To import the measured plot into this graph, right-click on the graph and choose *Import plot*. The OPEN FILE Window will appear.
12. Navigate to `tutorials_location\Irregular_Terrain_Tutorial\230` and open `r1-010-t2-o.230.plt`.
13. The plot, as seen in Figure 2.15, will appear. In the graphs context menu select *Plot properties...*, select *r1-010-t2-o.230.plt* from the list and modify the plot as desired and click OK. It is also possible to change the graph through its context menu by clicking *Properties*. Additional plots of measured and calculated data can be graphed through the same procedure.

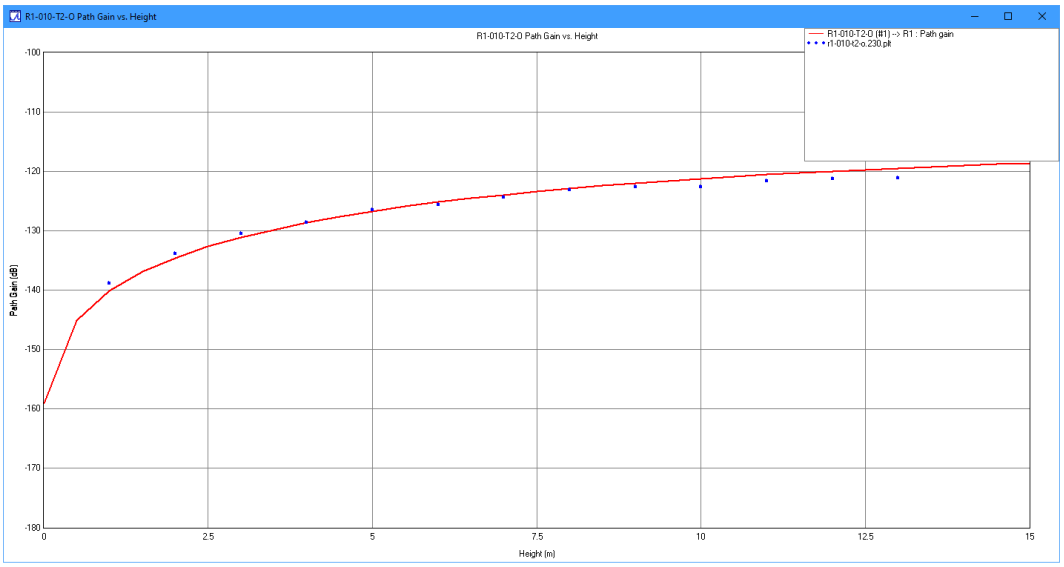


Figure 2.15: R1-010-T2-O measured and calculated results path gain vs. receiver height

Creating a Vertical Surface of Receivers

The vertical Receiver *Route* provided an accurate simulation of how the ITS data measurements were taken. To better understand propagation between a transmitter and receiver, it can be helpful to visualize

how the path gain varies along the terrain profile. A *Vertical Surface* extending from the transmitter to the receiver route can be used to calculate and display the path gain variations. Creating Vertical Surfaces of receivers is very similar to creating *Route*.

To create a *Vertical Surface*:

1. Click *File*→*New*→*Receiver Set*→*Vertical Surface*.
2. The GEOMETRY VIEW changes to 2D, with the receiver and transmitter sets still visible. Zoom in until the green transmitter is visible.

   ✓ If transmitters and/or receivers are not visible, use the *Project view*→*View* menu to make them visible.
3. Click in the middle of the transmitter point.
4. Pan the view until the red receiver set is visible. Click in the middle of the receiver set.
5. Right-click to terminate entry of the new *Vertical Surface* receiver set. The SPECIFY HEIGHTS Window appears.
6. Enter “0.0” m for the *Base height* and “200” m for the *Top height*. Click OK, and the TX/RX PROPERTIES Window appears.
7. Name the set “Rx Vertical Surface”.
8. Click on LAYOUT PROPERTIES and in the TX/RX LAYOUT PROPERTIES Window, change the *Spacing* to “15” m and the *Vertical spacing* to “15” m.
9. Click OK to exit the TX/RX LAYOUT PROPERTIES Window, and click OK again to exit the TX/RX PROPERTIES Window. The transmitter and receivers should resemble Figure 2.16.

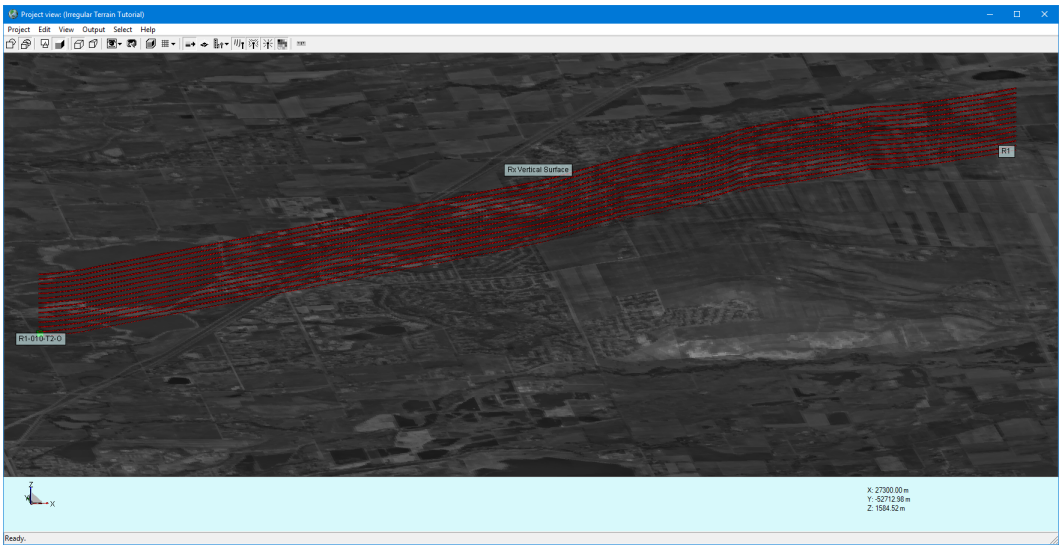


Figure 2.16: Vertical Surface of receivers created

10. Run the calculation by selecting RUN→*New* from the MAIN WINDOW. After the calculation finishes, view the output in the GEOMETRY VIEW.

(a) In the RESULTS tab, navigate to *Area: studyarea→Point to multipoint→Path gain→R1-010-T2-O→Rx Vertical Surface*. Right-click on *Rx Vertical Surface* and select *View* from the context menu. The output will appear inside the receiver set.

(b) Turn off the receiver boxes by clicking *View→Receivers*.

(c) Right-click on the scale bar and select *Range options→Manual scaling* and enter -110 as the *Minimum value* and -50 as the *Maximum value*.

(d) The output should look similar to Figure 2.17.

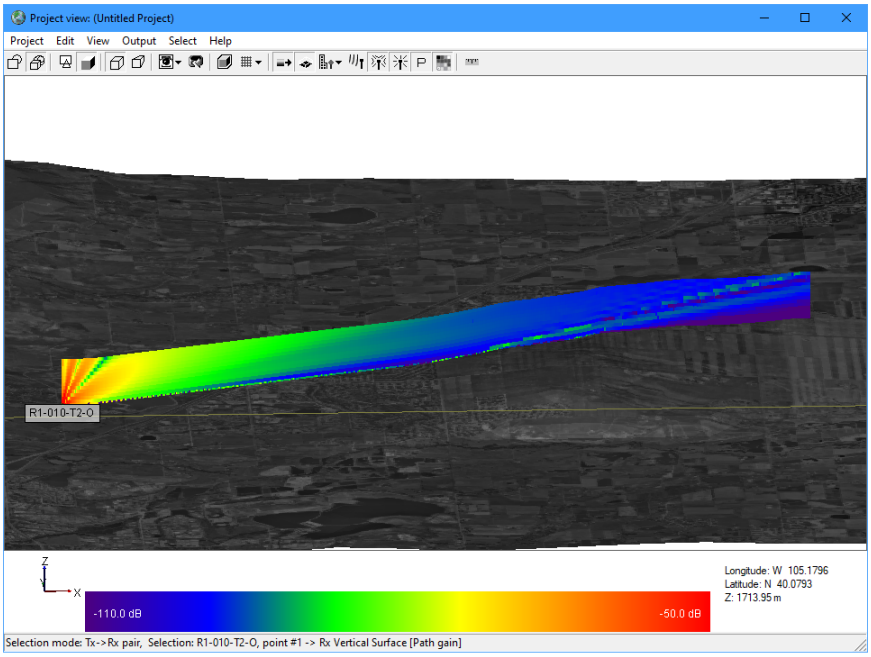


Figure 2.17: Path gain on the vertical surface of receiver

Chapter 3

Indoor Tutorial

This tutorial explores some of the features of Wireless InSite for making indoor propagation predictions. The scenario is based on the first example presented in [2] and provides direct comparison to measurements. Wall geometry and material data are given in Table 3.1 and Figure 3.1.

Table 3.1: Indoor Tutorial floor plan material properties

	900 MHz	1800 MHz
Wood	$\epsilon_r = 3.0$	$\epsilon_r = 3.0$
RC Wall	$\epsilon_r = 6.1 - j1.20$	$\epsilon_r = 6.2 - j0.69$
Brick Wall	$\epsilon_r = 5.1 - j0.20$	$\epsilon_r = 5.2 - j0.28$
Metal Structure	$\epsilon_r = 1.0 - j2e8$	$\epsilon_r = 1.0 - j1e8$
Ceiling/Floor	$\epsilon_r = 10.0 - j1.20$	$\epsilon_r = 7.9 - j0.89$

There are no dimensions given for the wall spacing and thickness, so these were estimated from the floor plan.

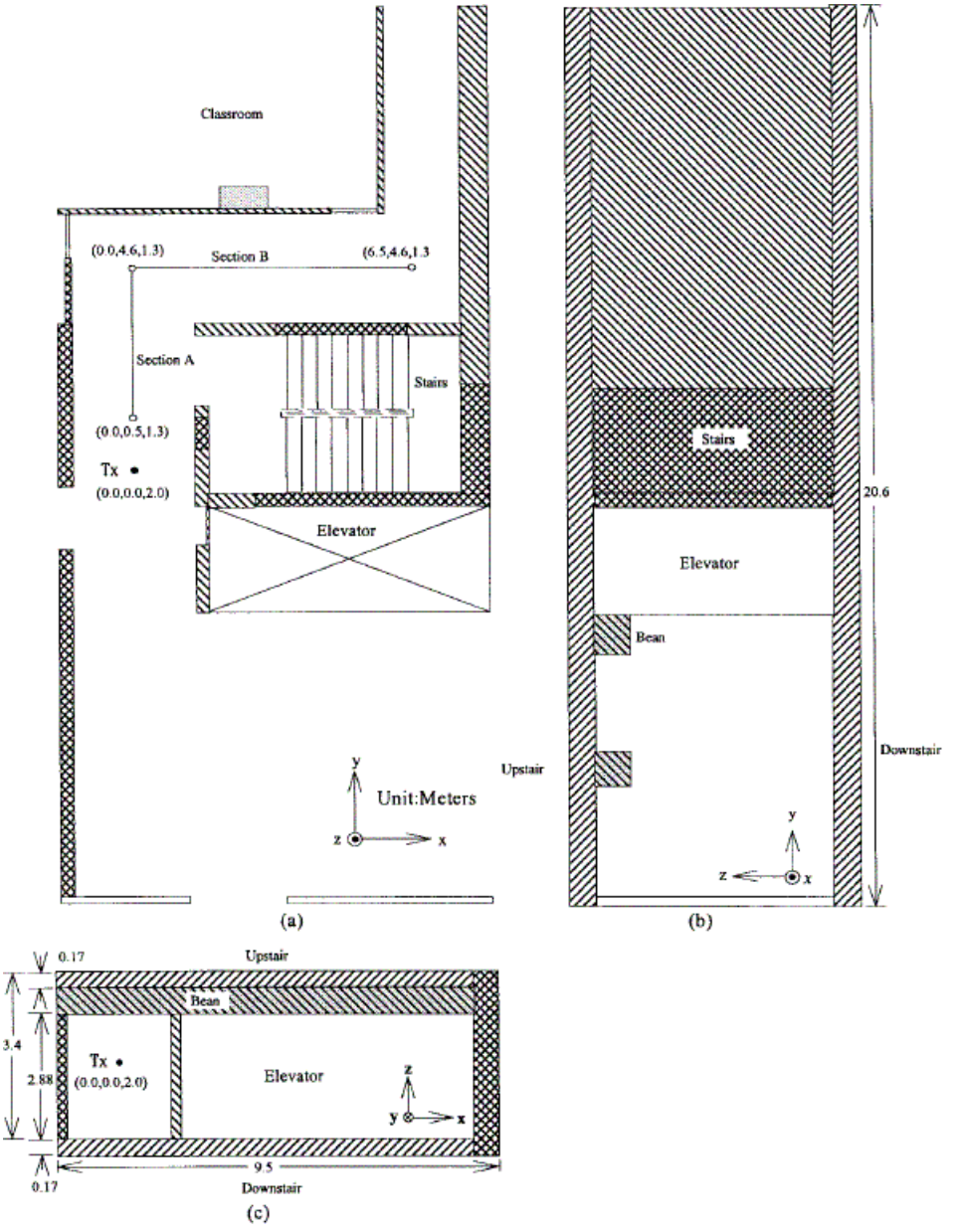


Figure 3.1: The original Indoor Tutorial floor plan

Loading the Floor Plan

Because of the complexity of the floor plan in this example, we will import a floor plan from our file library with the appropriate materials.

1. To begin, create a new project.
2. The floor plan file can be accessed by *File→Open→Geometry→Floor Plan*. For a typical installation, the file is located at `tutorials_location\Indoor_Tutorial\IndoorFloorPlan.flp`.
   - See page 3 for information about the location of the tutorial projects on your machine.

When viewed in 3D SOLID BODY modes in the GEOMETRY VIEW, the floor plan will look like the one seen in Figure 3.2.

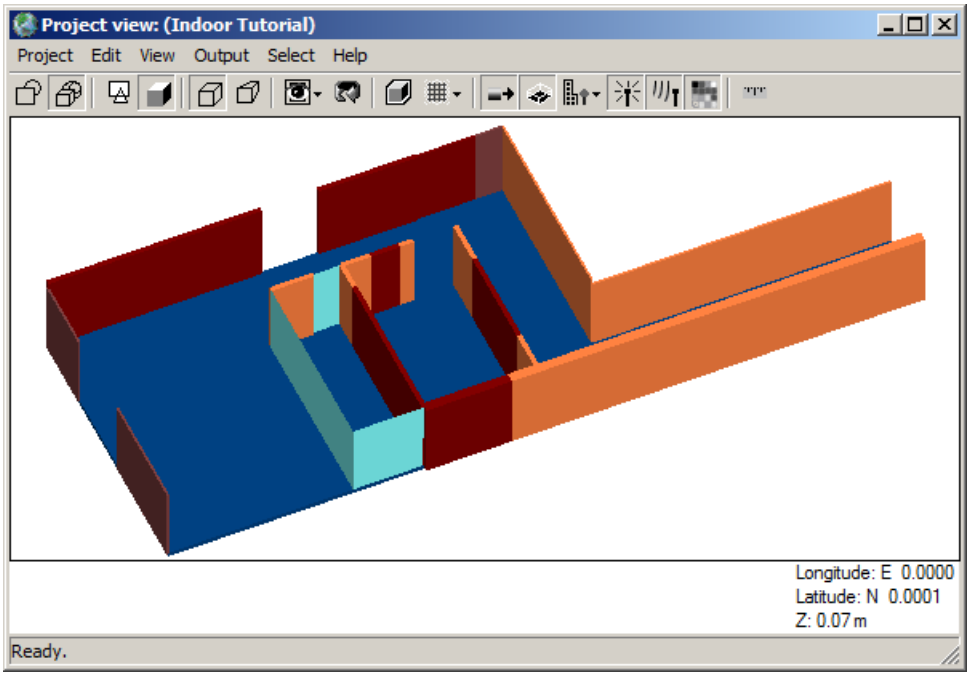


Figure 3.2: The imported floor plan

Define Antennas and Waveforms

First we will create a 900 MHz narrowband *Sinusoid* waveform.

1. Navigate to *File→New→Waveform* and choose *Sinusoid*. The SINUSOID PROPERTIES Window will appear, as seen in Figure 3.3.

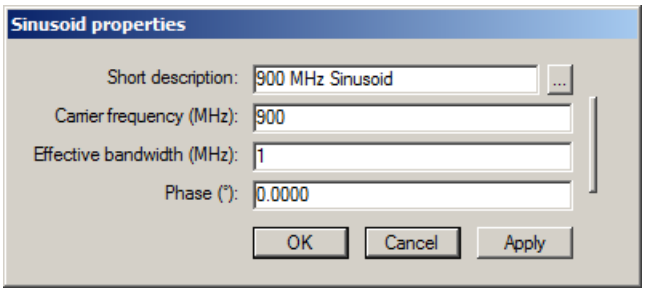


Figure 3.3: The Sinusoid Properties Window

2. In the *Short description* box, enter “900 MHz Sinusoid”.
3. In the *Carrier frequency* box, enter “900” MHz.
4. In the *Effective bandwidth* box, enter “1” MHz.
5. Ensure the *Phase* value is set to “0”.
6. Click OK.

Now that there is a waveform in the project, create a vertically-polarized *Linear Dipole* antenna.

1. Select *File→New→Antenna* and choose *Linear dipole*. The LINEAR DIPOLE ANTENNA PROPERTIES Window will appear, as seen in Figure 3.4.

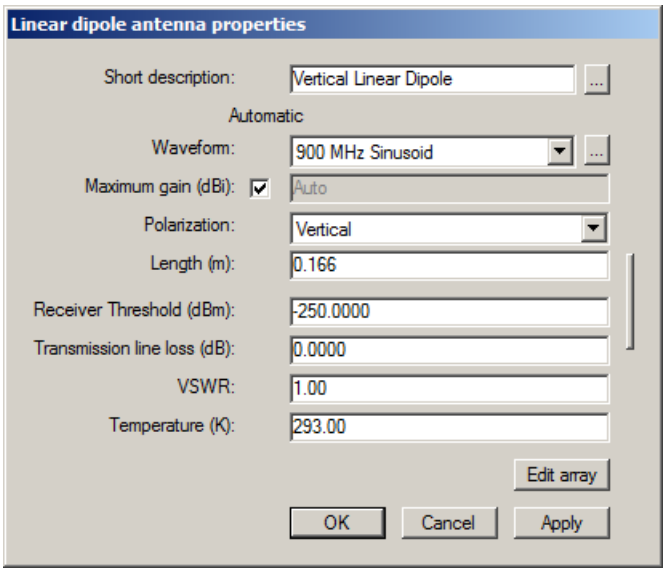


Figure 3.4: The Linear Dipole Antenna Properties Window

2. In the *Short description* field, enter “Vertical Linear Dipole”.
3. Choose “900 MHz Sinusoid” as the *Waveform*.

4. Select “Vertical” in the *Polarization* edit box.
5. Enter “0.166” m as the *Length* and click OK.

Creating the Tx Points and Rx Route

We are now ready to place TRANSMITTER and RECEIVER points at the locations shown in Figure 3.1.

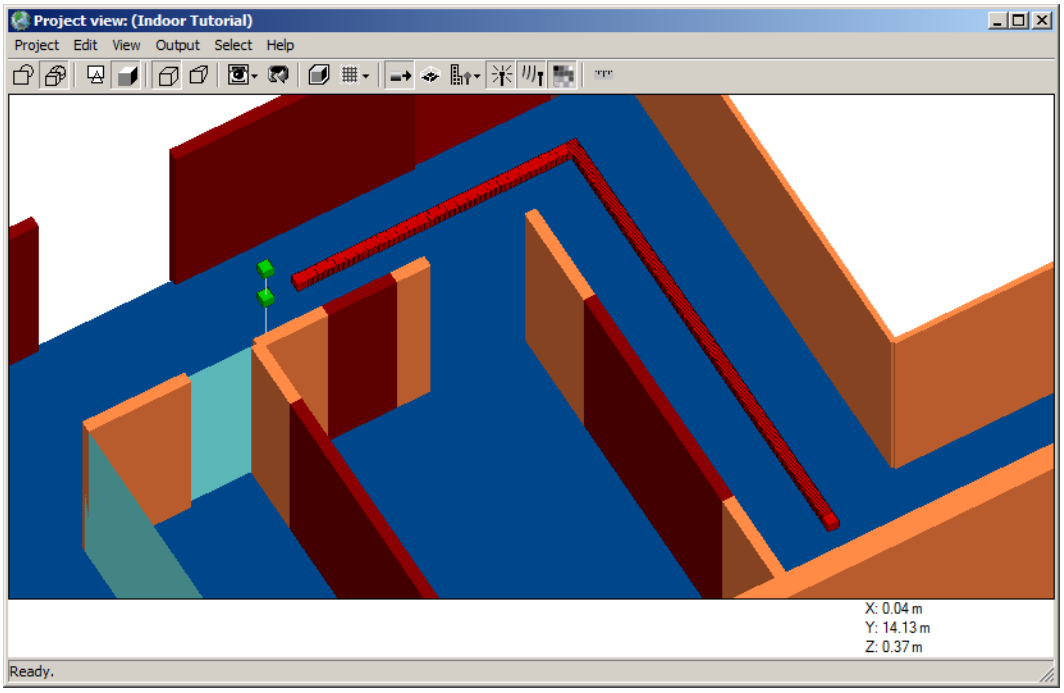


Figure 3.5: Transmitter Points (green) and Receiver Route (red) placed in the indoor tutorial

To create the TRANSMITTER points:

1. Click *File→New→Transmitter Set→Points*.
2. Click on a place in the middle of the floor plan hallway, and right-click to finish.
3. The TX/RX PROPERTIES Window will appear, as seen in Figure 3.6.

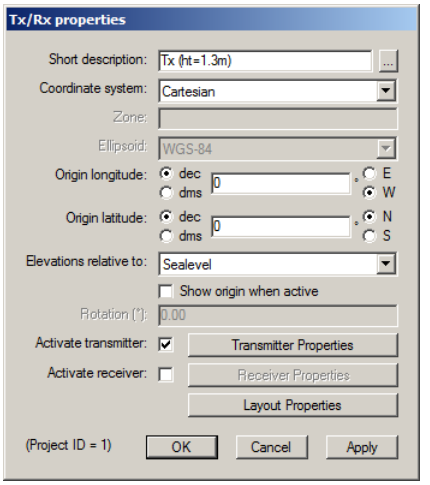


Figure 3.6: The Tx/Rx Properties Window

4. Enter “Tx (ht=1.3m)” as the *Short description*.
5. Enter “0” for both the *Origin longitude* and *Origin latitude*.
6. Click on TRANSMITTER PROPERTIES

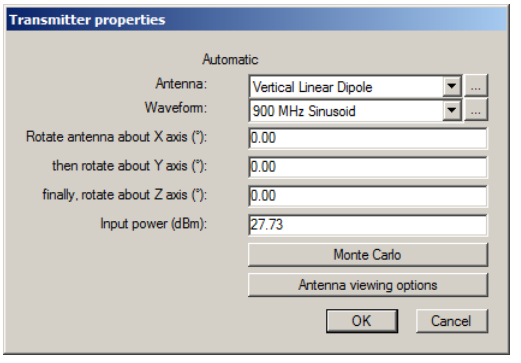


Figure 3.7: The Transmitter Properties Window

7. Select the “Vertical Linear Dipole” as the *Antenna*, which was created earlier.
8. Enter the calibrated received power found in the paper [2], or 27.73 dBm, as the *Input power*.
9. Click OK to exit the TRANSMITTER PROPERTIES window.
10. Click LAYOUT PROPERTIES

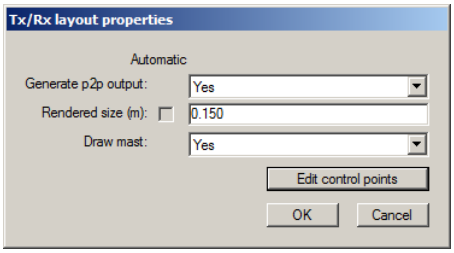


Figure 3.8: The Tx/Rx Layout Properties Window

11. Click EDIT CONTROL POINTS, double click on the point, and set X=1.5 m, Y=10 m, and Z=1.3 m.
12. Click OK twice to finish the creation of the TRANSMITTER point.

Create an additional TRANSMITTER by duplicating *Tx (ht=1.3m)*:

1. In the MAIN WINDOW, go to the Transmitters tab, and right-click on the entry for *Tx (ht=1.3m)*.
2. Select *Duplicate* from the context menu.
3. Double-click on the entry for *Copy of Tx (ht=1.3m)*.
4. The TX/RX PROPERTIES Window will appear.
5. Enter "Tx (ht=1.96m)" as the *Short description*.
6. Click LAYOUT PROPERTIES.
7. Click EDIT CONTROL POINTS, double click on the point, and set X=1.5 m, Y=10 m, and Z=1.96 m.
8. Click OK to exit the View\Edit Vertices window.
9. Click OK twice to close all properties windows.

Now we will create a *Route* of receivers, created from two sets. To capture fast fading at 900 MHz, the receiver spacing is made smaller than the default values.

1. Click *File*→*New*→*Receiver Set*→*Route*.
2. Click on a place just above the transmitter, and click again to set the endpoint at the end of the hallway. Right-click to finish.
3. The RECEIVER PROPERTIES Window will appear, as seen in Figure 3.9.

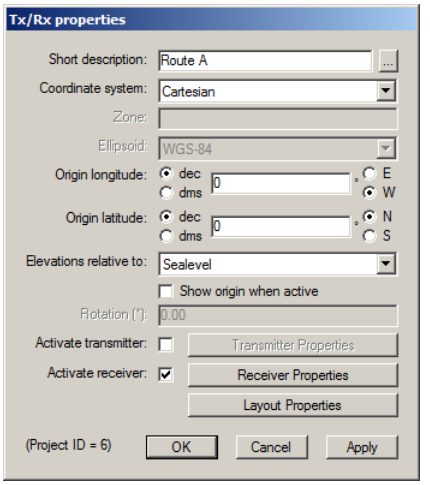


Figure 3.9: The Tx/Rx Properties Window

4. Enter “Route A” as the *Short description*.
5. Enter “0” for both the *Origin longitude* and *Origin latitude*.
6. Click RECEIVER PROPERTIES.

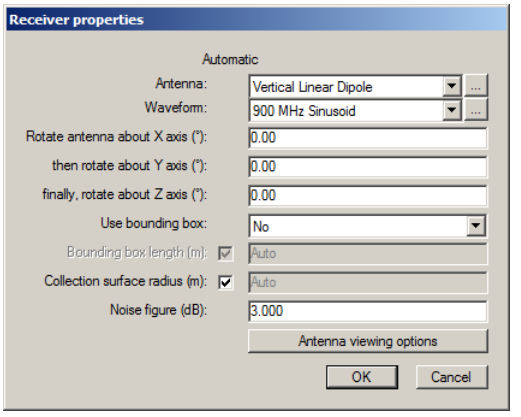


Figure 3.10: The Receiver Properties Window

7. Select the “Vertical Linear Dipole” as the *Antenna*, which was created earlier.
8. Click OK to exit RECEIVER PROPERTIES Window.
9. Click LAYOUT PROPERTIES.

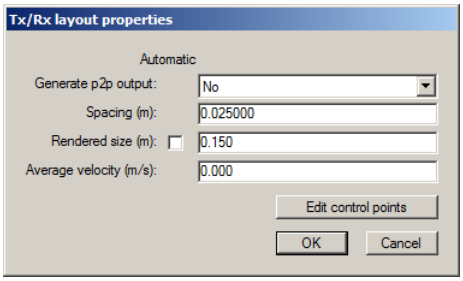


Figure 3.11: The Layout Properties Window for Route A.

10. Enter “0.025” m as the *Spacing*.
11. Click EDIT CONTROL POINTS.
    (a) For the first point, set X=1.5 m, Y=10.5 m, and Z=1.3 m.
    (b) For the second point, set X=1.5 m, Y=14.5 m, and Z=1.3 m.
12. Click OK twice to finish the creation of the first *Route*.

Following the same procedure, create Receiver “Route B”, extending from the end of “Route A” to the end of the hallway. The first control point is located at (1.5 m, 14.5 m, 1.3 m), and the second located at (8 m, 14.5 m, 1.3 m).

Defining the Study Area

We will now create a STUDY AREA which encloses the entire floor plan.

1. Click *File*→*New*→*Study area*.
2. Select *Fit to geometry* and click BEGIN. The STUDY AREA PROPERTIES Window will appear, as seen in Figure 3.12.

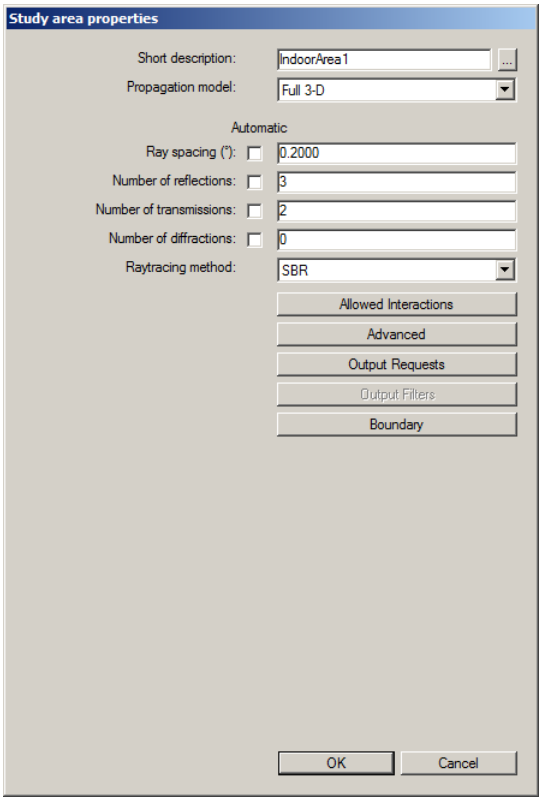


Figure 3.12: Study area properties for the Indoor Tutorial

3. Enter “IndoorArea1” as the *Short description*.
4. Select “Full 3-D” as the *Propagation model*.
5. Deselect the *Automatic* checkbox on *Ray Spacing* and set this to “0.2”.
6. Deselect the *Automatic* checkbox on *Number of reflections* and set this to “3”.
7. Deselect the *Automatic* checkbox on *Number of transmissions* and set this to “2”.
8. Deselect the *Automatic* checkbox on *Number of diffractions* and set this to “0”.
9. Click the ADVANCED button.
    - Deselect the *Automatic* checkbox on *Path loss threshold* and enter “-60” dB.
    - Deselect the *Automatic* checkbox on *Sum complex electric fields* and set it to “All”, and click OK.
10. Click OK to finish the creation of the STUDY AREA.

Running a Calculation

To start a calculation, click on RUN button in the MAIN WINDOWand select *New*. If the project has not been saved, Wireless InSite will be prompt you to do so. When the calculation starts, the SIMULATION LOG will open and provide information on the progress of the calculation.

Comparison to Measurements

The measured results were manually read from the figures in the paper and written to a text files with the extension `.plt`.

1. Under RESULTS, drill down to *Area:IndoorArea1→Point to multipoint→Received power→Tx (ht=1.3m)*. Right-click on *Route A* and select *Plot*.
2. Select the *Distance* as the x-axis and press OK. The received power plot appears.
3. To import the measured data, right-click on the plot and select *Import→Data →PLT*.
4. Open the plot:`tutorials_location\Indoor_Tutorial\measureddata\fig11ameasured.plt`. The graph should look similar to Figure 3.13.

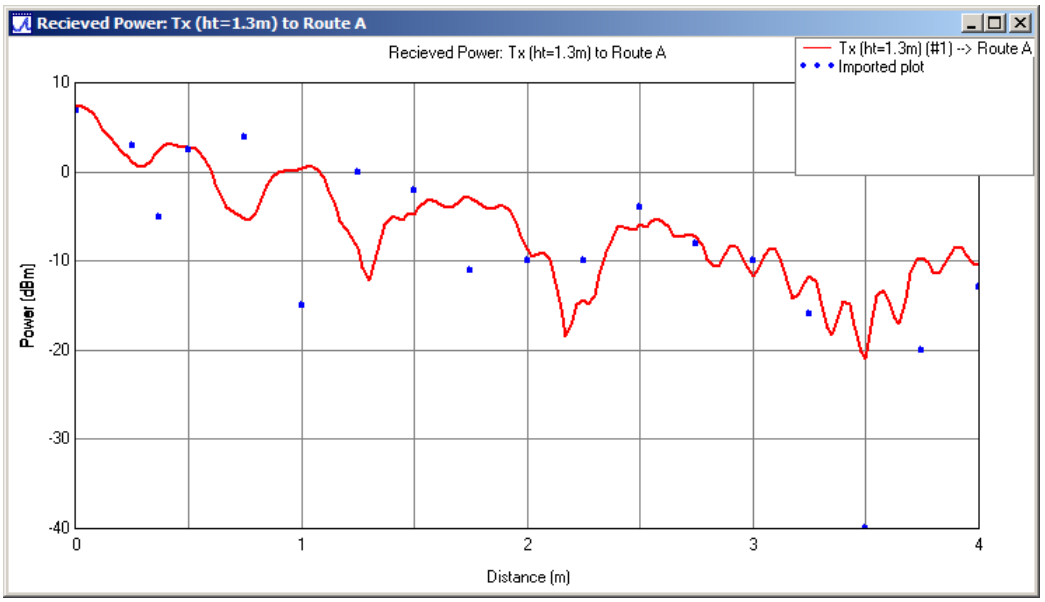


Figure 3.13: Comparison of Wireless InSite calculations to measurements for receiver Route A with a transmitter height of 1.3 m

Following the same procedure, create a graph of the received power from *Tx (ht=1.3m)* to Route B using the measured data stored in the file `fig11bmeasured.plt`. The graph should look similar to Figure 3.14.

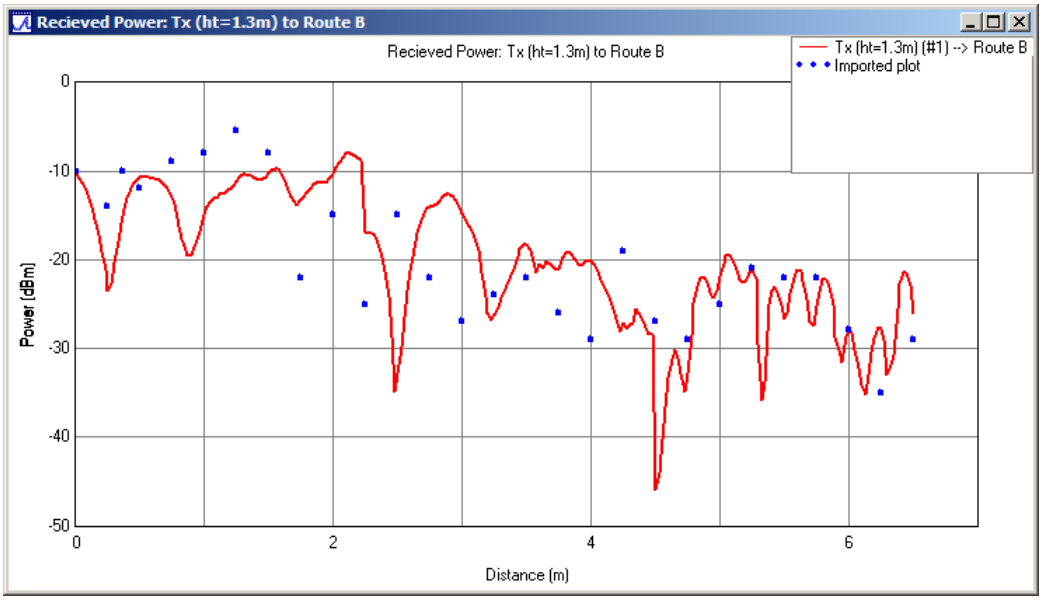


Figure 3.14: Comparison of Wireless InSite calculations to measurements for receiver Route B with a transmitter height of 1.3 m

Figures 3.15 and 3.16 can be replicated by plotting the received power vs distance for *Tx (ht=1.96m)* for routes A and B, respectively. Measurement data for Figure 3.15 is located in `fig12bmeasured.plt` and measurement data for Figure 3.16 is located in `fig12bmeasured.plt`.

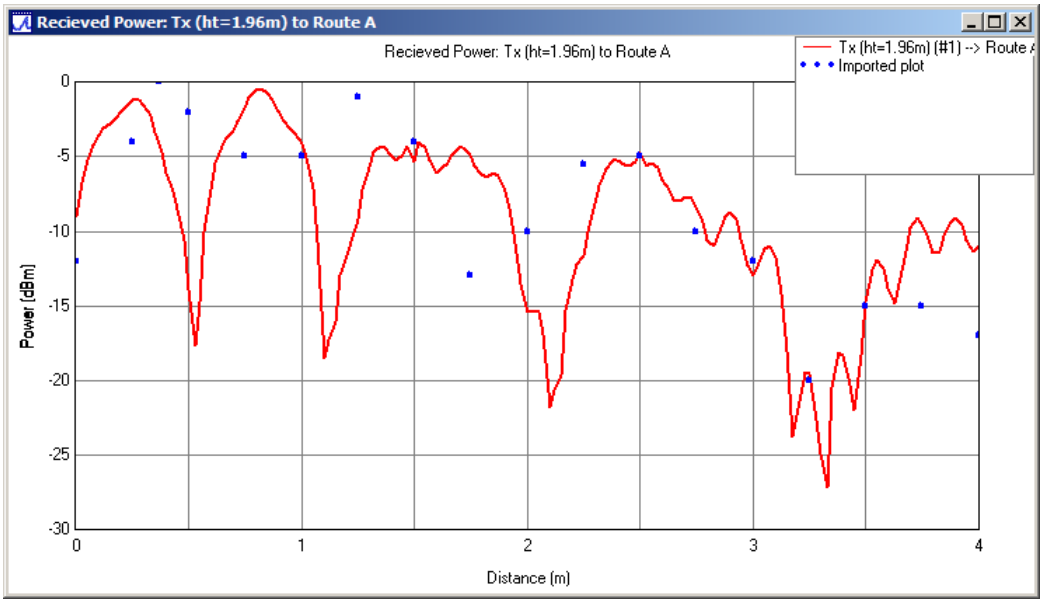


Figure 3.15: Comparison of Wireless InSite calculations to measurements for receiver Route A with a transmitter height of 1.96 m

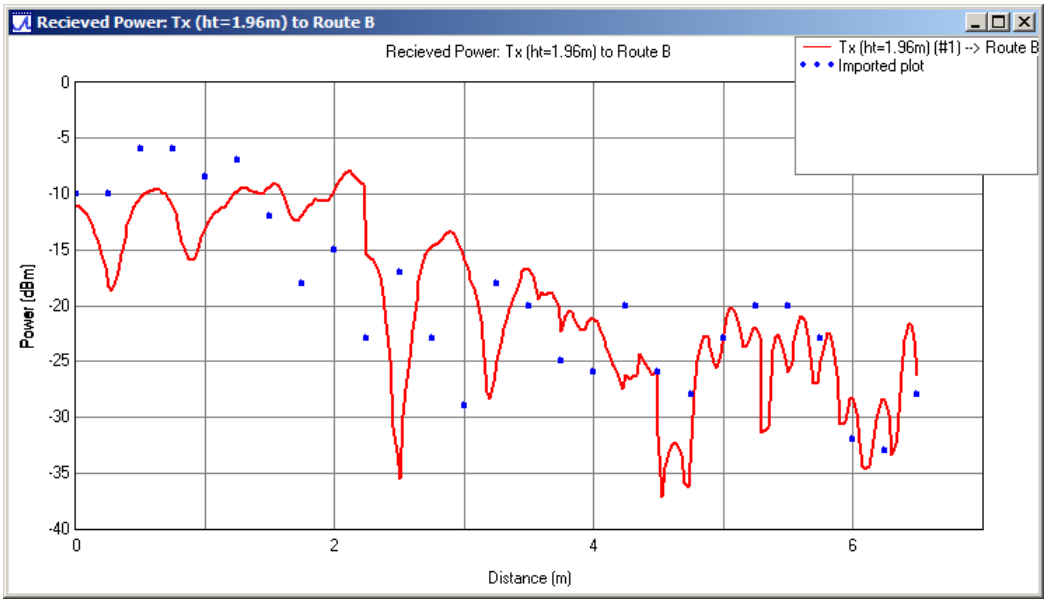


Figure 3.16: Comparison of Wireless InSite calculations to measurements along receiver Route B with a transmitter height of 1.96 m

Figures 3.13 through 3.16 show good overall agreement between Wireless InSite predictions and measurements data. Wireless InSite does not exactly reproduce the fast fading seen in the measurements, but this is because the exact wall locations and thickness are not explicitly reported in the paper. Even without precise geometry, Wireless InSite replicates the signal levels and general trends of the measurements.

Viewing Ray Path Output

Ray paths for a Tx/Rx pair are shown in Figure 3.17. Paths from the TRANSMITTER to a RECEIVER can viewed this by:

1. Under RESULTS, drill down to *Area:IndoorArea1→Point to multipoint→Propagation paths→Tx (ht=1.96m)*. Right-click on *Route B* and select *Load*.
2. Expand *Route B*, scroll down, and locate the entry for *Receiver #246: 15 Paths*.
3. Right-click on the entry for *Receiver #246: 15 Paths.* and select *View all paths*. The GEOMETRY VIEW should look similar to Figure 3.17.

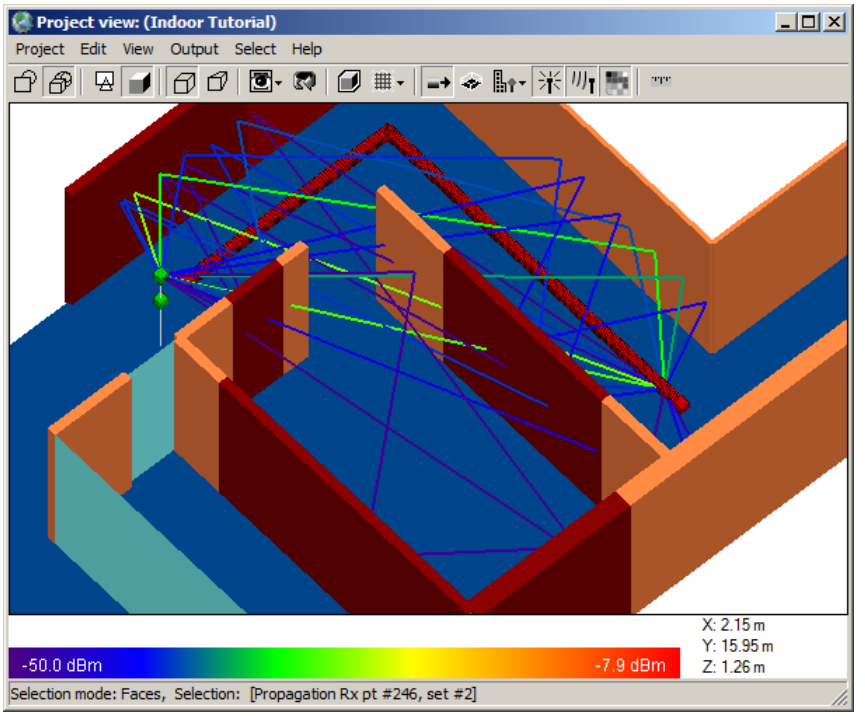


Figure 3.17: Propagation ray paths for the indoor tutorial

Creating a Grid of Receiver Points in the Hallway

Receiver *Grids* allow the user to view received power over large areas and to quickly recognize any fast-fading effects that may be present in the *Floor plan*.

To prepare for the calculation with the receiver *Grid*, first disable the receiver *Routes*. Right click on each Route in the RECEIVERS tab, and uncheck both *Visible* and *Active*. Additionally, disable the TRANSMITTER at height = 1.3 m by right-clicking on *Tx (ht=1.3m)* TRANSMITTER tab, and uncheck both *Visible* and *Active*.

For this example, two Receiver *Grids* were created to provide coverage around the original L-shaped receiver routes. At least 8 receivers per wavelength are required, however 10 receivers per wavelength is better. Given the 900 MHz WAVEFORM, a spacing of 0.03 m is appropriate.

1. Select *File→New→Receiver Set→XY Grid*.
2. Initiate the grid in the GEOMETRY VIEW by clicking at one corner of the desired grid location and dragging to the opposite corner, to create a grid covering the area seen in Figure 3.18.

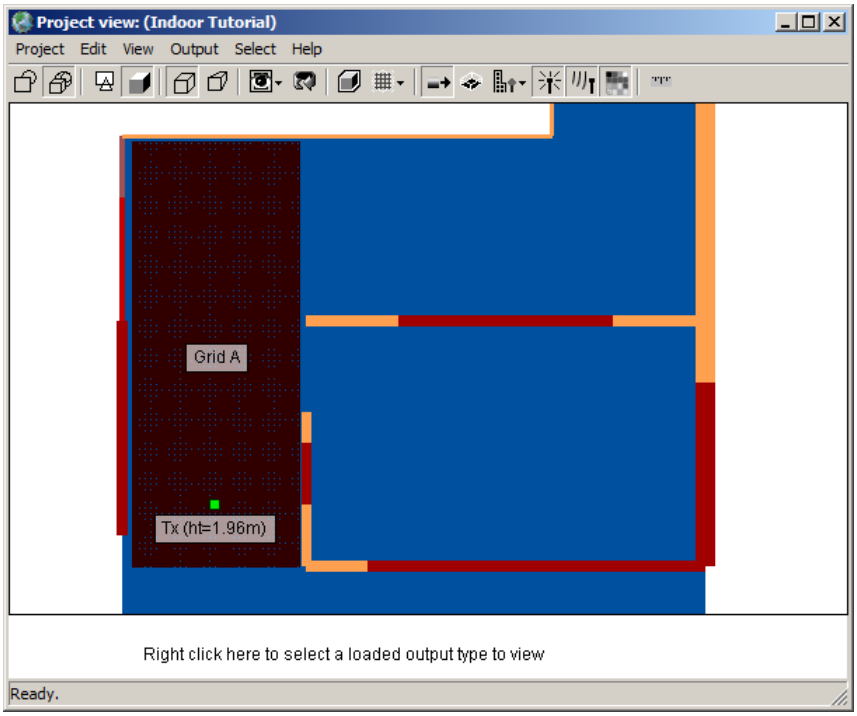


Figure 3.18: The area covered by Receiver Grid A

3. Enter “Grid A” as the *Short Description*.
4. Enter “0” for both the *Origin longitude* and *Origin latitude*.
5. Click LAYOUT PROPERTIES The TX/RX LAYOUT PROPERTIES window will appear, as seen in Figure 3.19.

.

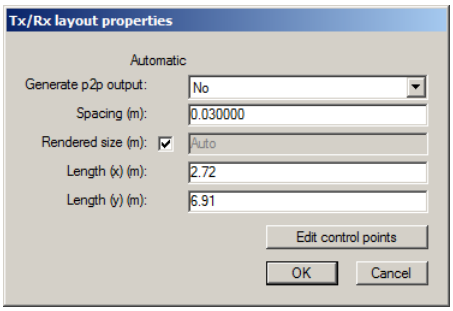


Figure 3.19: The Layout Properties Window for Grid A

6. Enter “0.03 m” as the *Spacing*.
7. Enter “2.72 m” as the *Length (x)* value.
8. Enter “6.91 m” as the *Length (y)* value.
9. Click EDIT CONTROL POINTS. Double-click on the control point and edit the location to be (0.18 m, 9.0 m, 1.3 m).
10. Click OK twice to finish the creation of Receiver Grid A.

Create Receiver Grid B following the same procedure, covering the area seen in Figure 3.20. (Grid B is directly adjacent to Grid A.) Set *Length (x)* to “6.45 m” and *Length (y)* to “2.8 m”. The control point value for Grid B should be set to (2.91 m, 13.11 m, 1.3 m)

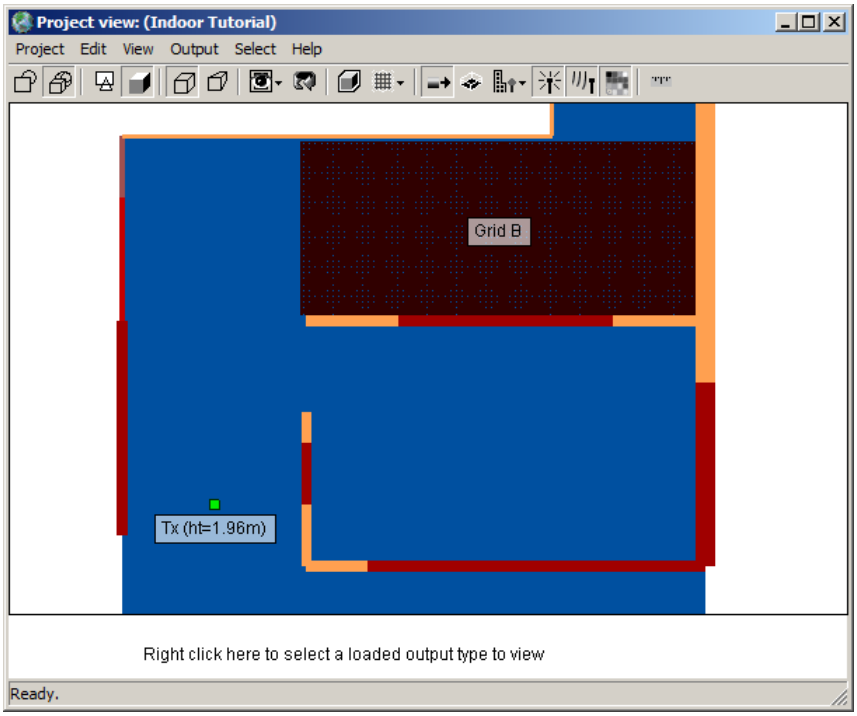


Figure 3.20: The area covered by Receiver Grid B

Running the Calculation with Receiver Grids

Run a new calculation with the Receiver *Grid* sets active, and then use the following process to view the results for the grids:

1. Select the RESULTS tab in the MAIN WINDOW and expand the *Study area* entry.
2. To view the received power, expand *Point to multipoint→Received power→Tx (ht=1.96m)*, and select *View* from the context menu for the receiver set *Grid A*. Each point in the receiver set will display a color corresponding to the received power at that point. Follow the same procedure for *Grid B*.

   ✓ To view the results more clearly, turn off the visibility of receivers and transmitters by clicking on the RECEIVER and TRANSMITTER icons at the top of the GEOMETRY VIEW.
3. In the GEOMETRY VIEW, right-click on the colored scale bar and choose *Range options* from the context menu. Select *Manual scaling*, setting a minimum of -30 dBm and a maximum of 10 dBm.

Figure 3.21 shows the received power throughout the hallway.

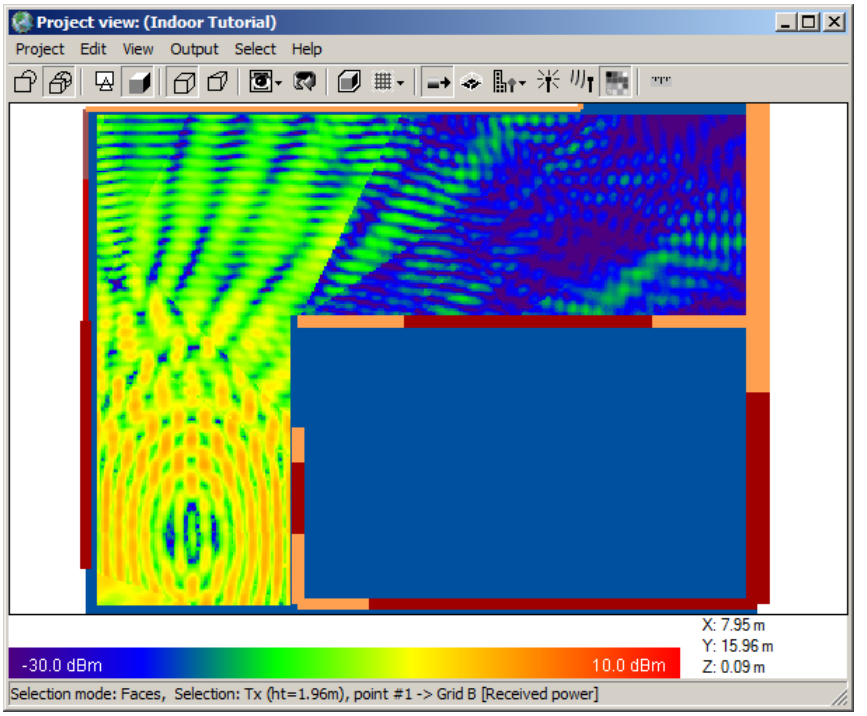


Figure 3.21: The received power on an XY Grid receiver

Figure 3.21 shows that received power can vary greatly over short distances due to constructive or destructive interference between rays reaching the RECEIVER locations. Relative locations of the walls, TRANSMITTERS and RECEIVERS determine the path lengths to a point, and therefore they must be accurate in order for Wireless InSite to calculate fast fading effects correctly.

Summary

The figures included with this example indicate the ability of Wireless InSite to organize, control, and display all the data needed for a radio propagation calculation.

Chapter 4

Outdoor Urban Propagation Prediction

This tutorial describes how to apply Wireless InSite's URBAN CANYON propagation model to predict path loss and other propagation characteristics in an urban area. The geographical area chosen for this tutorial is a section of Rosslyn, Virginia, shown in Figure 4.1. This map shows transmitter locations, street names and building heights relative to the local ground. Although there is more variation in the actual terrain, a flat ground approximation will be used that is sufficient for the purposes of this tutorial.

- Numerous path loss measurements have been made in this area, and more information about the area and the measurements can be found in [3].

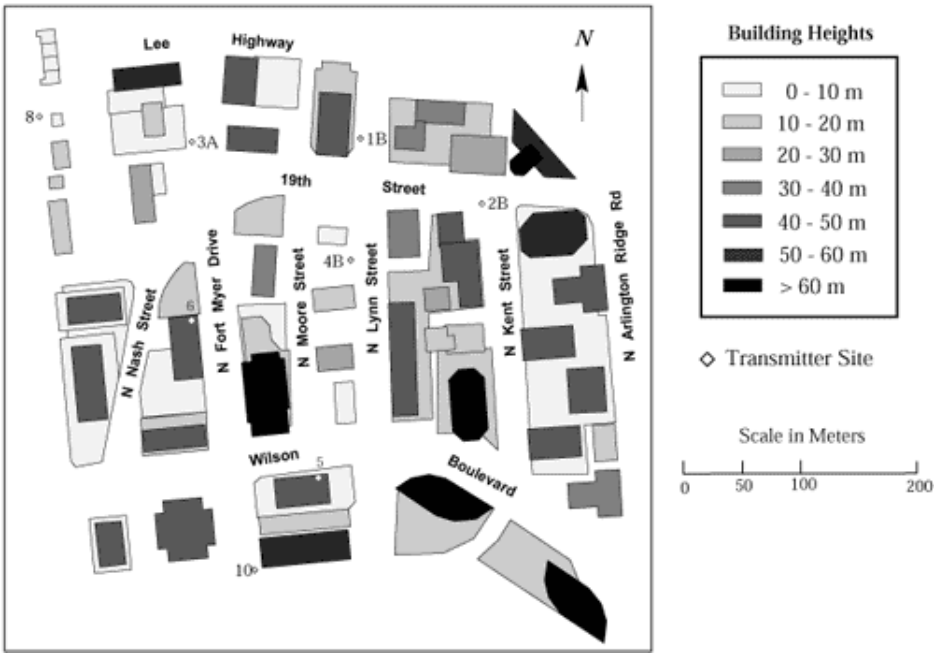


Figure 4.1: The Rosslyn, VA building geometry and transmitter locations

Creating a New Project

1. In the GEOMETRY VIEW, click *File→New→Project*. The PROJECT PROPERTIES Window will appear, as seen in Figure 4.2.

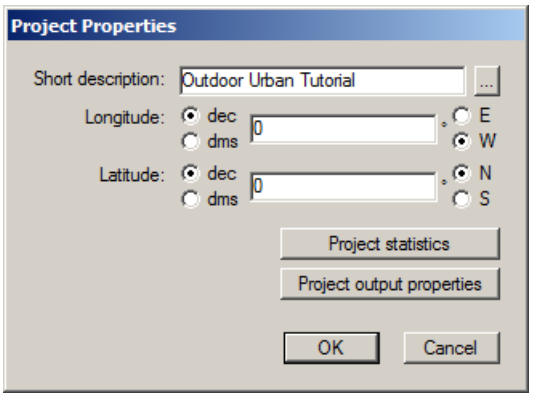


Figure 4.2: Project Properties Window

2. Enter a *Short description* of 80 characters or less.
3. Accept the defaults in the window and click OK.

Saving Your Files

It is a good idea to save your files periodically while working with Wireless InSite. To save your files:

1. In the MAIN WINDOW, click *File→Save Project as*.
2. Select a folder and provide a project name.

Loading a City File

Once you have created the project, you can load building data. To load an existing `*.city` file:

1. Click *File→Open→Geometry→City*.
2. Select the file `tutorials_location\Urban_Tutorial\rosslyn_va.city`.
   - ▶ See page 3 for information about the location of the tutorial projects on your machine.
3. Click OPEN.

✓ The above instructions refer to a `*.city` file in a directory other than the new project directory created for this tutorial. When the project is saved, files (*City*, *Terrain*, or IMAGE) that reside in directories other than the project directory will be copied to the project directory. From that point forward, the project will use the copy in the project directory. This allows a project directory to be saved, archived, and restored with the full assurance that all needed files are present.

❗ Changes to files in their original location do not affect the copies in each respective project directory.

The GEOMETRY VIEW, when viewed in SOLID BODY 3D mode, should look similar to Figure 4.3.

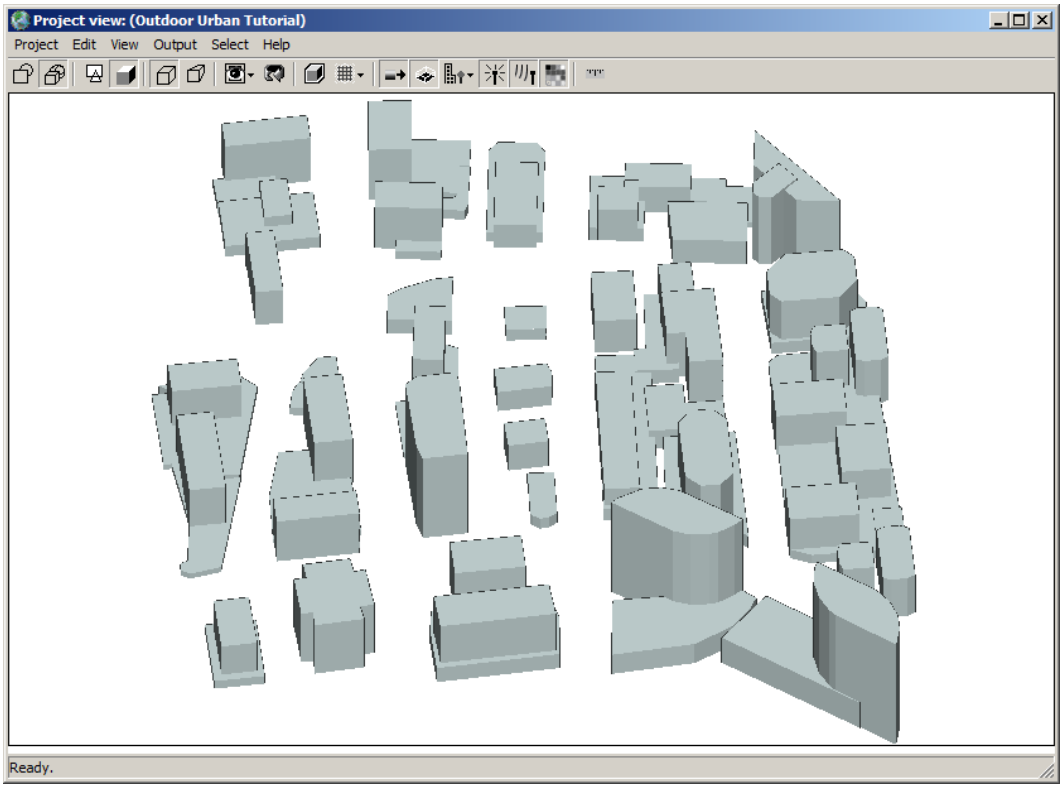


Figure 4.3: The Imported Rossyln City

Loading a Terrain File

To load the existing *Terrain* file:

1. Click *File*→*Open*→*Geometry*→*Terrain*.
2. Select the file `tutorials_location\Urban_Tutorial\Rosslyn_Flat_Terrain.ter`.
3. Click OK.

The GEOMETRY VIEWshould now look similar to Figure 4.4.

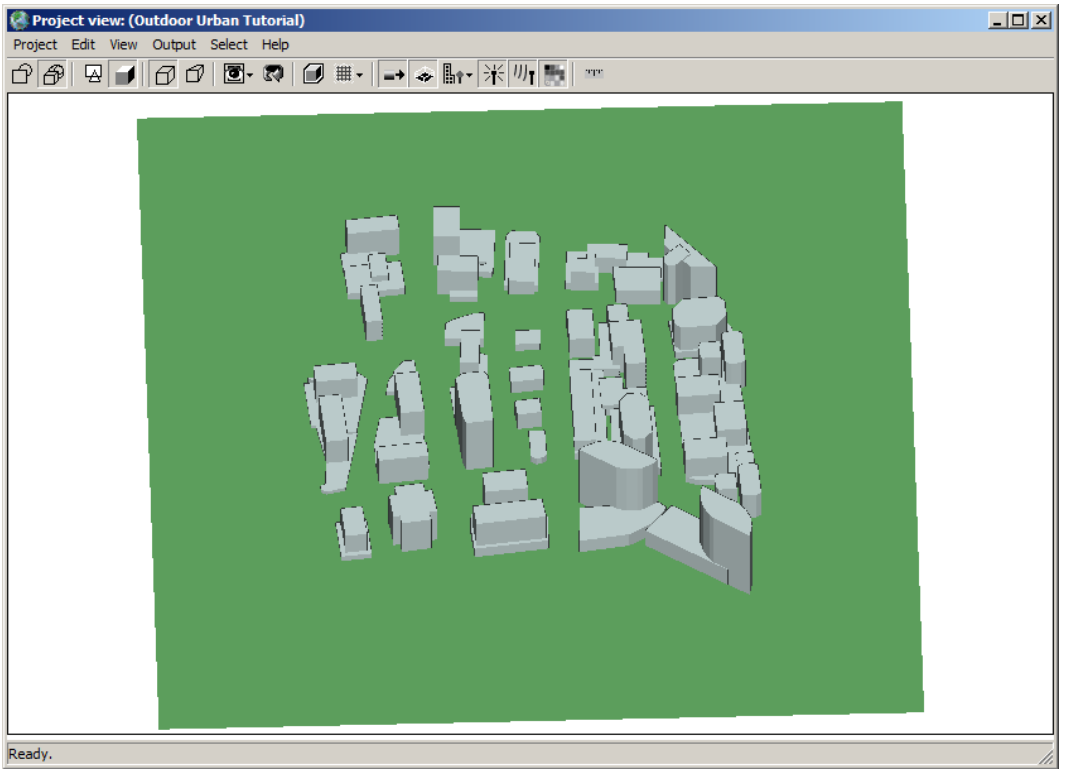


Figure 4.4: The Imported Terrain under Rosslyn

Creating a Waveform

Create a narrowband sinusoidal 908 MHz waveform with a bandwidth of 5 MHz.

1. Click *File→New→Waveform*.
2. Select a waveform type of *Sinusoid*. Click OK.
3. The WAVEFORM PROPERTIES Window will appear. Enter a *Short description* of “908-5 MHz”.
4. Enter a *Carrier frequency* of “908” MHz and an *Effective Bandwidth* of “5” MHz.
5. Click OK

Creating an Antenna

Now create a short vertical dipole with a $sin^2(\theta)$ radiation pattern.

1. Click *File→New→Antenna* to bring up the CREATE NEW ANTENNA Window.
2. Choose an *Antenna type* of “Short dipole” and click OK.
3. The ANTENNA PROPERTIES Window for the short dipole appears, as seen in Figure 4.5.

4. Enter a *Short description* of “Vertical dipole”.
5. Leave the *Maximum gain* automatic option checked (the maximum gain will be set to 1.76 dBi).
6. Set the *Polarization* to “Vertical”.
7. Leave the other fields at their default settings, and click OK.

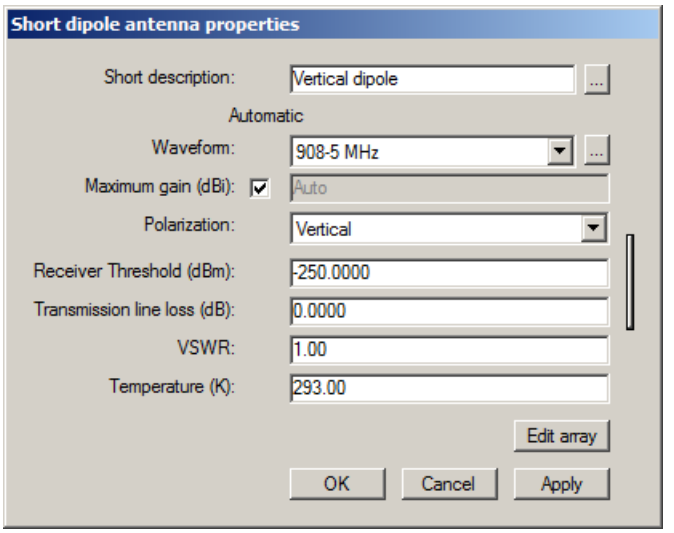


Figure 4.5: The Antenna Properties Window extended to show the antenna pattern

Creating Receiver Sets

Follow this procedure to create a route along Lynn Street starting at the north side of Wilson Blvd and ending at the south side of Lee Hwy (refer to Figure 4.1 or Figure 4.7).

1. Click *File*→*New*→*Receiver Set*→*Route*.
2. Define the *Route* by clicking on two or more points along Lynn St. To make the Route match the measured data, click the points from south to north, starting at Wilson Boulevard and ending at Lee Highway. (Clicking from north to south would reverse the plotted data.)
3. Right-click and the TX/RX PROPERTIES Window will appear.
4. Enter “Lynn St” as the *Short description*.
5. Click RECEIVER PROPERTIES

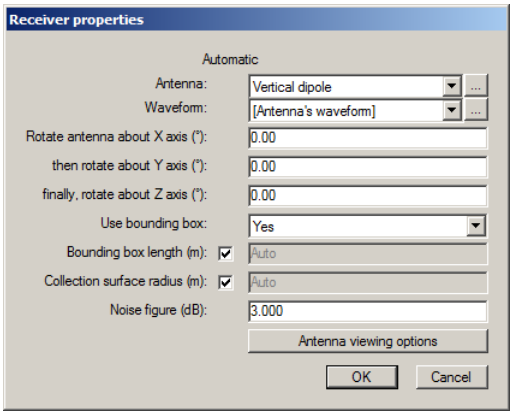


Figure 4.6: Receiver Properties Window

6. Select the “Vertical Dipole” as the *Antenna*, which was created earlier.

   The antenna’s WAVEFORM, previously selected as the 908-5 MHz waveform, is selected by default.

7. Click OK twice to complete the recevier route setup.

The resulting Receiver Route is displayed in the GEOMETRY VIEW, as seen in Figure 4.7.

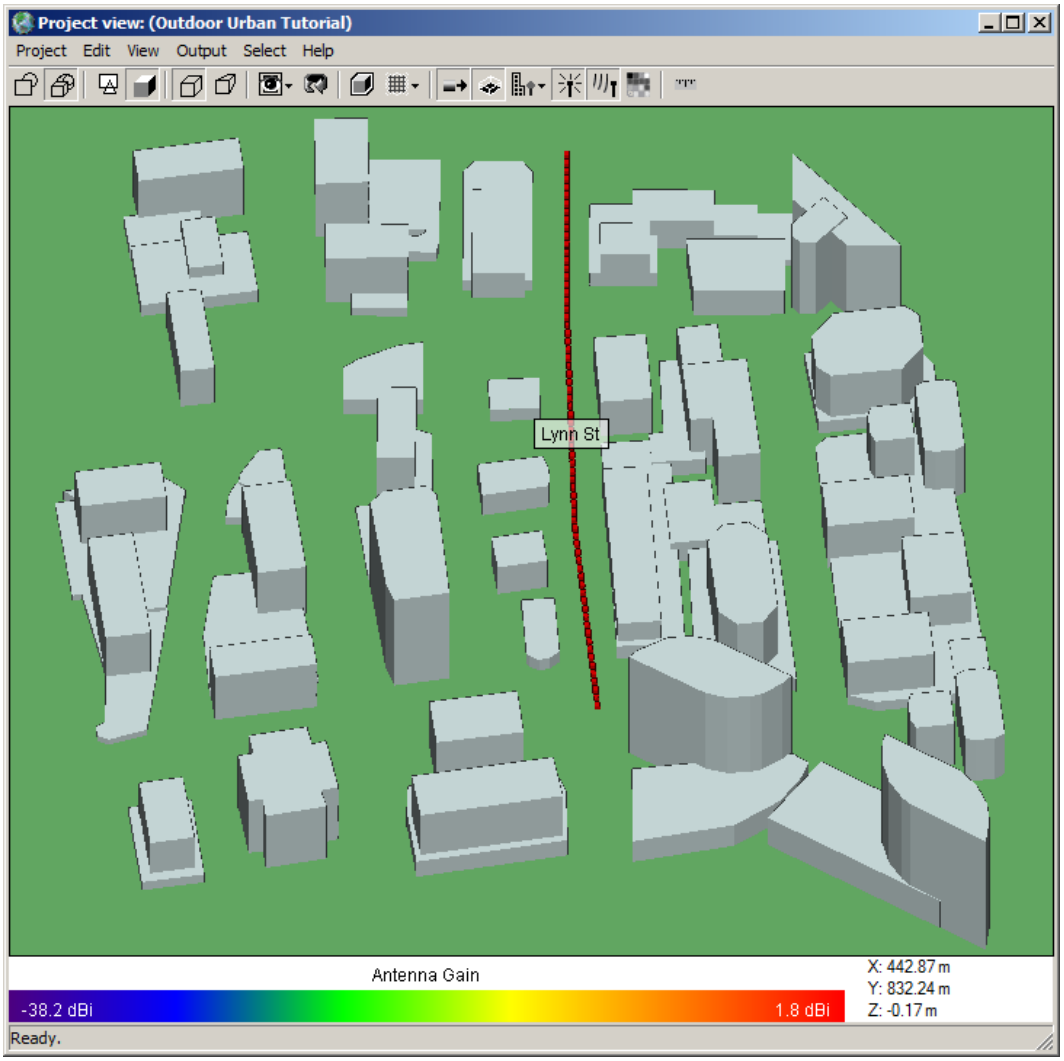


Figure 4.7: The Lynn St. Receiver Route

Creating Transmitter Sets

The creation of TRANSMITTER sites is very similar to the creation of RECEIVER locations. This section will describe how to place transmitters at sites 2B and 4B (shown in Figure 4.1 or Figure 4.8). Each transmitter site will be placed on a separate *Points* set.

1. Click *File*→*New*→*Transmitter Set*→*Points*.
2. In the GEOMETRY VIEW, right-click in the area where the coordinates are displayed (in the bottom-right corner) and choose *Lat/Long*→*Decimal degrees*.
3. In the GEOMETRY VIEW, click anywhere close to Longitude = 77.0702W, Latitude = 38.8967N (near the intersection of 19th and Kent). Right-click to finish.

   ✓ If the GEOMETRY VIEW is rotated, you can realign the global axis by clicking RESET and selecting *Sky view*. Left and right correspond to changes in longitude, while up and down correspond to changes in latitude, making it easier to locate a specific position.
4. In the TX/RX PROPERTIES Window, enter a *Short description* of "Tx 2B".
5. Manually set the longitude and latitude to the coordinates given in step 3.
6. Click TRANSMITTER PROPERTIES
7. The "Vertical dipole" antenna and the associated 908-5 MHz WAVEFORM should be selected by default.
8. Click OK to exit the TRANSMITTER PROPERTIES window.
9. Click LAYOUT PROPERTIES to access the TX/RX LAYOUT PROPERTIES window.
10. Click the EDIT CONTROL POINTS button.
11. Double click on the entry in the VIEW/EDIT VERTICES Window and manually set the height of the transmitter (Z-value) to 10 m.
12. Click OK to exit all open windows.

Repeat the process, changing the short description to "Tx 4B", to create another Transmitter *Point* at site 4B, at Longitude = 77.0714W, Latitude = 38.8963N. The GEOMETRY VIEW with transmitter sites 2B and 4B in green is shown in Figure 4.8.

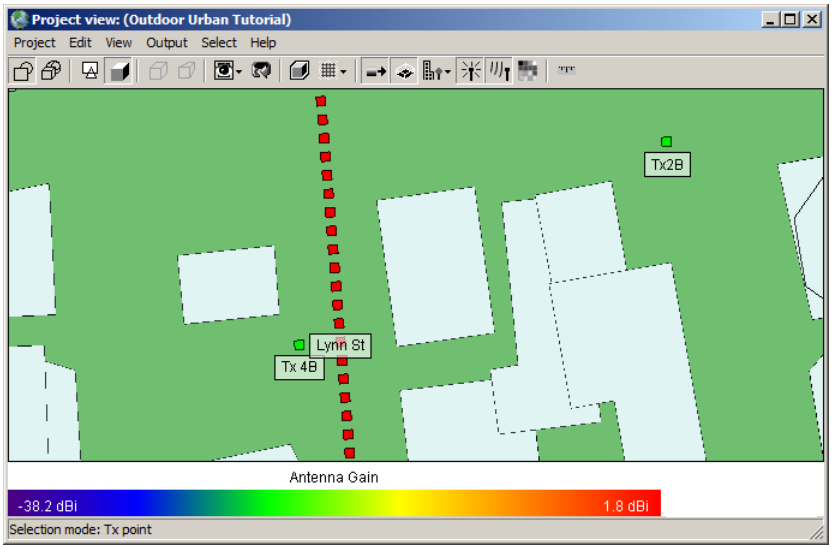


Figure 4.8: The Project View with Transmitters 2B and 4B

Creating a Study Area and Selecting a Propagation Model

To create a STUDY AREA manually:

1. Click *File→New→Study area*.
2. Select *Specify location and size* and click BEGIN.
3. Create a rectangular study area enclosing the buildings by clicking on the corner points, as seen in Figure 4.9.

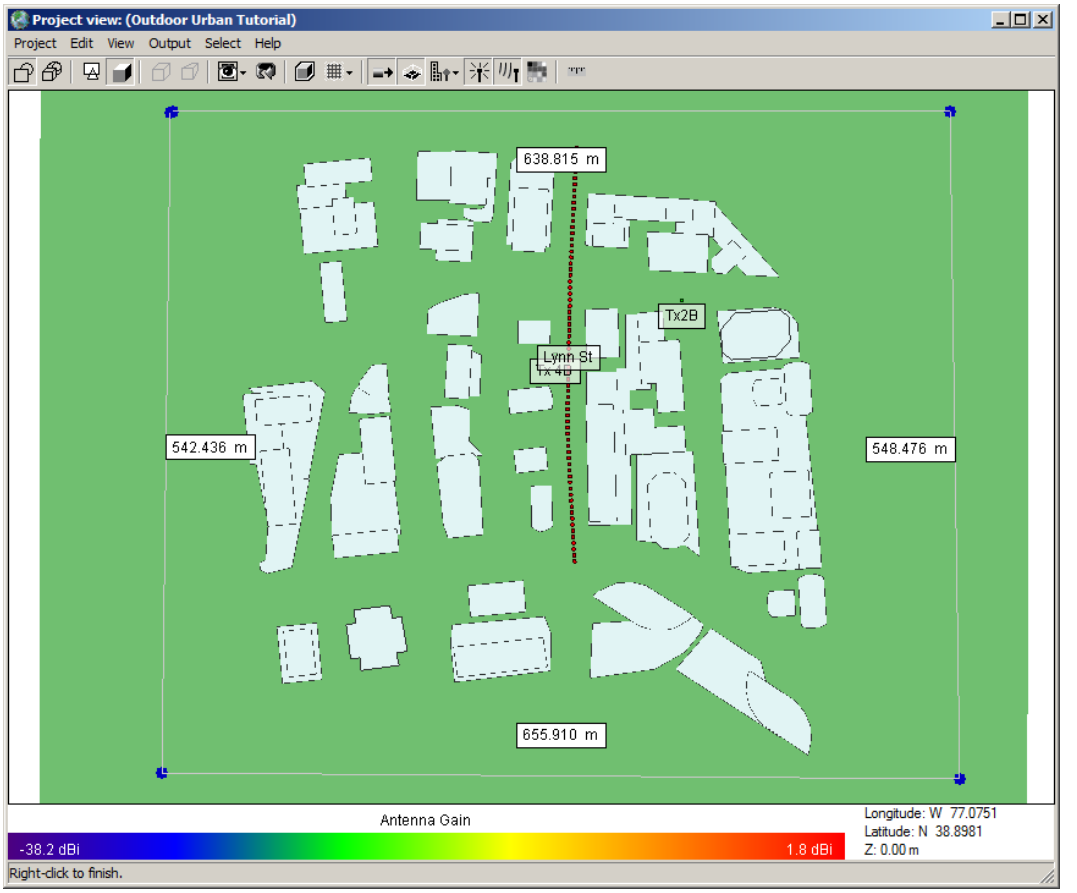


Figure 4.9: Defining the Study area boundary

4. Right-click to finish, which connects the first and last points.
5. Leave the *Base height* at “0” m, and enter “100” m as the *Top height*.
6. Click OK.
7. In the STUDY AREA PROPERTIES Window, enter “RosslynArea1” as the *Short description*.
8. Select “Urban canyon” as the *Propagation model*.
9. Deselect the *Automatic* checkbox on *Number of reflections* and set this to “6”.
10. Deselect the *Automatic* checkbox on *Number of diffractions* and set this to “1”.
11. Click OK.

Requesting Output

1. Double click the REQUESTED OUTPUT button in the STUDY AREA PROPERTIES window to access the REQUESTED OUTPUT CATEGORIES list.
2. Use the check boxes to select all desired output. Ensure that *Free space path loss*,*Path loss/gain* and *Propagation paths* are checked.

Running a New Calculation

To start a calculation, click on RUN→*New* in the MAIN WINDOW. If you have not saved all changes made to this point in your Wireless InSite project, you will be prompted to do so. The SIMULATION LOG opens and shows the output of the calculation engine, indicating its current status, as seen in Figure 4.10.

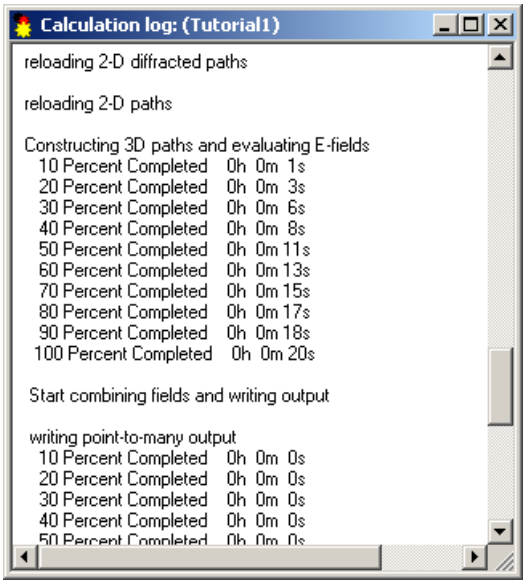


Figure 4.10: The Calculation Log Window

Viewing Output

Viewing Path Loss

1. In the PROJECT TREE, expand *Study areas and Output*→*Area: RosslynArea1*→*Point to multipoint*→*Path loss*→*Tx 2B*.

   If you have already modified the Output Tree sorting criteria in the output preferences, then the exact path to the output file may differ, but will still include all of the information in the path given above.

   ▶ See the Overview in the Reference Manual for more on output preferences.

2. Select *View* from the context menu for *Lynn St.* to see a color display of path loss along the Receiver *Route* in the GEOMETRY VIEW.

3. To turn off receivers in the GEOMETRY VIEW, click on *View→Receivers*. This should turn off the display of RECEIVERS and allow the path loss colors to display more clearly, as seen in Figure 4.11.

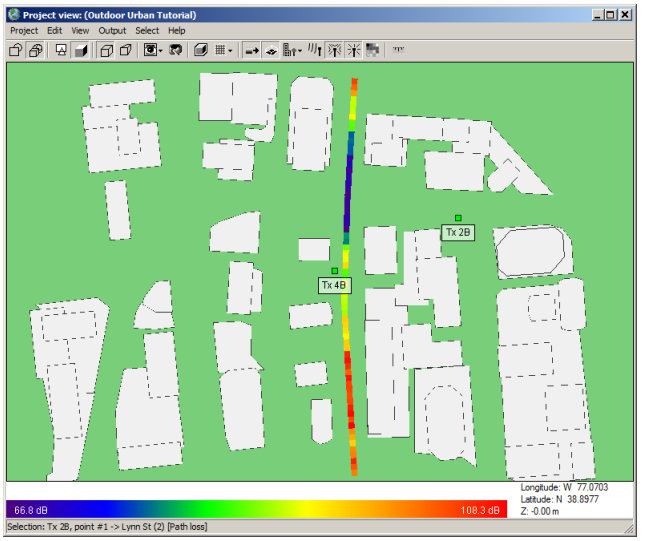


Figure 4.11: The path loss display for Lynn St.

4. Restore the *Receivers* view in the GEOMETRY VIEW and turn off the *Path loss* view from the PROJECT TREE.

Viewing Propagation Paths

There are several methods for viewing ray paths for a specific Tx/Rx pair. Here is one example:

1. In the PROJECT TREE, expand *Study areas and Output→Area: RosslynArea1→Point to multipoint→Propagation paths→Tx 2B*.

2. Right click on Lynn St. and select *Load*.

3. Expand the Lynn St. output file entry in the tree.

4. Select *View all paths* from the context menu for Receiver #20 (or any other desired point). The paths are displayed in the GEOMETRY VIEW, as seen in Figure 4.12.

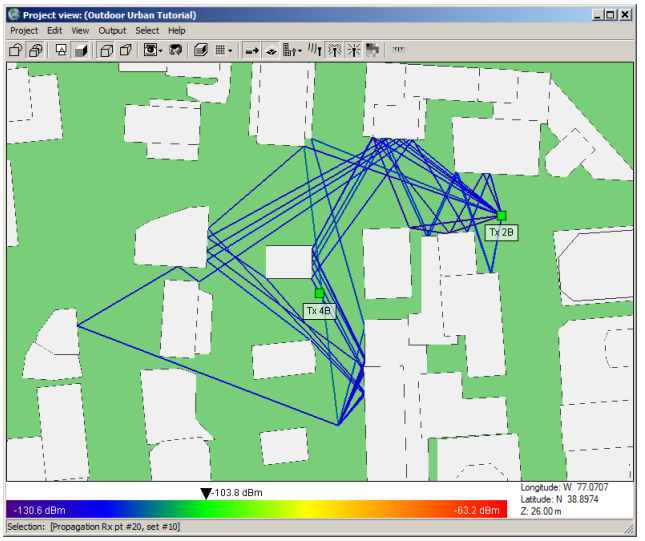


Figure 4.12: Propagation paths for Receiver #20

Creating Line Plots

The following steps will create a line plot comparing the predicted path loss to the path loss in free space.

In this discussion, "graph" refers to the figure that includes the title, axes, legends, and data, and "plot" refers to a particular data set being plotted on the graph. Using this terminology, one graph may contain several different plots.

1. In the PROJECT TREE, open *Study areas and output→Area: RosslynArea1→Point to multipoint→Path loss→Tx 2B*.
2. Select *Plot* from the context menu for *Lynn St.*
3. Select the *Distance* option and then click OK.
4. When the PLOT PROPERTIES Window appears, click OK.
5. In the PROJECT TREE, expand *Study areas and output→Area: RosslynArea1→Point to multipoint→Free space path loss with antenna pattern→Tx 2B*.
6. Select *Plot* from the context menu for *Lynn St.*
7. Select the *Distance* option and then click OK.
8. Select *Add to existing graph*.
9. Select *Path loss vs. Distance*.
10. When the PLOT PROPERTIES Window appears, click OK.
11. Click OK.

To change the range of the Y-axis:

1. Select *Adjust scale* from the plots' context menu.

2. Select *Y axis→Manual minimum*.
3. Set minimum Y value to 60 and click OK.
4. Using the same procedure, set the *Y axis→Manual maximum* value to 120.
5. Using the same procedure, set the *X axis→Manual maximum* value to 350. Your plot should look similar to the one shown in Figure 4.13.

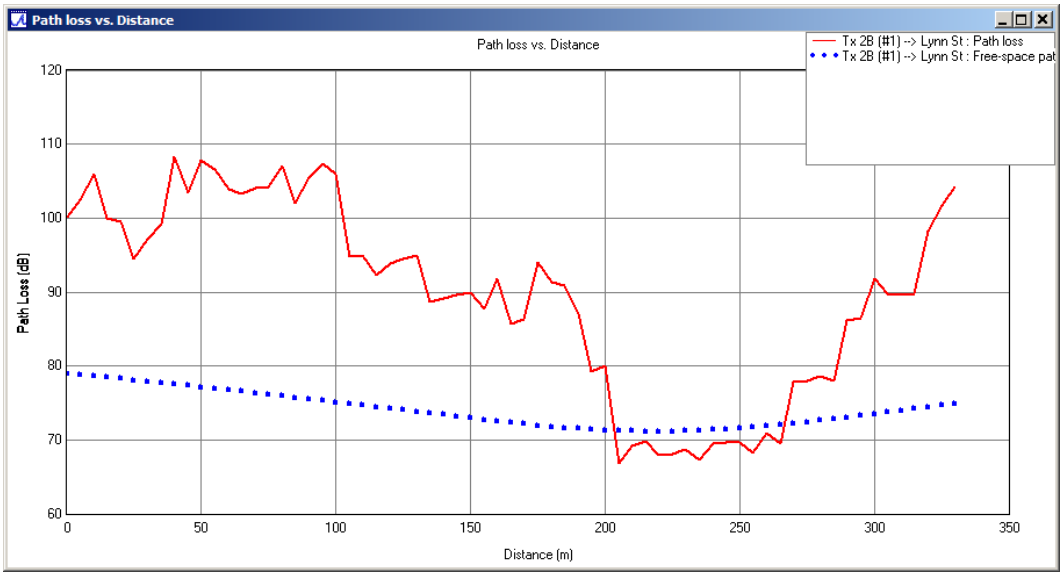


Figure 4.13: Path Loss vs. Distance for Lynn St.

To add a plot of the measured path loss to the graph, select *Import→Data →PLT* from its context menu. Navigate to `tutorials_location\Urban_Tutorial`, and select the files `lynn.nr.t2b.f908.plt` and `lynn.nl.t2b.f908.plt`. These files contain measurements of the path loss at 908 MHz, taken while driving north along the right-hand and left-hand sides of the street.

✓ A graph can be saved as an image file by selecting *Save to JPEG* from its context menu.

▶ See page 3 for information about the location of the example projects on your machine.

Chapter 5

Co-Siting of WiMax Transmitters in Ottawa

This tutorial describes how to apply Wireless InSite's FULL 3D propagation model to predict path loss and other propagation characteristics in Ottawa. This map shows transmitter locations, street names and building heights relative to the local ground.

Creating a New Project

1. In the GEOMETRY VIEW, click *File*→*New*→*Project*. Name the project "Ottawa Cosite" and click OK.

Loading a City File

Once you have created the project, you can load building data. To load an existing `*.city` file:

1. Click *File*→*Open*→*Geometry*→*City*.
2. Select the file `tutorials_location\CoSite_Tutorial\ottawa.city`.
3. Click OPEN.
   - See page 3 for information about the location of the example projects on your machine.

The GEOMETRY VIEW, when viewed in SOLID BODY 3D mode, should look similar to Figure 5.1.

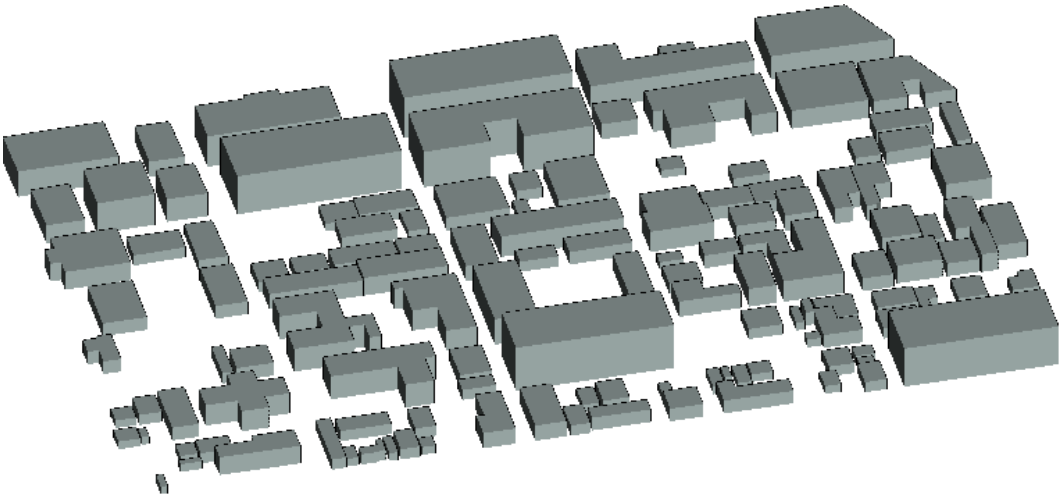


Figure 5.1: The Imported Ottawa City

Loading a Terrain File

To load the existing *Terrain* file:

1. Click *File*→*Open*→*Geometry*→*Terrain*.
2. Select the file `tutorials_location\CoSite_Tutorial\ottawa.ter`.
3. Click OK.

The GEOMETRY VIEW should now look similar to Figure 5.2.

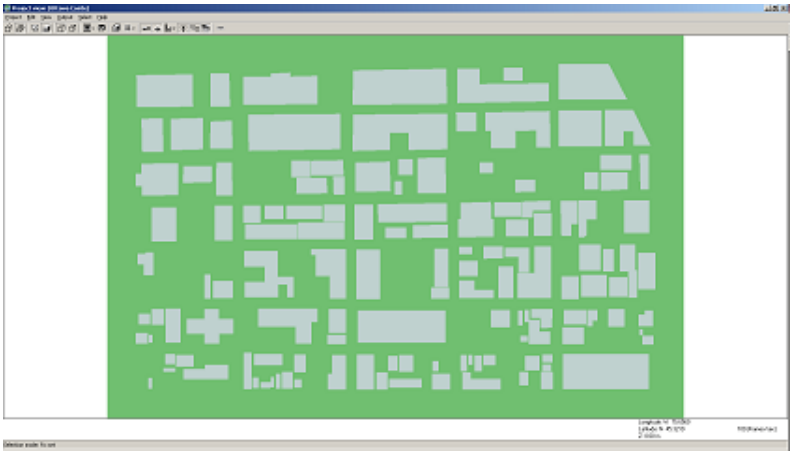


Figure 5.2: The Imported Terrain under Ottawa

Creating a Waveform

Create a narrowband sinusoidal 450 MHz waveform with a bandwidth of 1 MHz.

1. Click *File→New→Waveform*.
2. Select a waveform type of *Sinusoid*. Click OK.
3. The WAVEFORM PROPERTIES Window will appear. Enter a *Short description* of "Narrowband (450 MHz, 1 MHz)".
4. Enter a *Carrier frequency* of 450 MHz and an *Effective Bandwidth* of 1 MHz.
5. Click OK.

Creating an Antenna

Create a vertically polarized half-wave dipole antenna (for the TRANSMITTERS).

1. Click *File→New→Antenna* to bring up the CREATE NEW ANTENNA Window.
2. Choose an *Antenna type* of "Half-wave dipole" and click OK.
3. The ANTENNA PROPERTIES Window for the dipole appears, as seen in Figure 4.5.

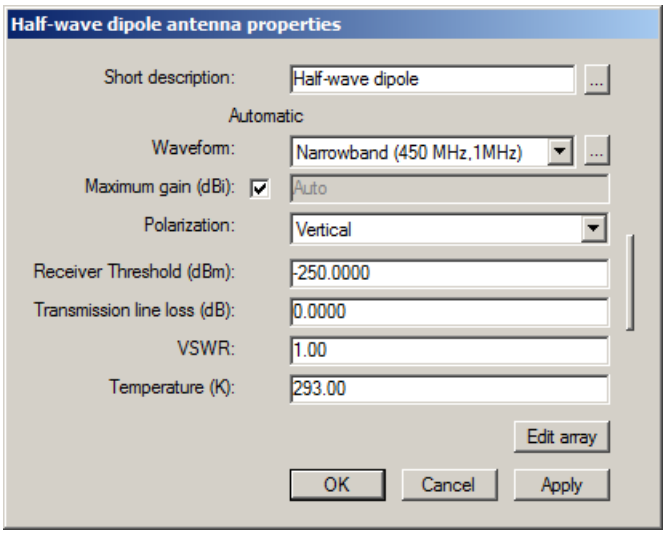


Figure 5.3: The Antenna Properties Window for the Half-Wave Dipole Antenna

4. Enter a *Short description* of "Half-wave dipole".
5. Leave the *Maximum gain* automatic option checked.
6. Set the *Polarization* to "Vertical".
7. Leave the other fields at their default settings, and click OK.

Now create a vertical isotropic antenna (for the RECEIVERS).

1. Click *File*→*New*→*Antenna* to bring up the CREATE NEW ANTENNA Window.
2. Choose an *Antenna type* of "Isotropic" and click OK.
3. The ANTENNA PROPERTIES Window for the dipole appears. Enter a *Short description* of "Vertical isotropic".
4. Leave the *Maximum gain* automatic option checked.
5. Set the *Polarization* to "Vertical".
6. Leave the other fields at their default settings, and click OK.

Creating Transmitter Sets

This section will describe how to place a TRANSMITTER in each of the four sectors of the *City*. Each transmitter site will be placed on a separate *Points* set.

1. Click *File*→*New*→*Transmitter Set*→*Points*.
2. In the GEOMETRY VIEW, right-click in the area where the coordinates are displayed (in the bottom-right corner) and choose *Lat/Long*→*Decimal degrees*.
3. In the GEOMETRY VIEW, click anywhere close to Longitude = 75.6561W, Latitude = 45.3205N. Right-click to finish.
4. In the TX/RX PROPERTIES Window, enter a *Short description* of "NE Sector".
5. Manually set the longitude and latitude to the coordinates given in step 3.
6. Click the TRANSMITTER PROPERTIES button.
7. The "Half-wave dipole" antenna should be selected.
8. Enter 30 dBm as the *Input power*.
9. Click OK to close the TRANSMITTER PROPERTIES Window.
10. Click LAYOUT PROPERTIES button.
11. Click the EDIT CONTROL POINTS button.
12. Double click on the entry in the VIEW/EDIT VERTICES Window and manually set the height of the transmitter (Z-value) to 26 m.
13. Click OK to exit all open windows.

Repeat the process for the following three transmitters:

- *Short description*: "SE Sector", *Longitude* = 75.6561W, *Latitude* = 45.3179N, *Control Point* Z*-value* = 14 m.
- *Short description*: "SW Sector", *Longitude* = 75.6633W, *Latitude* = 45.3182N, *Control Point* Z*-value* = 17 m.
- *Short description*: "NW Sector", *Longitude* = 75.6634, *Latitude* = 45.3206N, *Control Point* Z*-value* = 29 m.

The GEOMETRY VIEW with the four TRANSMITTER sites in green is shown in Figure 5.4.

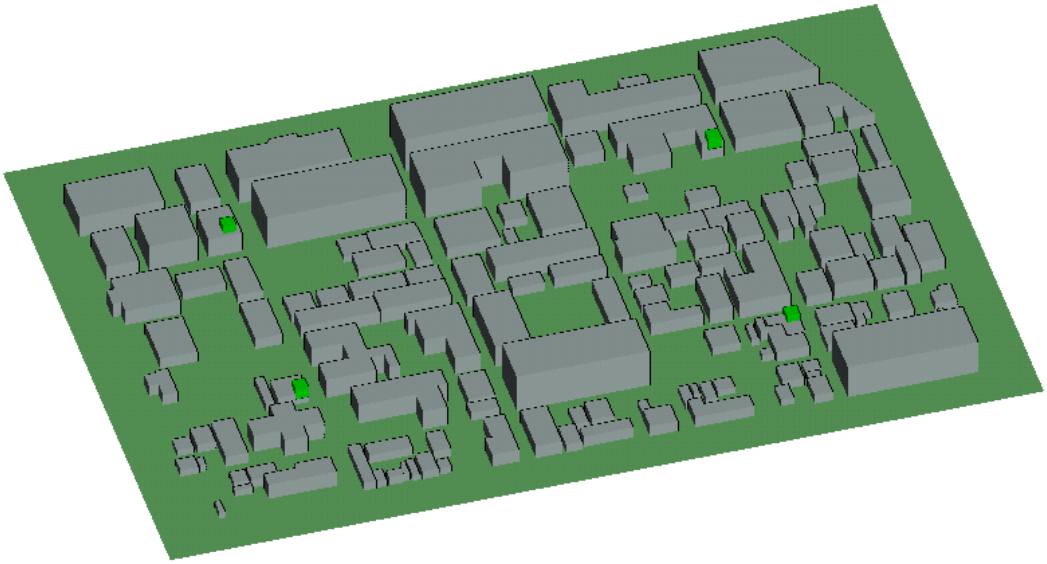


Figure 5.4: The Project View with transmitters at the four corners of Ottawa

Creating Receiver Sets

Follow this procedure to create a *Grid* of receivers across the city.

1. Click *File*→*New*→*Receiver Set*→*XY Grid*.
2. Define the *Grid* by clicking on one corner of the city, and dragging to the opposite corner of the city. Release the mouse button to finish.
3. The TX/RX PROPERTIES Window will appear.
4. Enter “Rx Coverage Grid” as the *Short description*.
5. Click RECEIVER PROPERTIES.
6. Select the “Vertical isotropic” as the *Antenna*, which was created earlier.
7. Click OK to exit the RECEIVER PROPERTIES Window.
8. Click LAYOUT PROPERTIES.
9. Define the *Spacing* as 5 m.
10. Click the EDIT CONTROL POINTS button. Verify that the Z-value of the control point is 2 m.
11. Click OK to exit all open windows and finish the creation of the receiver *Grid*.

The resulting Receiver *Grid* is displayed in the GEOMETRY VIEW, as seen in Figure 5.5.

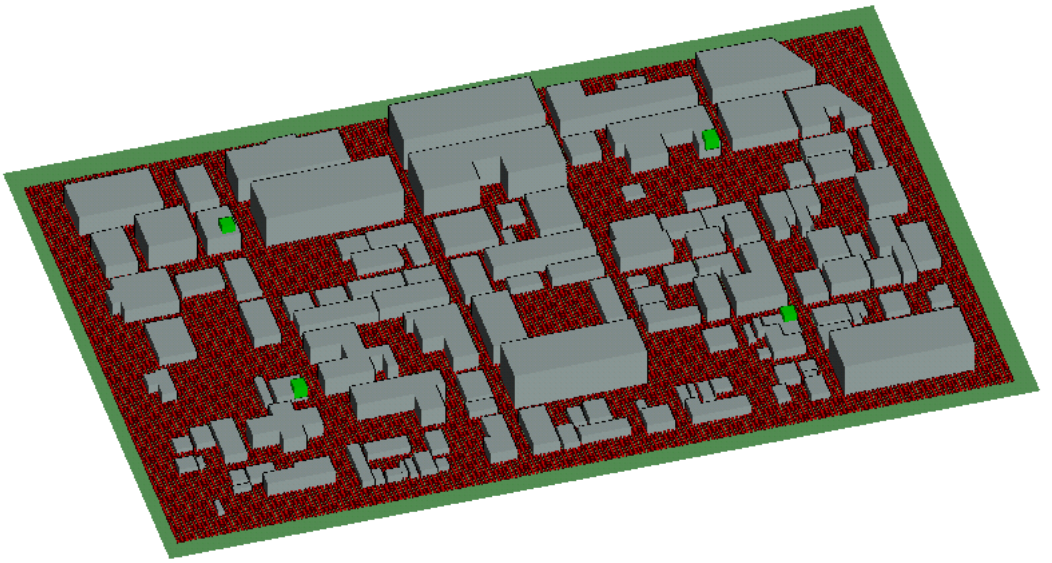


Figure 5.5: The Ottawa receiver grid

Creating a Study Area and Selecting a Propagation Model

To create a STUDY AREA:

1. Click *File→New→Study area*.
2. Select *Fit to geometry* and click BEGIN.
3. In the STUDY AREA PROPERTIES Window, enter “Full_3d_R3_D1” as the *Short description*.
4. Select “Full 3-D” as the *Propagation model*.
5. Deselect the *Automatic* checkbox on *Ray spacing* and set the spacing to “0.25°”.
6. Deselect the *Automatic* checkbox on *Number of reflections* and set this to “3”.
7. Deselect the *Automatic* checkbox on *Number of transmissions* and set this to “0”.
8. Deselect the *Automatic* checkbox on *Number of diffractions* and set this to “1”.
9. Ensure the *Raytracing method* is set to “SBR”.
10. Click OK to finish the creation of the STUDY AREA.

Requesting Output

Transmitter Point to Receiver Set Output

Output is created for each channel between active transmitter points and each active receiver set in the project.

1. Double click the REQUESTED OUTPUT button in the STUDY AREA PROPERTIES window to access the REQUESTED OUTPUT CATEGORIES list.
2. Ensure that the *Received power* field is checked.

Requesting Consolidated Output

Consolidated output takes into account the effects of multiple transmitter points at each receiver point location.

1. Click *File→New→Communication systems*.
2. Enter a *Short description* of "Complete".
3. Click SELECT ALL under the TRANSMITTER and RECEIVER set tables.
4. Ensure that only the checkboxes for *Generate Carrier to Interferer Ratio output* and *Generate Strongest Transmitter output* are selected.
5. Click THROUGHPUT.
6. Ensure that the *Generate Throughput output* checkbox is selected.
   ✓ Throughput calculations require strongest transmitter power to be generated and will activate this request when it is selected.
7. Click OK.

Running a New Calculation

To start a calculation, click on RUN→*New* in either the MAIN WINDOW or the GEOMETRY VIEW. If you have not saved all changes made to this point in your Wireless InSite project, you will be prompted to do so.

Viewing Output

Viewing Received Power

1. In the PROJECT TREE, expand *Study areas and Output→Area: Full_3d_R3_D1→Point to multipoint→Received Power→NE Sector→Rx Coverage Grid*.
2. Select *View* from the context menu for *Rx Coverage Grid* to see a color display of the power coverage from the NE Sector transmitter.
3. To turn off receivers in the GEOMETRY VIEW, click on *View→Receivers*. This should disable the display of RECEIVERS and show the path loss colors more clearly.
4. To see the coverage map more clearly, change the color bar scale. In the GEOMETRY VIEW, select *Range Options→Manual Scaling* from the color bar context menu. Set the *Minimum value* to "-80" dBm and the *Maximum value* to "-20" dBm.

5. Repeat steps 1 and 2 to view the power coverage from the SE Sector, SW Sector, and NW Sector transmitters. The coverage maps for all four transmitters are seen in Figure 5.6.

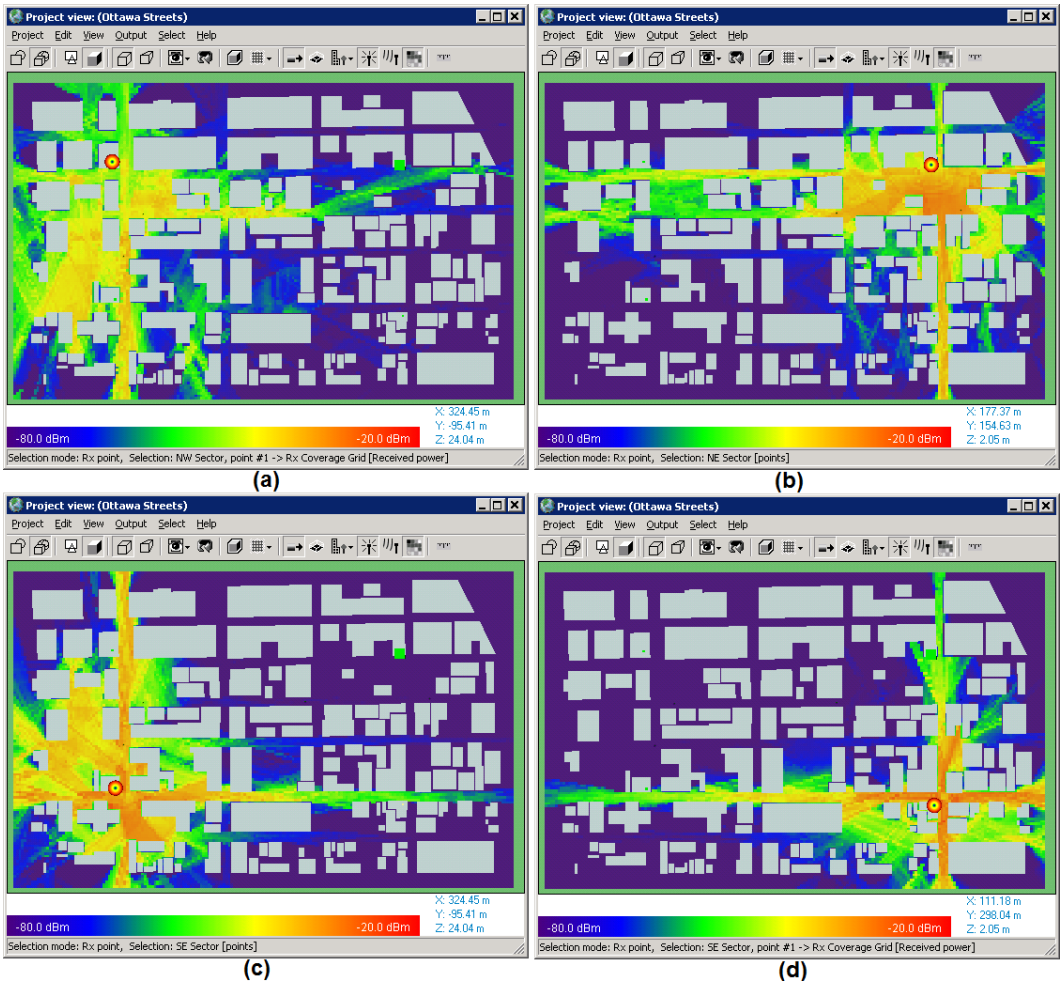


Figure 5.6: The power coverage maps for (a) NW Sector transmitter, (b) NE Sector transmitter, (c) SW Sector transmitter, and (d) SE Sector transmitter

Viewing the Strongest Transmitter

The *Strongest transmitter* is the TRANSMITTER point that delivers the most *received power* to each RECEIVER location.

1. In the PROJECT TREE, expand *Study areas and Output→Area: Full_3d_R3_D1→Communication systems→complete→Strongest transmitter→Combined Tx's→Rx Coverage Grid*.

2. Select *View* from the context menu for *Rx Coverage Grid* to see a color display of the strongest transmitter map, as seen in Figure 5.7.

- ✓ When the mouse is held over the scale bar colors a tooltip will appear indicating which transmitter point the color represents.

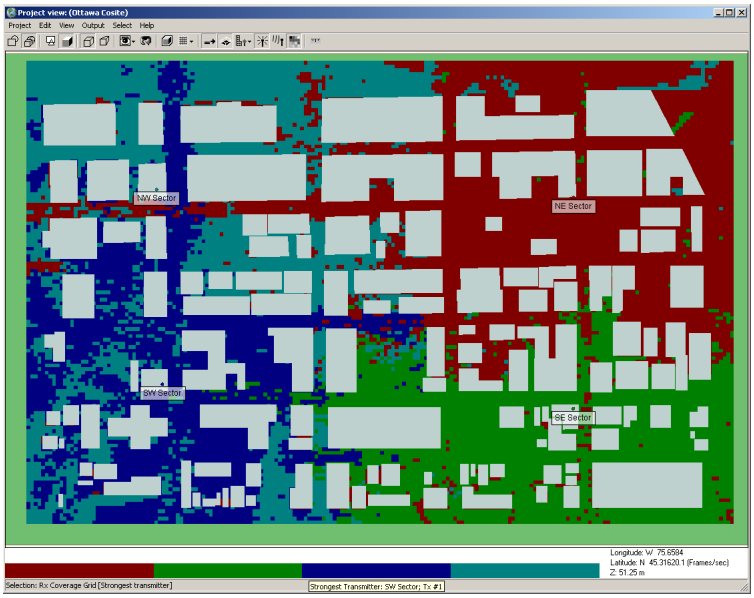


Figure 5.7: The strongest transmitter map

Viewing the Carrier to Interferer Ratio

The *Carrier to Interferer Ratio* gives the ratio of the power from the intended transmitter to all other active transmitters. A positive value indicates a region where the intended transmitter provides signal power that is greater than the sum of the power of all other interfering transmitters.

1. In the PROJECT TREE, expand *Study areas and Output→Area: Full_3d_R3_D1→Communication systems→complete→Carrier to interferer ratio→NE Sector→Rx Coverage Grid*.
2. Select *View* from the context menu for *Rx Coverage Grid* to see a color display of the carrier to interferer ratio map.
3. Repeat this procedure for each remaining sector.
4. Modify the scale bar to show the areas where the power of the main transmitter for each sector is stronger than the combined power of the other interfering transmitters.
   (a) Select *Properties* from the scale bar context menu while viewing *Carrier to Interferer Ratio* output. To ensure that *Carrier-interferer ratio* is the current viewing mode, right-click on the scale bar and select *Output type→Carrier-interferer ratio*.
   (b) Select the checkbox *Use discrete colors*.
   (c) Select “3” in the *Number of colors* option menu.
   (d) Click on the three color boxes to set them to be Green, Yellow and Red, in that order.

(e) Deselect the checkbox *Uniform color spacing*. At this point, fields will appear to the right of the color boxes to indicate the ranges of each color.

(f) Set the values for the *Ranges* to “20” and “0”.

(g) Click OK.

5. The four coverage maps are shown in Figure 5.8.

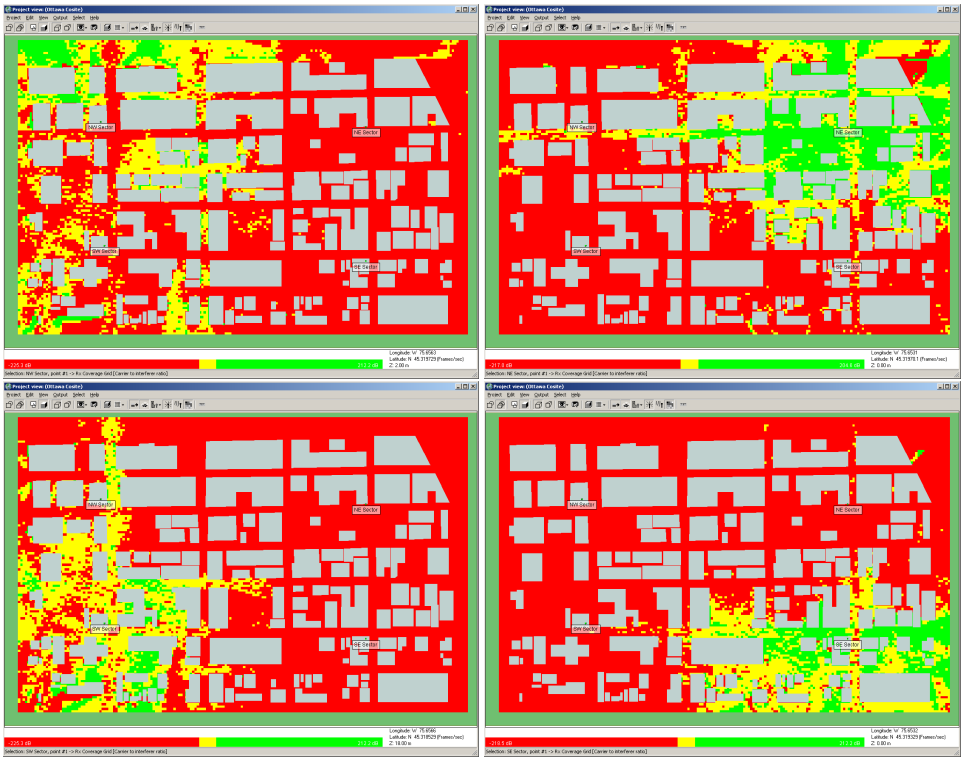


Figure 5.8: Clockwise from top left, the carrier to interferer ratio maps for NW, NE, SE, and SW Sectors.

Viewing the Throughput

Throughput is a measurement of the expected transmission rate at any given location within the RECEIVER grid.

1. In the PROJECT TREE, expand *Study areas and Output→Area: Full_3d_R3_D1→Communication systems→complete→Throughput→Combined Tx's→Rx Coverage Grid*.

2. Select *View* from the context menu for *Rx Coverage Grid* to see a color display of the throughput map, as seen in Figure 5.9.

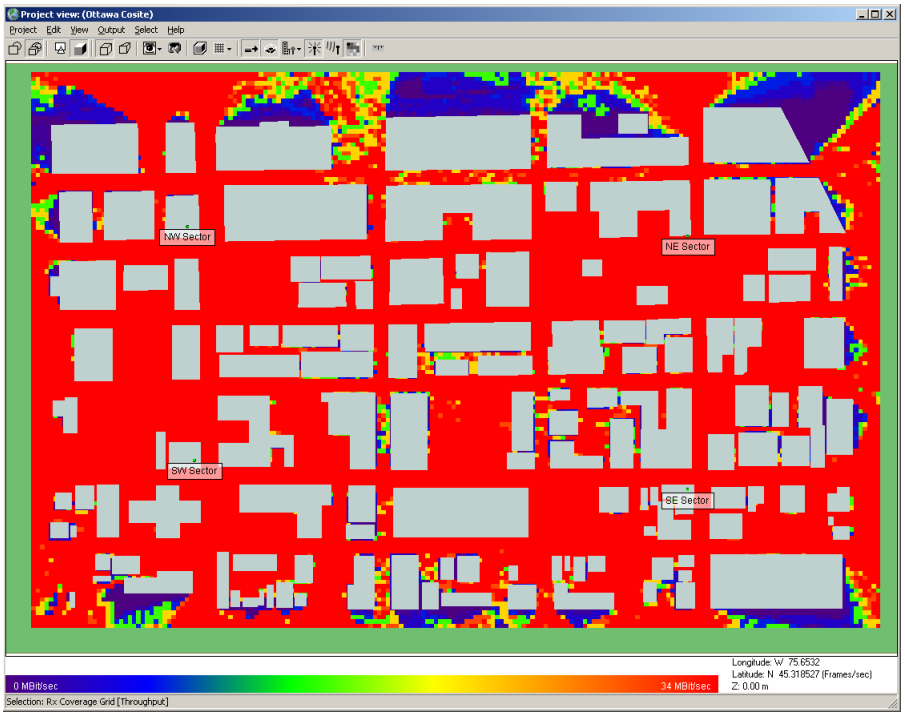


Figure 5.9: The throughput map

Creating Output Without Interference Source

This section describes how to create a request for output that shows the effects of removing the NE Sector TRANSMITTERas an interference source to the other transmitters in the project by using a COMMUNICATION SYSTEM to filter the output for analysis.

Generating Filtered Output Using a Communication System

First create a new COMMUNICATION SYSTEM that will remove the effects of the “NE Sector” transmitter:

1. Click *File→New→Communication systems*.
2. Enter a *Short description* of “Without NE Sector effects”.
3. Click SELECT ALL under the transmitter and receiver set tables.
4. Deselect the checkbox next to “NE Sector” in the transmitter table.
5. Ensure that only the checkboxes for *Generate Carrier to Interferer Ratio output* and *Generate Strongest Transmitter output* are selected.
6. Click THROUGHPUT.
7. Ensure that the *Generate Throughput output* checkbox is selected.

8. Click OK.

To create the output from the COMMUNICATION SYSTEM, click on RUN→*Comm. system analysis* in either the MAIN WINDOW or the GEOMETRY VIEW. If you have not saved all changes made to this point in your Wireless InSite project, you will be prompted to do so.

✓ COMMUNICATION SYSTEM output is a post-processing analysis of the output generated by the FULL 3D model. Once the model has run and generated the required inputs for the COMMUNICATION SYSTEM analysis, it only needs to rerun if there are changes to other elements in the project that would change the models' output values. This time-saving RUN mode can significantly reduce the amount of time required to create output from one or more COMMUNICATION SYSTEMS.

Viewing Output Without NE Interferer

The steps for viewing the output with the *NE Sector* TRANSMITTER are the same as given for the *complete* COMMUNICATION SYSTEM. The only difference is that instead of expanding the *complete* COMMUNICATION SYSTEM, expand the *without NE Sector effects* COMMUNICATION SYSTEM when going through the procedures.

Chapter 6

Modeling Parameter Uncertainty using Monte Carlo

This tutorial describes how to use Wireless InSite's PARAMETER UNCERTAINTY capability to assess the impact that the conductivity of building materials has on the received power along a route through an urban scene. A uniform distribution for the conductivity of city's concrete building material is defined, and 10 Monte Carlo (MC) runs are made using this distribution. Wireless InSite's X3D model calculates the minimum, maximum, mean and median received power based on the Monte Carlo iterations of the conductivity's distribution definition.

To set up a PARAMETER UNCERTAINTY analysis, the user must specify:

In STUDY AREA:

- Set Monte Carlo to active
- Set Monte Carlo simulation parameters

Material:

- Define a Monte Carlo Material
- Assign a Monte Carlo distribution to a Material parameter

Opening Example Project

1. Open the Parameter Uncertainty Tutorial project.
2. In the MAIN WINDOW Window, select *File→Open→Project*
3. Browse to the `tutorials_location\Parameter_Uncertainty_Tutorial\rosslyn_area.setup` on the hard drive and open it.

▶ See page 3 for information about the location of the example projects on your machine.

Creating a Receiver Route

1. Create a new RECEIVER route by *Right Click→New→Receiver Set→Route*.
2. Draw a route by clicking in three places as indicated below. Right Click to finish.

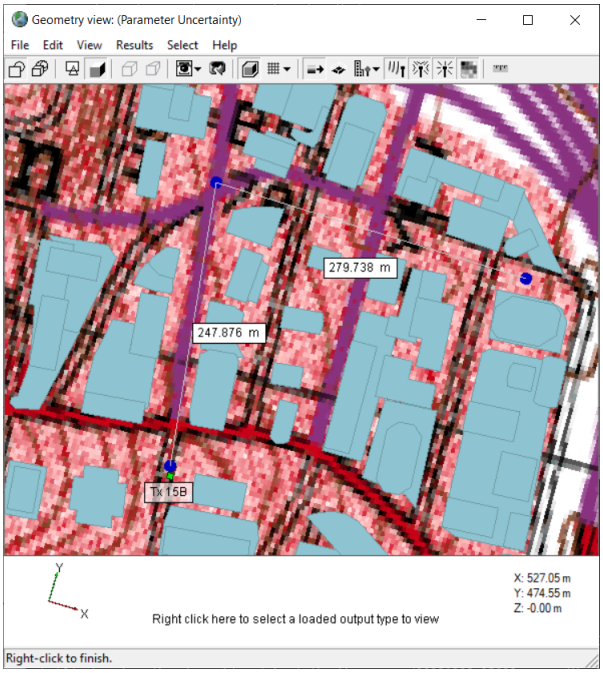


Figure 6.1: Receiver Route.

3. The RECEIVER PROPERTIES window will open up.
4. Leave all values as default values, and click OK.
5. Navigate to the TRANSCEIVERS tab to confirm the active TRANSMITTER and RECEIVER sets.

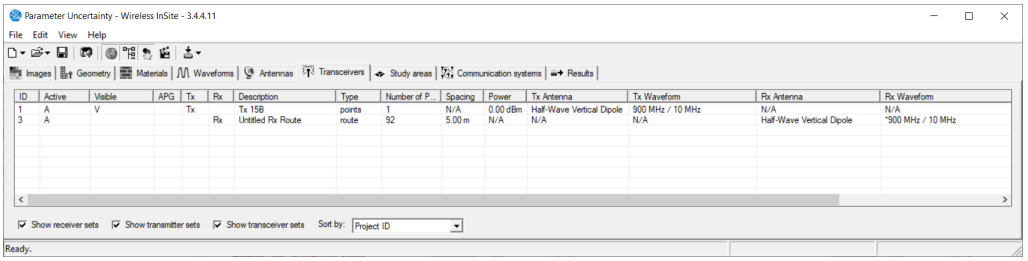


Figure 6.2: Transceiver Tab.

Creating a Monte Carlo Material

1. *Right Click→New→Material*.
2. A message will come stating Select a Geometry to which the new material will be added; click OK.
3. Select the City Geometry; click OK.
4. Select *Monte Carlo variable parameter single-layer* from the drop down menu; click OK.
5. Enter ”MC Variable Conductivity” in the *Short description*.

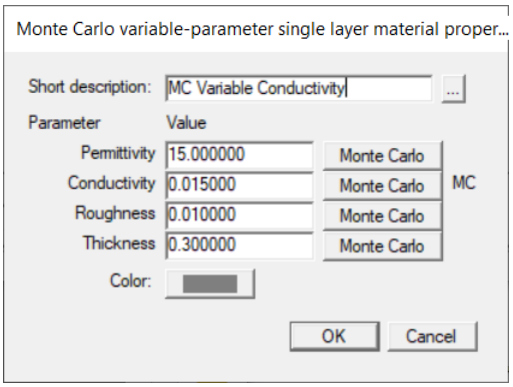


Figure 6.3: Monte Carlo Variable Parameter Material.

6. Click MONTE CARLO next to *Conductivity*.
7. Change *Minimum Conductivity* to “0.0005”
8. Change *Maximum Conductivity* to “0.05”
9. Click OK.

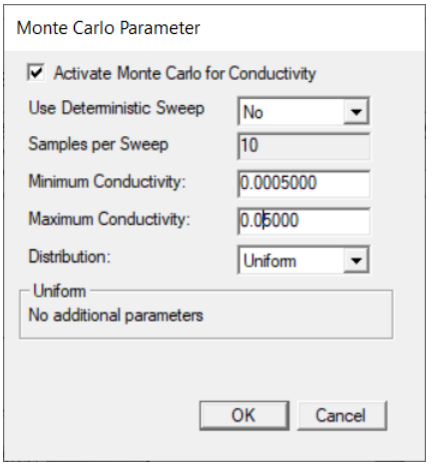


Figure 6.4: Variable Conductivity Parameters.

Assigning a MC Material to City

1. Navigate to GEOMETRY Tab.
2. *Right Click→Geometry→Change Material*
3. Select the MC material.

Setting MC Calculations in X3D Studyarea

1. *Right Click→New→Study Area*
2. The STUDY AREA creation window will pop up. Select Fit to geometry, and click BEGIN.
3. The STUDY AREA PROPERTIES window will pop up. Change *Short Description* to X3D_MC.
4. Leave interactions as default values.
5. Click MONTE CARLO PROPERTIES.

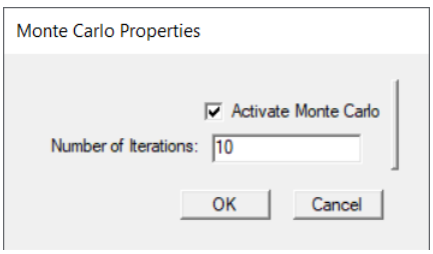


Figure 6.5: Monte Carlo Properties Window.

6. Click OK.

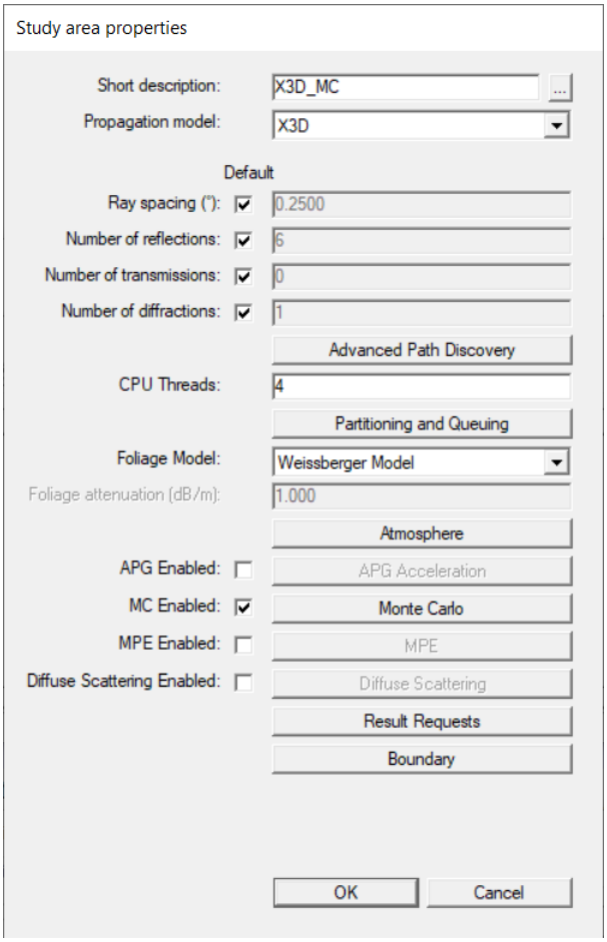


Figure 6.6: Study Area Properties Window.

7. Save the project and then run the simulation.

Plotting Monte Carlo Output

1. Navigate to the RESULTS tab.
2. Click '+' next to Area:X3D_MC
3. Click '+' next to Point to multipoint
4. Click '+' next to Minimum received power
5. Click '+' next to Tx15B
6. *Right Click on Untitled Rx Route→Plot*

7. Select Receiver number, and click OK.
8. Select To a new graph, and click OK.
9. Change the Legend to ”Minimum Received Power”, and click OK.
10. *Right Click→Edit Labels→Title*
11. Change title to ”MC Received Power”, and click OK.
12. Repeat this process for Mean and Maximum received power.

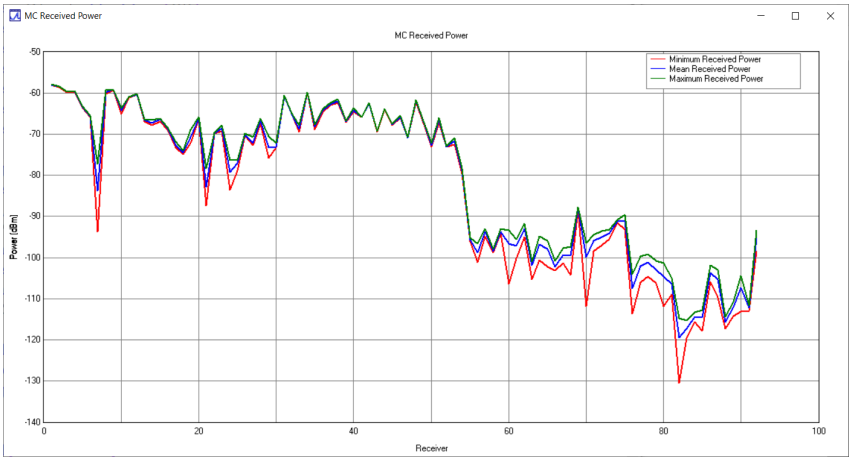


Figure 6.7: MC Plotted Results.

✓ Note: Depending on the precise location of the RECEIVER route, the results might differ slightly.

Chapter 7

MIMO Tutorial

This tutorial describes how to set up and run a simulation using multiple-input multiple-output (MIMO) antennas in Wireless InSites X3D propagation model. It also shows how to use the COMMUNICATION SYSTEM Analyzer to post-process propagation results to calculate Throughput and Signal-to-Interference-Plus-Noise for a MIMO system.

Creating a New Project

1. Click *File→New→Project*. The PROJECT PROPERTIES window will appear.
2. Enter a *Short description* and click OK.

Loading a Terrain File

To load the existing terrain file:

1. Click *File→Open→Geometry→Terrain*.
2. Select `tutorials_location\MIMO_Tutorial\rosslyn_DTED.ter`
3. Click OPEN.

Loading an Image File

To load an image file over the terrain:

1. Click *File→Open→Image*.
2. Select the file at `tutorials_location\MIMO_Tutorial\OneFoot_Potomac_DCVA_2013-Small2.tif`
3. The IMAGE PROPERTIES window appears. Enter the following values to place it in the scene:
   (a) Set the *coordinate system* to “Cartesian”.

(b) *Lower Left corner X (m)*: 278

(c) *Lower Left corner Y (m)*: 212

(d) *Length (X axis) (m)*: 766

(e) *Length (Y axis) (m)*: 1450

The GEOMETRY VIEW should now look similar to Figure 7.1.

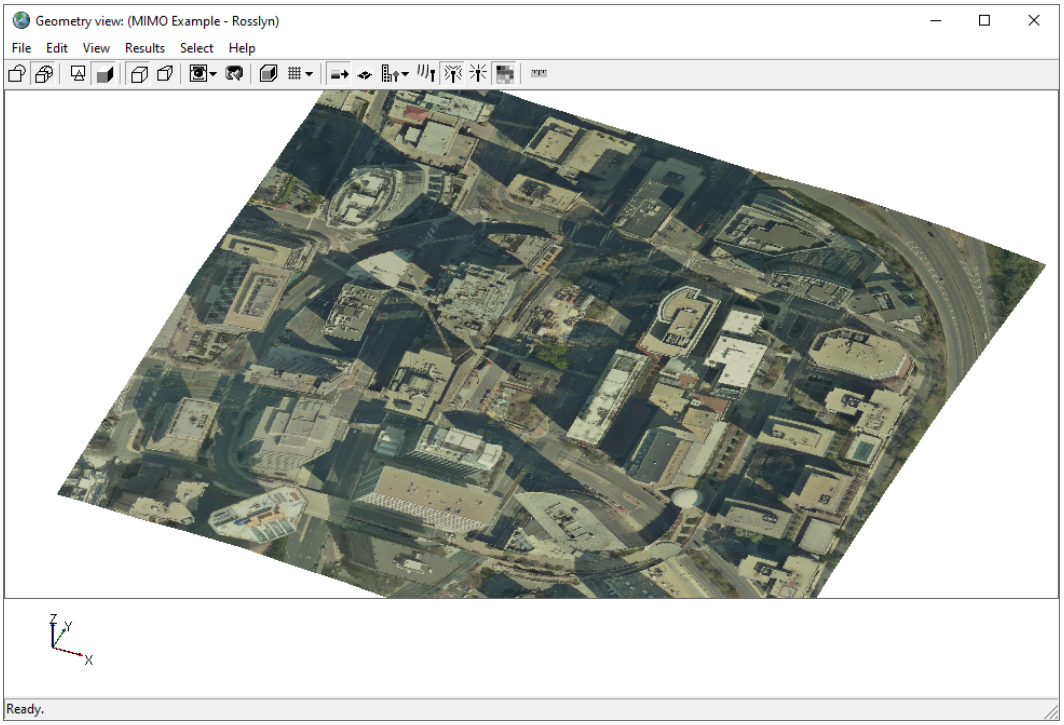


Figure 7.1: Rosslyn terrain with the image overlay.

Loading a City File

Once the project has been created, building data can be loaded. To load an existing `*.city` file:

1. Click *File*→*Open*→*Geometry*→*City*.
2. Select `tutorials_location\MIMO_Tutorial\rosslyn.city`
3. Click OPEN.

The GEOMETRY VIEW, when viewed in 3D SOLID BODY mode, should look similar to Figure 7.2.

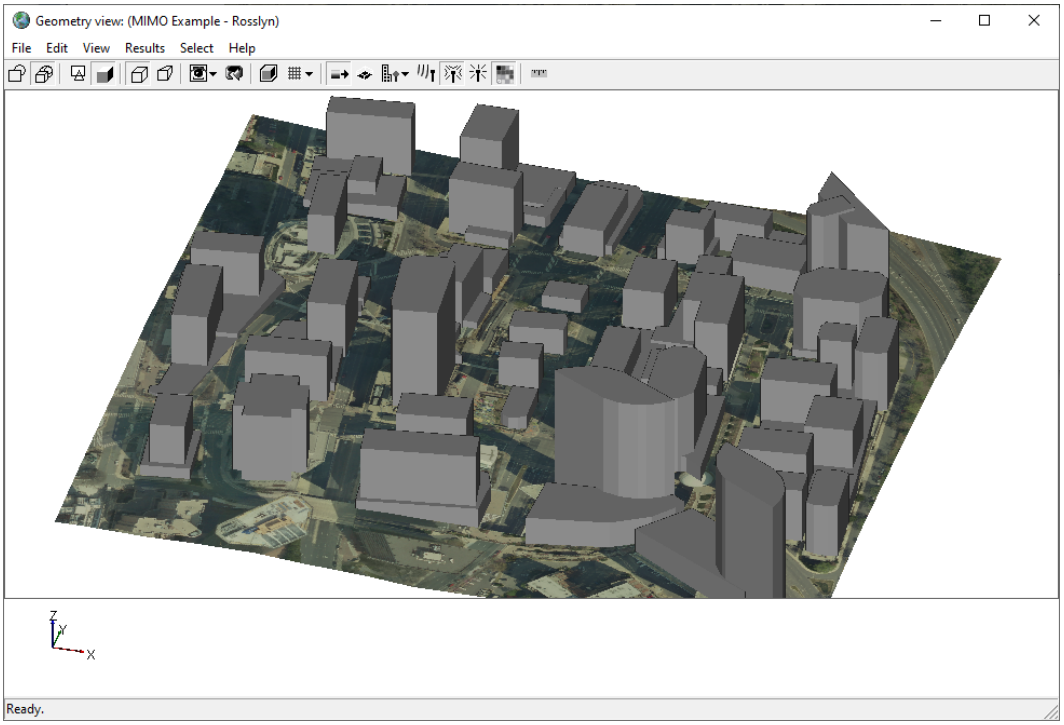


Figure 7.2: Rosslyn city geometry placed on the terrain.

Creating a Waveform

Create a narrowband sinusoidal 28 GHz waveform with a bandwidth of 100 MHz.

1. Click *File→New→Waveform*.
2. Select *Sinusoid* from the drop-down menu and click OK.
3. The WAVEFORM PROPERTIES window will appear. Enter a *Short description* of "28 GHz Narrowband".
4. Enter a *Carrier frequency* of "28000" and an *Effective bandwidth* of "100 MHz".
5. Click OK.

Creating a MIMO Antenna

Create a dual-polarized 4x4 MIMO antenna array for the TRANSMITTER.

1. Click *File→New→Antenna*.
2. Choose *MIMO* from the drop-down menu and click OK.
3. The MIMO ANTENNA PROPERTIES window appears, as seen in Figure 7.3.

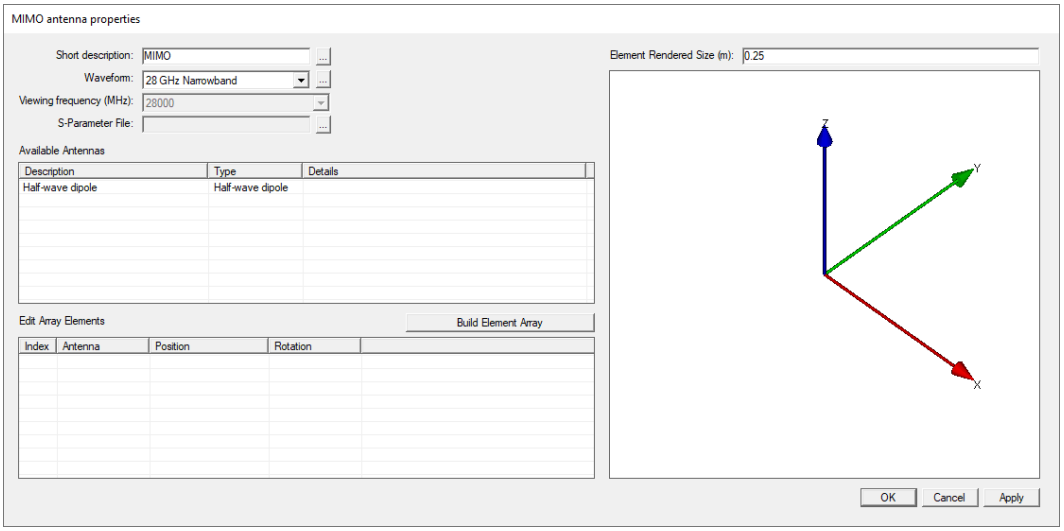


Figure 7.3: A view of the empty MIMO ANTENNA PROPERTIES window.

4. Enter a *Short description* of “MIMO 4x4 Dual-Pol (28GHz)”.
5. Select “28 GHz Narrowband” from the *Waveform* drop-down menu.
6. Click the BUILD ELEMENT ARRAY button to open the MIMO ARRAY BUILDER window.
7. Select “Half-wave dipole” from the *Antenna* drop-down menu.
8. Select the *Wavelengths* radio button to set the unit of spacing between array elements to wavelengths of the specified WAVEFORM (1 = lambda).
9. Enter the follwing values for the *Array Layout*:
   (a) *X # Elements*: 1
   (b) *X Spacing*: 1
   (c) *Y # Elements*: 4
   (d) *Y Spacing*: 1
   (e) *Z # Elements*: 4
   (f) *Z Spacing*: 1
10. Click OK to add the new element array to the *MIMO* antenna.
11. Click the BUILD ELEMENT ARRAY button again to begin adding another array.
12. Duplicate the settings of the previous array.
13. Enter “90” degrees in the *Rotate antenna about X axis* field.
14. Under the *Generation Mode* section, ensure that “Add Layout” is selected to specify that the new element array is to be added to the existing elements of the *MIMO* antenna.

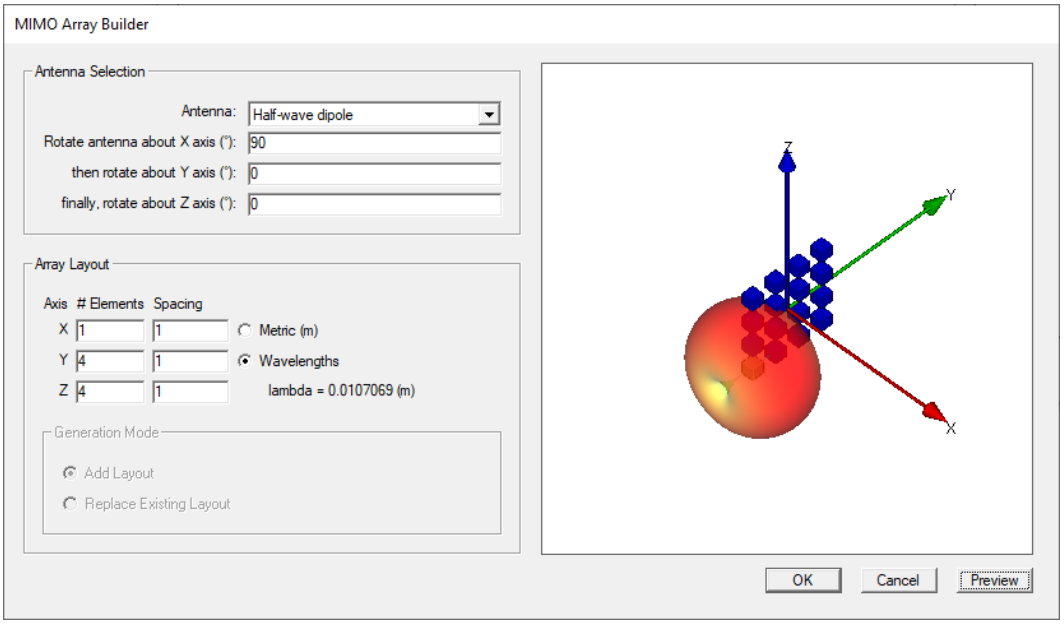


Figure 7.4: The MIMO ARRAY BUILDER window.

15. Click OK on the MIMO ARRAY BUILDER window. The *MIMO* antenna should now look as it does in Figure 7.5.
16. Click OK on the MIMO ANTENNA PROPERTIES window to finish creating the *MIMO* antenna.
17. Save the project.

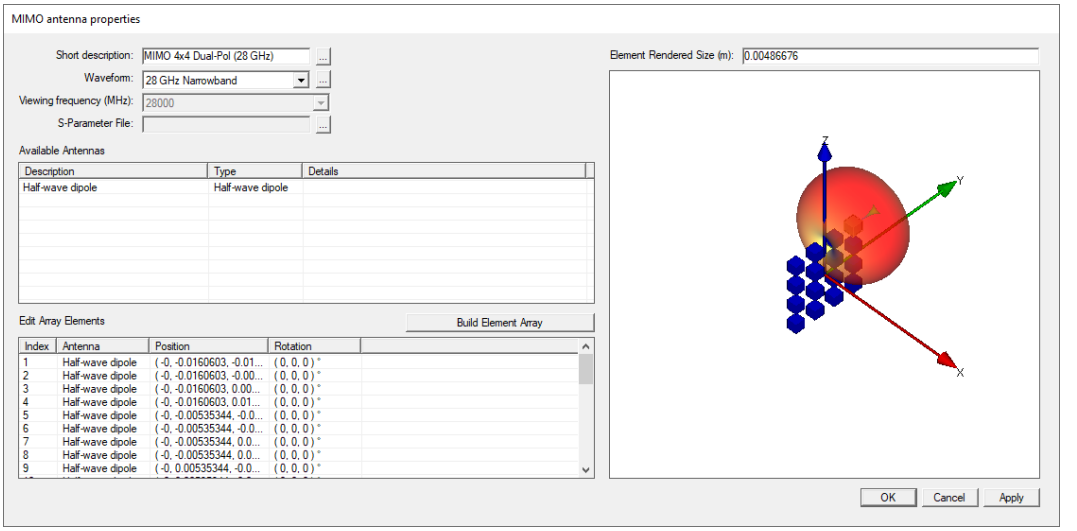


Figure 7.5: The finished MIMO antenna in the MIMO ANTENNA PROPERTIES window.

Importing a MIMO Antenna from XFdtd®

To import a *MIMO* antenna generated by XFdtd® for the RECEIVERS:

1. Click *File→Import→XF MIMO antenna specification*.
2. Select `tutorials_location\MIMO_Tutorial\2x2-WI-MIMO-SingleFreq\index.xml`
3. The location, orientation, and antenna pattern of each element in the *MIMO* antenna will be imported along with its accompanying WAVEFORM.

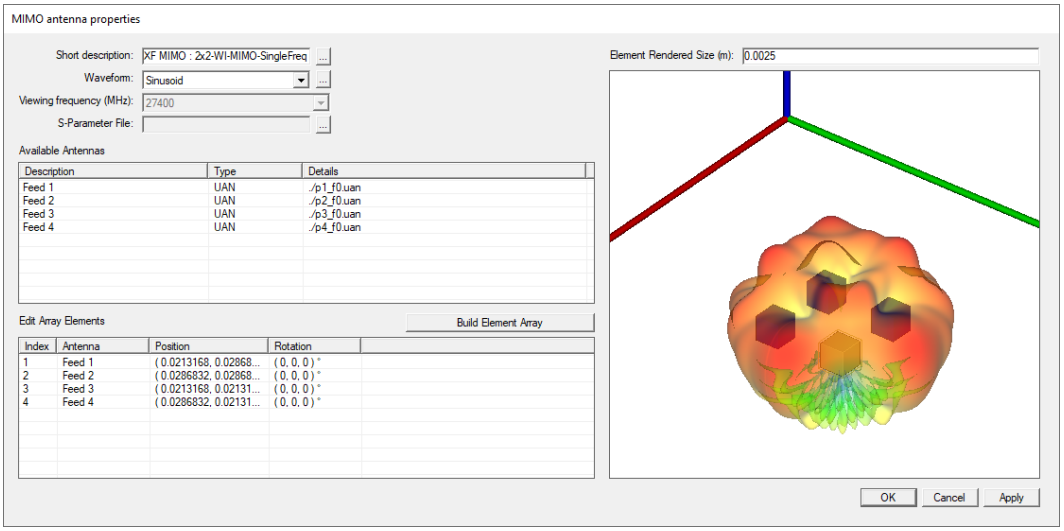


Figure 7.6: An XF-generated MIMO antenna imported into Wireless InSite.

Creating a Transmitter

In this project, one TRANSMITTER point will act as a *MIMO* base station. To create the TRANSMITTER:

1. Click *File*→*New*→*Transmitter set*→*Points*.
2. In the GEOMETRY VIEW, click anywhere close to Cartesian coordinates X=641 m, Y=435 m. Right-click to finish.
3. In the TX/RX PROPERTIES window, enter a *Short description* of "Base Station 28GHz 4x4 MIMO".
4. Click the TRANSMITTER PROPERTIES button.
   (a) Select "MIMO 4x4 Dual-Pol (28GHz)" as the antenna.
   (b) Select "28 GHz Narrowband" as the waveform.
   (c) Select "Sequential" as the coordinate system.
   (d) Enter "90" degrees in the *Last about Z axis* field.
   (e) Enter "30" dBm in the *Input power* field.
   (f) Click OK to return to the TX/RX PROPERTIES window.
5. Click the LAYOUT PROPERTIES button.
   (a) In the TX/RX LAYOUT PROPERTIES window, click the EDIT CONTROL POINTS button.
   (b) Double-click the control point and set *Z (m)* = "10".
   (c) Click OK.
6. Click OK to exit all open windows.

Creating a Receiver Route

The *MIMO* Base Station will be transmitting to a route of *MIMO* RECEIVERS used to simulate a mobile device moving through the scene. To create the *MIMO* *Route*:

1. Click *File*→*New*→*Receiver Set*→*Route*.
2. Starting from the southwest, click along a path to create a route similar to the one seen in Figure 7.7. Right-click to finish.
3. In the TX/RX PROPERTIES window, enter a *Short description* of "Mobile Device Route".
4. Click the RECEIVER PROPERTIES button.
   (a) Select "XF MIMO : 2x2-WI-MIMO-SingleFreq @ 27400 MHz" as the ANTENNA.
   (b) Select "Sinusoid", the waveform imported alongside the XF MIMO antenna, as the WAVEFORM.
   (c) Click OK to return to the TX/RX PROPERTIES window.
5. Click the LAYOUT PROPERTIES button.
   (a) Set the *Spacing* to "1" meter.
   (b) Make sure the *Enable APG for X3D* checkbox is checked in order to enable the use of adjacent path generation on the *Route* to accelerate X3D calculations.
   (c) Click OK.
6. Click OK to exit all open windows.

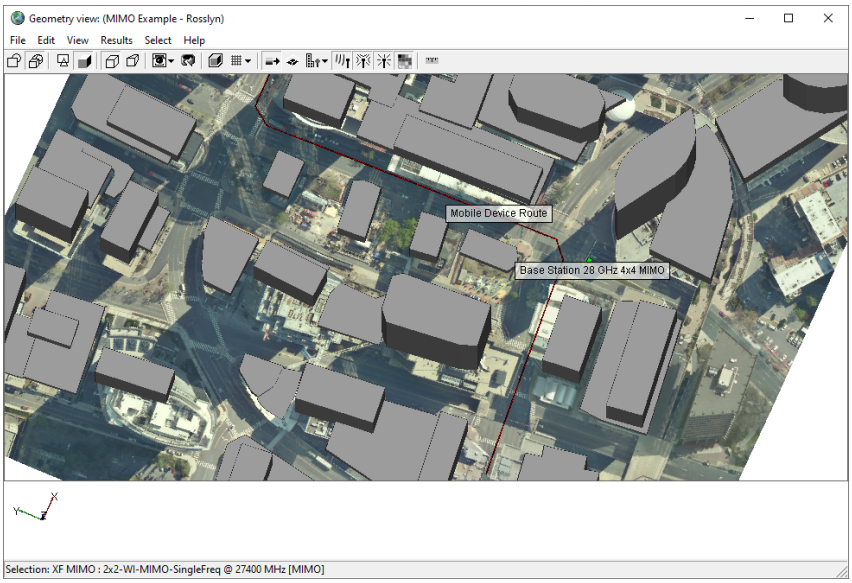


Figure 7.7: A route of receivers running through the scene.

Creating a Study Area

To create a STUDY AREA:

1. Click *File→New→Study area*.
2. Select *Fit to geometry* and click BEGIN.
3. In the STUDY AREA PROPERTIES window, enter "MIMO Base Station" as the *Short description*.
4. Select "X3D" as the *Propagation model*.
5. Deselect the *Automatic* checkbox on *Number of reflections*, and set this to "6".
6. Deselect the *Automatic* checkbox on *Number of transmissions*, and set this to "0".
7. Deselect the *Automatic* checkbox on *Number of diffractions*, and set this to "1".
8. Select the *APG Enabled* checkbox and click the APG ACCELERATION button.
9. In the APG ACCELERATION PROPERTIES window, enter the following values:
   (a) *Adjacency Distance (m)*: 5
   (b) *Paths to Consider*: 50
   (c) Click OK.

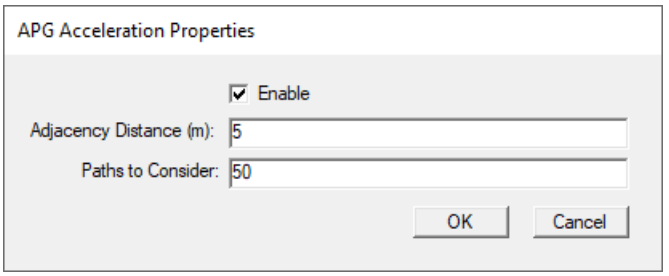


Figure 7.8: The APG ACCELERATION PROPERTIES window.

10. Click OK to finish creating the STUDY AREA.

Running a Calculation

To start a calculation, click RUN→*New* in the MAIN WINDOW. If you have not saved all changes made to this point in the Wireless InSite project, you will be prompted to do so.

Viewing MIMO Channel Output

The MULTI-CHANNEL RESULTS BROWSER window provides three methods of processing and viewing output generated by *MIMO* antennas. This provides the ability to select specific sections of output to generate or view based on Tx/Rx set, UTD instance, and individual points and elements within sets. This

prevents the need to navigate through a potentially overwhelming amount of data, especially when using MIMO antennas with a large number of elements.

To browse the results generated by a *MIMO* antenna in X3D:

1. Click on the RESULTS tab in the MAIN WINDOW.
2. Expand the output tree: *Area: MIMO Base Station→Multi-Channel Data*.
3. Right-click on "Multi-Channel Data" and select "Browse Results". This will open the MULTI-CHANNEL RESULTS BROWSER window.

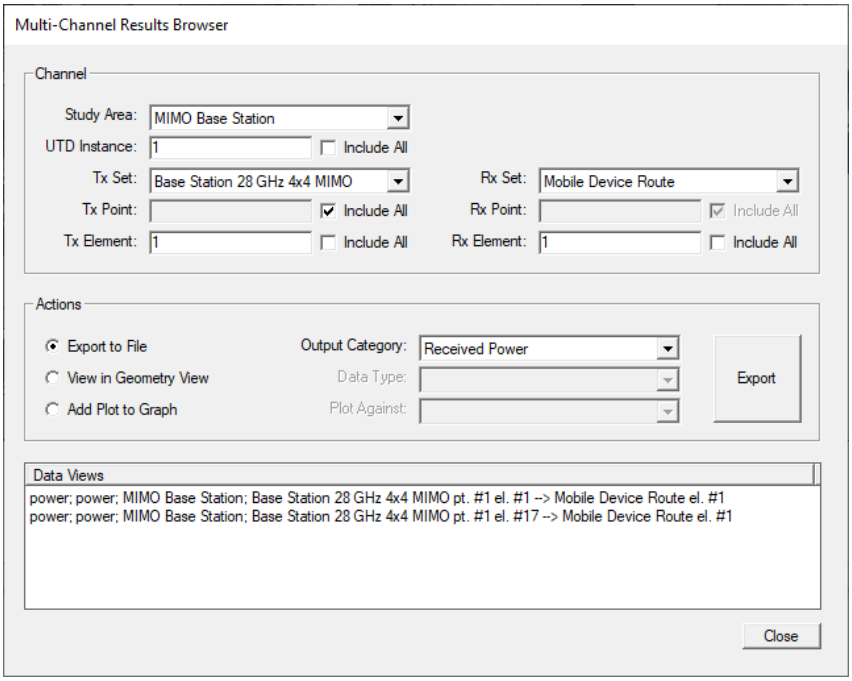


Figure 7.9: The MULTI-CHANNEL RESULTS BROWSER window.

To process *MIMO* data via the MULTI-CHANNEL RESULTS BROWSER window:

- Method 1: Export to File
  1. Check the *Include All* checkbox to the right of the *Tx Element* field to select all elements of the MIMO transmitter.
  2. The *Rx Element* field should read "1".
  3. In the *Actions* section, select "Export to File".
  4. Select "Received Power" from the *Output Category* drop-down menu.
  5. Click the EXPORT button.
  6. CSV files containing received power data for Element 1 of the receiver route from each element of the MIMO transmitter will have been generated in `project_location\MIMO Base Station\power`.

- Method 2: View in Project View

  1. Uncheck the *Include All* checkbox to the right of the *Tx Element*. The *Tx Element* field should read “1”.
  2. The *Rx Element* field should read “1”.
  3. Select “View in Project View” under the *Actions* section.
  4. Select “Received Power” in both the *Output Category* and *Data Type* drop-down menus.
  5. Click the VIEW button.
  6. Return to the ANALYZE CHANNEL DATA window.
  7. Repeat the process for *Tx Element* “17”.
  8. Note the differences in power along the receiver route (Figure 7.10).

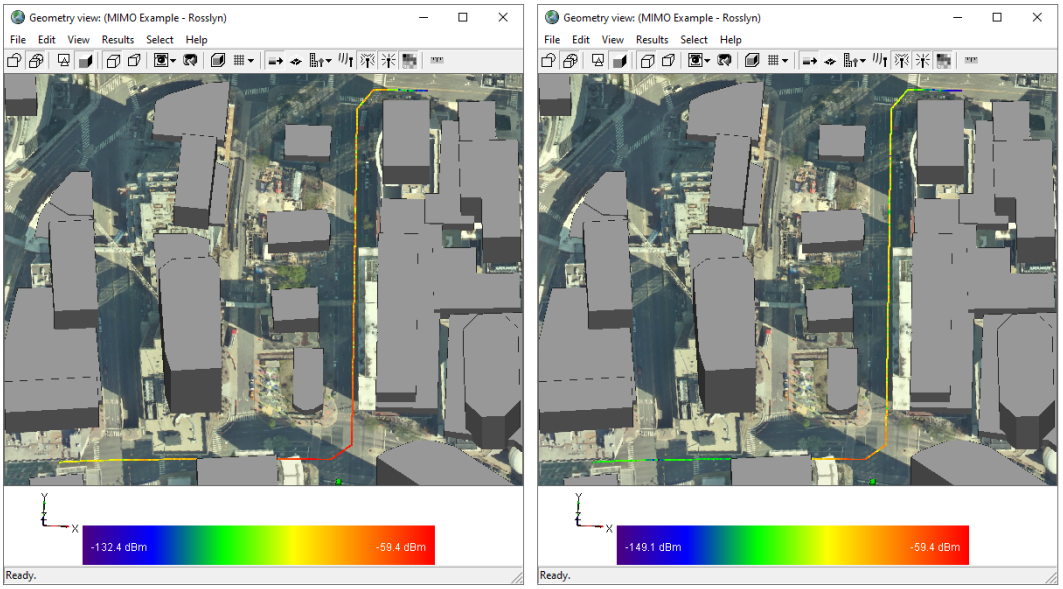


Figure 7.10: Power received by Element 1 of the receiver route from Elements 1 (left) and 17 (right) of the MIMO transmitter.

- Method 3: Add Plot to Graph

  1. Check the *Include All* checkbox to the right of the *Tx Element* field to select all elements of the MIMO transmitter.
  2. The *Rx Element* field should read “1”.
  3. Select “Add Plot to Graph” under the *Actions* section.
  4. Select “Received Power” from the *Output Category* drop-down menu.
  5. Select “Received Power” from the *Data Type* drop-down menu.
  6. Select “Receiver Number” from the *Plot Against* drop-down menu.

7. Click the PLOT button.

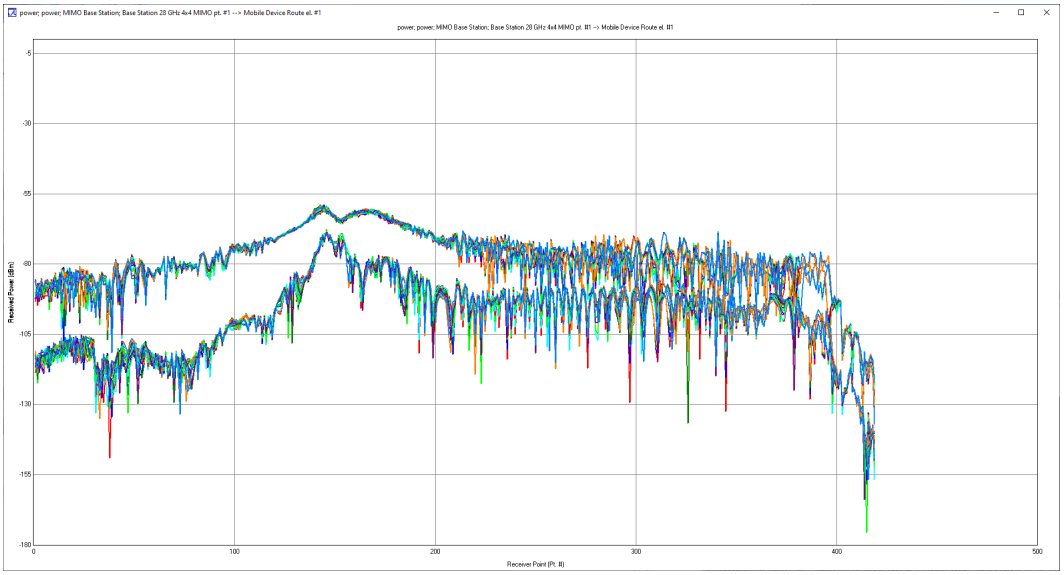


Figure 7.11: The Mobile Device Route's received power from each element of the MIMO transmitter.

8. Return to the MULTI-CHANNEL RESULTS BROWSER window.
9. Uncheck the *Include All* checkbox next to the *Tx Element* field. The field should read "1".
10. Check the *Include All* checkbox next to the *Rx Element* field.
11. Click the PLOT button.

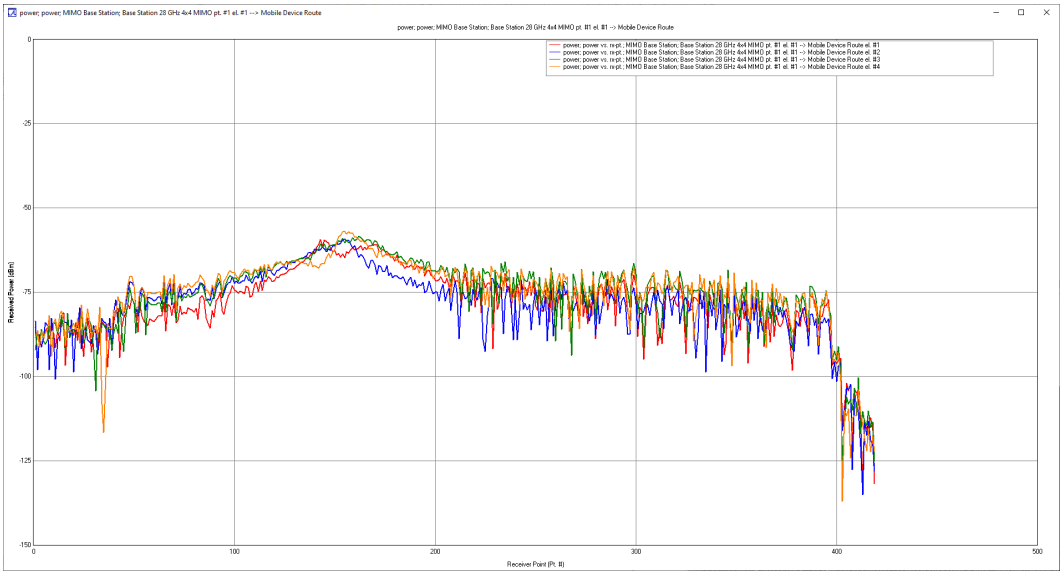


Figure 7.12: Power received by every element of the Mobile Device Route from Element 1 of the MIMO transmitter.

Post-Processing of Calculations using the Communication System Analyzer

Wireless InSite's Communication System Analyzer will be used to post-process propagation results to calculate:

- Signal-to-Interference-Plus-Noise ratio along a receiver route
- System throughput of a 5G NR MIMO device

At the MIMO transmitter, the maximum ratio transmission (MRT) method will be used to beamform to each of the receivers along the route. At the MIMO receiver, maximal ratio combining will be used to model the receiver diversity. Both methods use the channel vectors of the H-matrix to adjust the magnitudes and phases of the MIMO antenna elements in order to maximize the total received power.

Performing MIMO Comms Analysis: Signal-to-Interference-Plus-Noise Ratio (SINR)

To calculate SINR along a receiver route:

1. Select *File→New→Communication systems* to add a Communication System to the project.
2. Enter "SINR Analysis" as the *Short description*.
3. In the upper portion of the Communication System properties window, place check marks next to the transmitter and receiver to include them in the analysis.
4. In the *Analysis Type* section, click the radio button next to *Interference and Receiver Summary Files Only*.
5. In the *MIMO Method* section, click on *Beamforming / Diversity*.

6. For *Tx Beamforming / Precoding* to *Max Ratio Trans (MRT)*.
7. For *Rx Diversity Combining*, select *Max Ratio Combining*.
8. Enter *100* for the *Signal Bandwidth (MHz)* in the *Signal Parameters* section. When the *Signal Bandwidth* field is updated, the *Noise* updates to provide the global noise level in the project.
9. Click *OK* to save the Communication System settings.

Figure 7.13 shows the COMMUNICATION SYSTEM properties window will the settings described above. To run a Communication System Analysis, choose RUN→*Comm. system analysis* in the MAIN WINDOW.

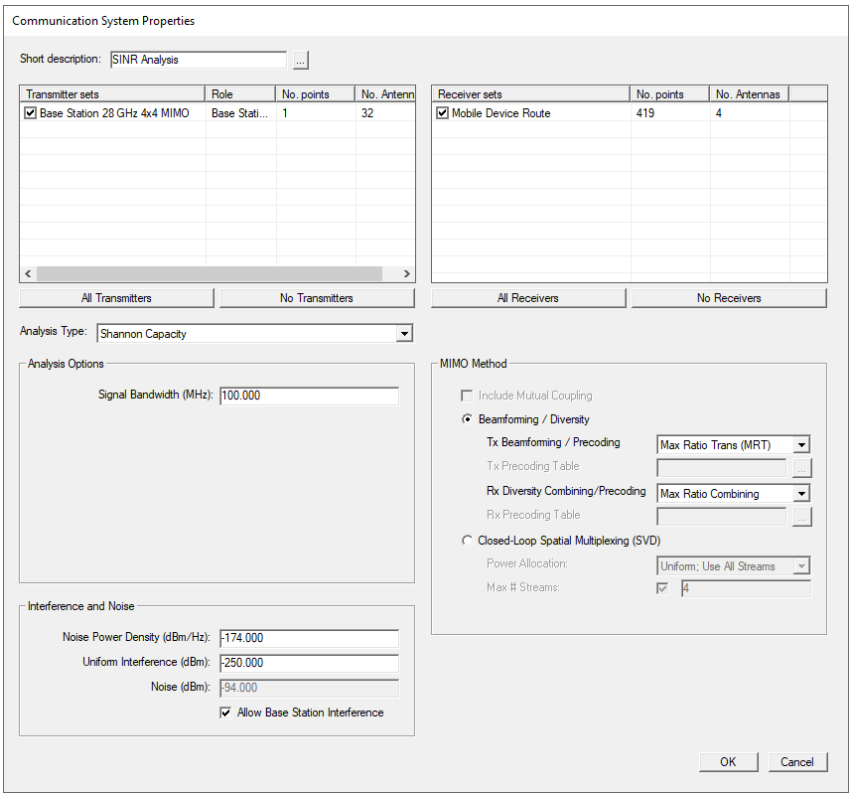


Figure 7.13: Communication System properties window for signal-to-interference-plus-noise ratio calculation.

After the calculation has completed, the SINR results can be plotted by:

1. Click on RESULTS in the MAIN WINDOW.
2. Expand *Area: MIMO Base Station→Communication systems→SINR Analysis→Interference and Noise→Base Station 28 GHz 4x4 MIMO→Mobile Device Route*.
3. Select *Plot* from the context menu.
4. For *Select Data* choose the *SINR* option and then click OK.
5. When the CHOOSE GRAPH window appears, verify *To a new graph* is selected and click OK.

The resulting plot of SINR vs distance for the mobile device should resemble Figure 7.14.

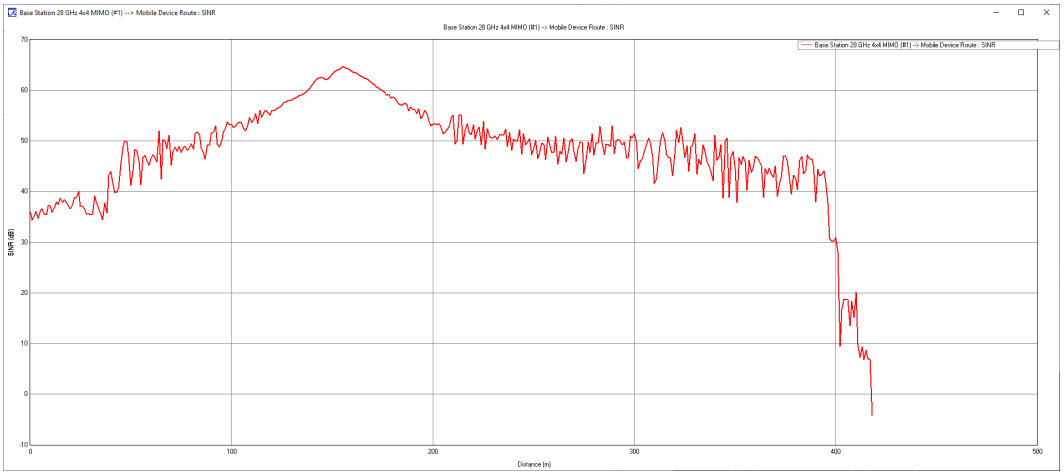


Figure 7.14: SINR along route from the MIMO transmitter to mobile device.

Performing MIMO Comms Analysis: Throughput

A Wireless InSite COMMUNICATION SYSTEM is also capable of calculating throughput between MIMO transmitters and receivers. To do so for this system, first duplicate the COMMUNICATION SYSTEM used for the SINR analysis. In the MAIN WINDOW, navigate to the COMMUNICATION SYSTEM tab, then right click on the existing COMMUNICATION SYSTEM (SINR Analysis) and select *Duplicate* from the context menu. Right-click on the duplicated COMMUNICATION SYSTEM and choose *Properties* to modify the COMMUNICATION SYSTEM :

1. Enter “5G NR Throughput Analysis” in the *Short Description* field.
2. Select *Throughput* in the *Analysis Type* section by clicking the adjacent radio button.
3. Select *User-defined Table* in the *Wireless Access Method* section.
4. To specify the user-defined throughput file, click the “…” button next to the *Throughput File* field.
5. Navigate to `tutorials_location\MIMO_Tutorial\5GNR_100MHz_3GPPSpec.wam` and select the .wam file
6. Click *OK* to save the Communication System settings.

Figure 7.15 shows the COMMUNICATION SYSTEM properties window will the settings described above. To run a Communication System Analysis, choose RUN→*Comm. system analysis* in the MAIN WINDOWwindow.

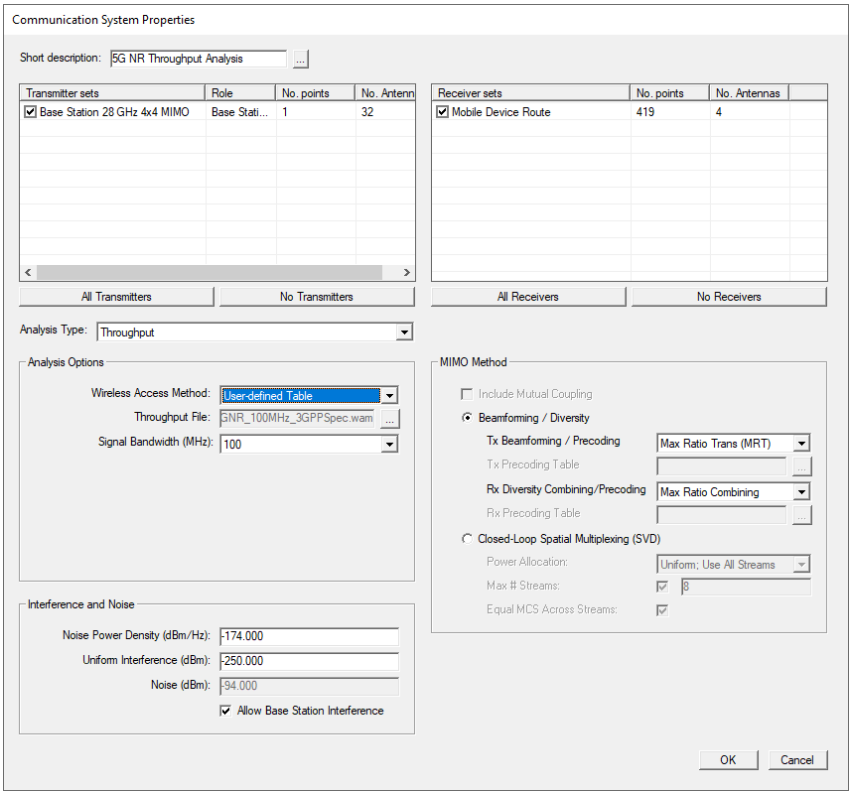


Figure 7.15: Communication System properties window for throughput analysis.

After the calculation has completed, the throughput results can be plotted by:

1. Select RESULTS in the MAIN WINDOW.
2. Expand *Area: MIMO Base Station→Communication systems→5G NR Throughput Analysis→Throughput→Base Station 28 GHz 4x4 MIMO→Mobile Device Route*.
3. Right click on *Mobile Device Route* and select *Plot* from the context menu.
4. For *Select Data* make sure the *Throughput* option is selected and then click OK.
5. When the CHOOSE GRAPH Window appears, verify *To a new graph* is selected and click OK.

The resulting plot of throughput vs distance for the mobile device should resemble Figure 7.16.

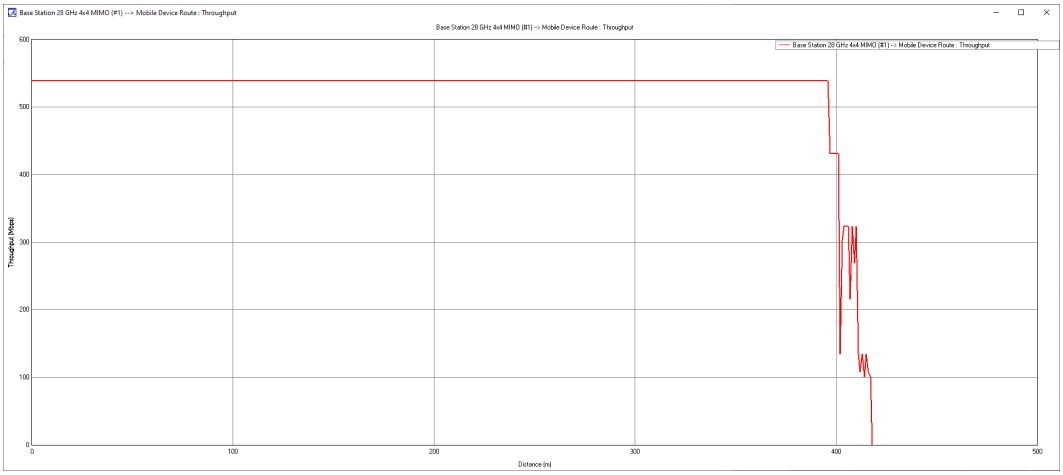


Figure 7.16: Throughput for the Mobile Device along a route from the MIMO transmitter.

Chapter 8

Maximum Permissible Exposure Prediction

MPE Tutorial

The following tutorial covers setting up a scenario with both Controlled and Uncontrolled MPE calculations. The calculations use the IEEE standard for MPE. The required inputs are listed below along with step-by-step instructions how to set up a complete scenario from start to finish.

To set up an MPE analysis, the user must specify:

In Study Area:

- IEEE Controlled
- Uncontrolled
- User Defined Thresholds

Transmitter Set:

- Trajectory Set
- Single Point Point Set

Waveform:

- ▶ Please refer to the Waveform Chapter of the Reference Manual for full explanation on each of these waveform types.
- Sinusoid, Blackman or Gaussian
- Carrier frequency
- Pulse Repetition Frequency (PRF)
- Pulse width

Importing Terrain

1. Create a new project in Wireless InSite.
2. In the MAIN WINDOW Window, select *File*→*Open*→*Geometry* →*Terrain*
3. Browse to the `tutorials_location\Urban_Tutorial\Rosslyn_Flat_Terrain.ter` on the hard drive and open it.

   ▶ See page 3 for information about the location of the tutorial projects on your machine.

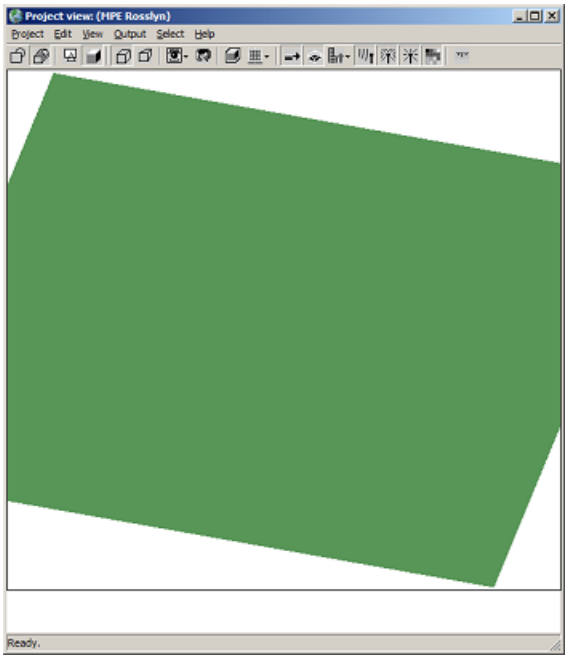


Figure 8.1: Terrain File Imported.

Importing City File

1. Click *File*→*Open*→*Geometry*→*City*
2. Select the `tutorials_location\Urban_Tutorial\rosslyn_va.city` city file.
3. Click OK.

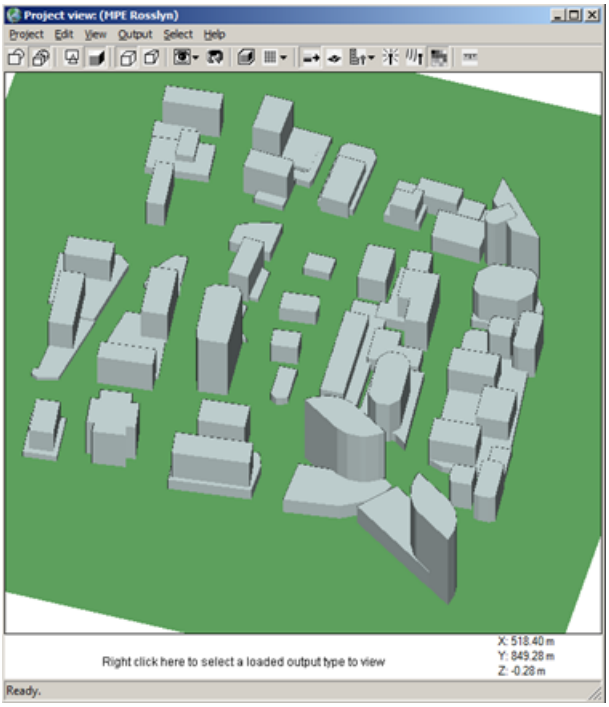


Figure 8.2: Rosslyn City File Imported.

Save Project

1. *File→Save Project as*
2. Save the project on your hard drive
3. Save the city and terrain file when prompted.

Creating Waveform

1. *Right Click→New→Waveform*
2. Select Sinusoid.
3. Change the *Short Description* to “100 GHz-Sinuoid”.
4. Change the *Carrier Frequency* to “100000 MHz”.
5. Click OK.

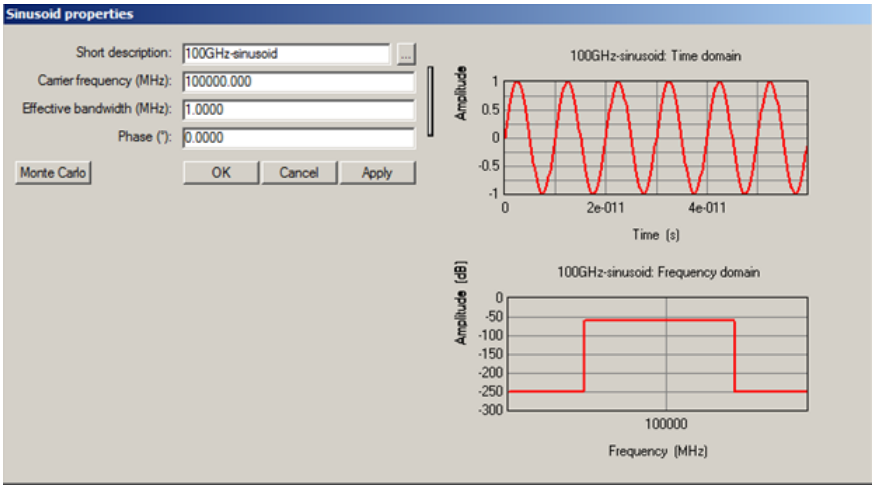


Figure 8.3: Waveform Properties Window.

Creating Transmitter Set

1. Create a Tx Trajectory; *Right Click→New→Transmitter Set→Trajectory*

   ✓ MPE works with Trajectories or Point TRANSMITTER sets

2. Define the trajectory set by tracing the route of the trajectory within the GEOMETRY VIEW . Click at each check of direction. Right click to finish.

   (a) In the GEOMETRY VIEW, right-click in the area where the coordinates are displayed (in the bottom-right corner) and choose *Cartesian*.

   (b) Set the first point by clicking anywhere close to X =635 m, Y =680 m

   (c) Set the turning point by clicking anywhere close to X =490 m, Y =680 m

   (d) End the trajectory by clicking anywhere close to X =520 m, Y =570 m, and right-click to finish.

3. The TX/RX PROPERTIES window will open up.

4. Change the *Short description* to “Tx Trajectory 15m -100GHz very slow”.

5. Click the TRANSMITTER PROPERTIES button.

6. Set *Input Power* to 100dBm.

7. Click OK to exit the TRANSMITTER PROPERTIES window.

8. Click the LAYOUT PROPERTIES button which will bring up the TX/RX LAYOUT PROPERTIES window where the spacing, start time, and velocity can be set.

9. Set the spacing to determine the number of points within the Trajectory to 15m.

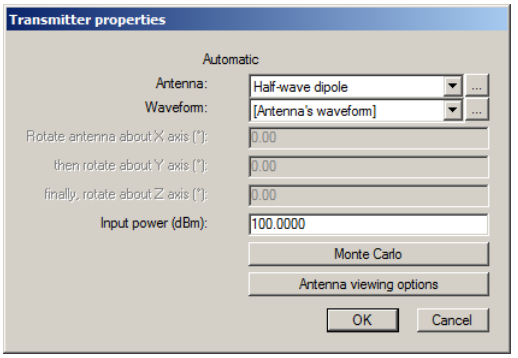


Figure 8.4: Tx Properties Window.

10. Change *Start Time* to “0.00”.
11. Change *Velocity* to “0.001” m/s.
12. Click OK.
13. Click OK.

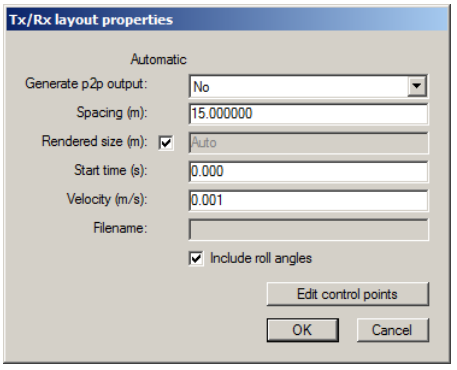


Figure 8.5: Tx/Rx layout Properties window.

Creating Receiver Set

1. *Right Click→New→Receiver Set→XY Grid*
2. Draw a square to encompass the trajectory set.
3. Change *Short description* to “RX Grid”.
4. Click the RECEIVER PROPERTIES button.
5. Change *Antenna* to “Half-wave dipole”.
6. Click OK.
7. Click LAYOUT PROPERTIES.
8. Change *Spacing* to “5m”.
9. Click OK.

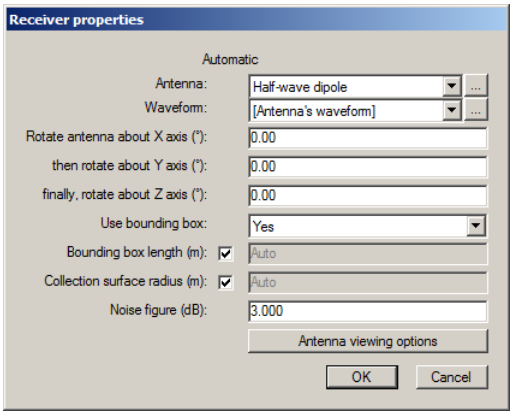


Figure 8.6: Receiver Properties Window.

The set up should look something like the scene in Figure 8.7.

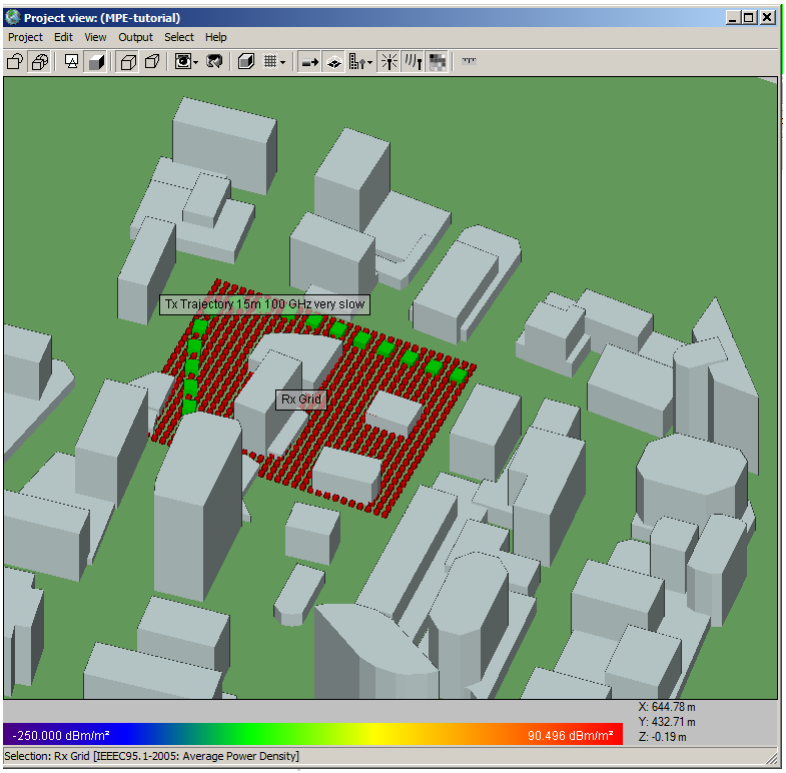


Figure 8.7: MPE Trajectory and Receiver grid

Setting MPE Calculations in X3D Study Area

1. Ensure there is one active Tx Set and Rx Set within the scene.
2. *Right Click→New→Study Area*
3. The Study area creation window will pop up. Select Fit to geometry.
4. Click BEGIN.
5. The Study area properties window will pop up.
6. Change Short Description to Controlled_100GHz_100dBm_0_001ms.
7. Change the Propagation model to X3D. Everything else should be Grayed out with Auto.
8. Unclick the check box next to reflections and enter 6 in the box.
9. Unclick the check box next to transmissions and enter 1 in the box.
10. Unclick the check box next to diffractions and enter 0 in the box.

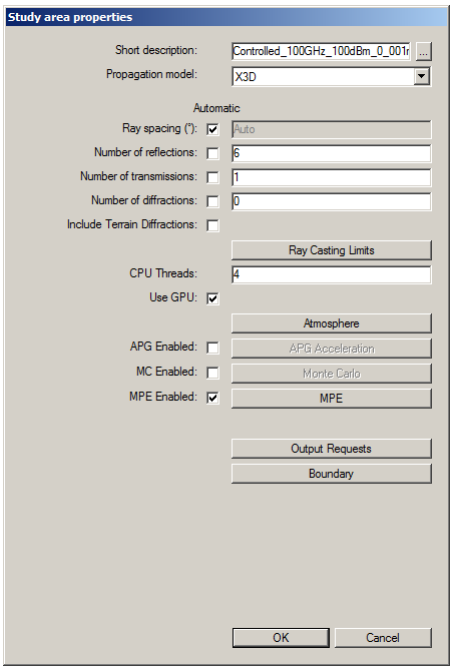


Figure 8.8: Studyarea Properties Window.

11. Within the Study area properties window, click on the MPE Properties button.
12. This brings up the MPE Properties window.
13. Set the MPE to be active by checking the box next to MPE Active
14. Select IEEE Controlled environment.

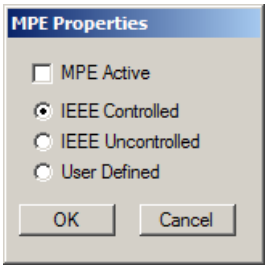


Figure 8.9: Controlled MPE Properties Window.

15. To create an Uncontrolled MPE calculation study area, follow steps 2-12.
    - It is recommended to change the name of the study area to Uncontrolled_100GHz_100dBm_0_001ms.
16. When the MPE properties window comes up, select IEEE Uncontrolled.

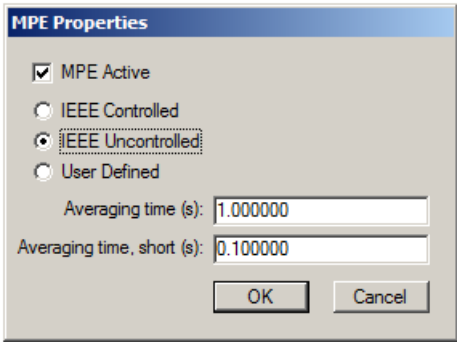


Figure 8.10: Uncontrolled MPE Properties Window.

17. Click OK.
18. Click OK.

Running a New Calculation

To start a calculation, click on RUN→*New* in the MAIN WINDOW. If you have not saved all changes made to this point in your Wireless InSite project, you will be prompted to do so.

Viewing MPE Output

1. Navigate to the Output tab.
2. Expand the Controlled_100GHz_100dBm_0_001ms study area by clicking on the '+' next to the name.
3. Click the '+' next to Point to multipoint.
4. All of the calculated MPE quantities will be displayed.
5. Click the '+' by the IEEEC95.1-2005: Average Power Density in Short Interval.
6. Click the '+' next to the Tx Trajectory 15m -100GHz - slow, and click the '+' next to Point #0.

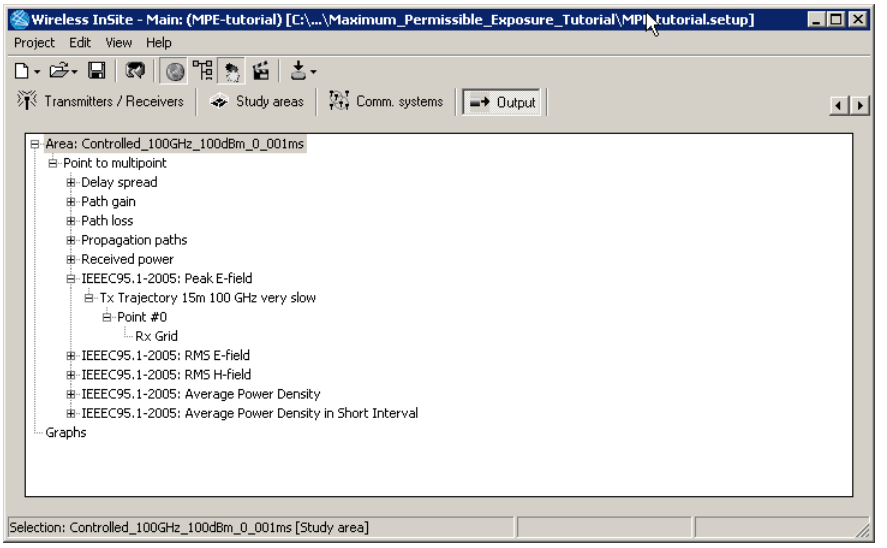


Figure 8.11: Output Tab.

7. Right click on RX Grid, and select View.

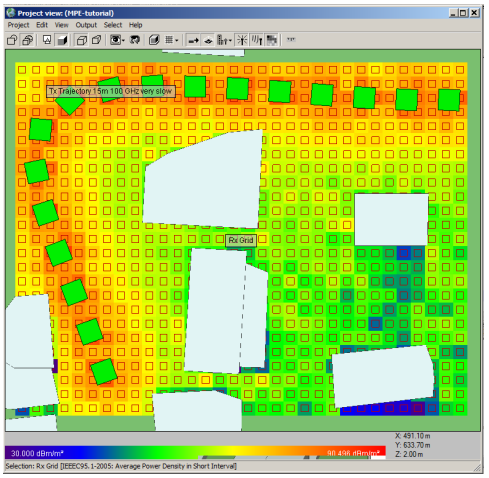


Figure 8.12: Average Power Density in Short Interval dBm/m^2

8. Right click on the RX set, and select View MPE Threshold to view the percent of the IEEE threshold that was reached at each receiver point.

   ✓ White regions mean there is either no simulated data available or there is no threshold to evaluate. Green means the simulated value is 50

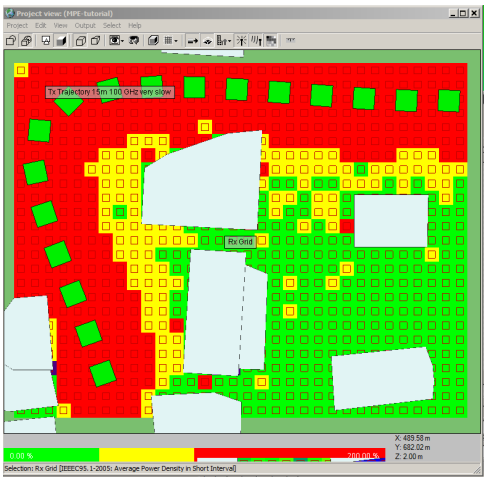


Figure 8.13: Percent Threshold of Average Power Density in Short Interval

9. To view the aggregate threshold reached, based on all IEEE quantities, Right click on the RX set, and select View Aggregated MPE Thresholds.

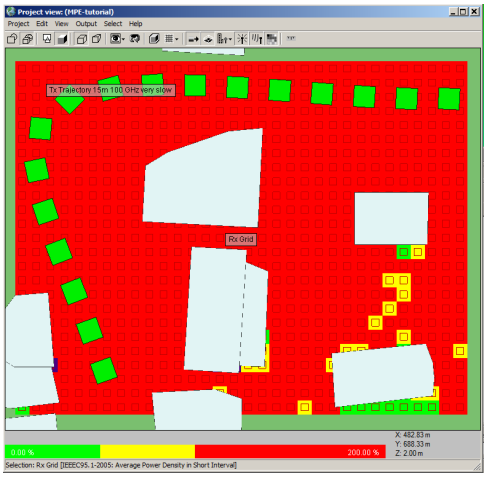


Figure 8.14: Aggregated MPE Thresholds

Chapter 9

Modified COST Building Penetration Model

Modified COST Building Penetration Model

Wireless InSite's Wireless InSite's Modified COST Building Penetration Model capability models outdoor-to-indoor propagation without requiring the building's interior details to be explicitly defined. The following tutorial describes how to set up and run a simulation using the modified COST building penetration model capability.

To use the Building Penetration Model capability, the user must:

- Enable the Building Penetration Model in the Extensions window
- Define an Building Penetration receiver grid in the project
- Run the simulation with the X3D propagation model

Enabling the Modified COST Building Penetration Model

Prior to using Modified COST Building Penetration Model, it must first be activated in the Extensions tab of the Preferences window.

1. In the MAIN WINDOW, click *Edit→Preferences*.
2. Click on the *Extensions* tab at the top of the window.
3. Select the checkbox for *Modified COST Building Penetration Model*
4. Click OK.

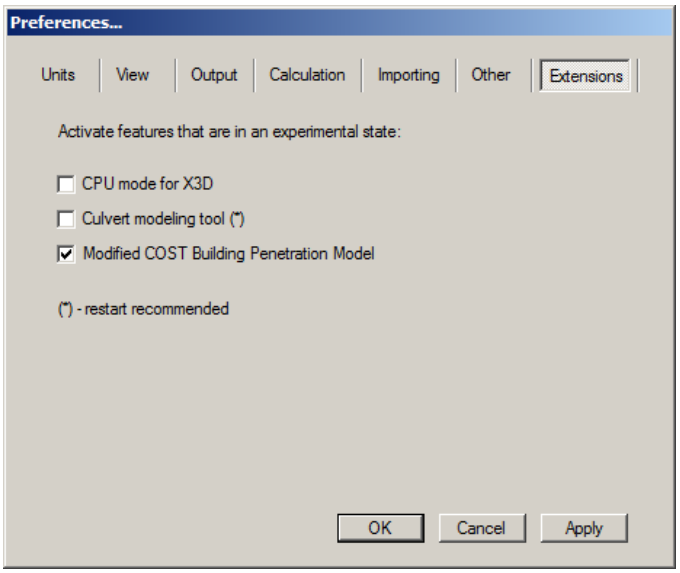


Figure 9.1: Extension window for activating the building penetration model capability

Creating a New Project

1. In the GEOMETRY VIEW, click *File→New→Project*. Name the project "Building Penetration Model Tutorial" and click OK.

Loading a City File

Once you have created the project, you can load building data. To load an existing `*.city` file:

1. Click *File→Open→Geometry→City*.
2. Select the file `tutorials_location\Building_Penetration_Model_Tutorial\highlands.city`.
   - See page 3 for information about the location of the tutorial projects on your machine.
3. Click OPEN.

Loading a Terrain File

To load the existing *Terrain* file:

1. Click *File→Open→Geometry→Terrain*.
2. Select the file `tutorials_location\Building_Penetration_Model_Tutorial\highlands.ter`.
3. Click OK.

The GEOMETRY VIEWshould now look similar to Figure 9.2.

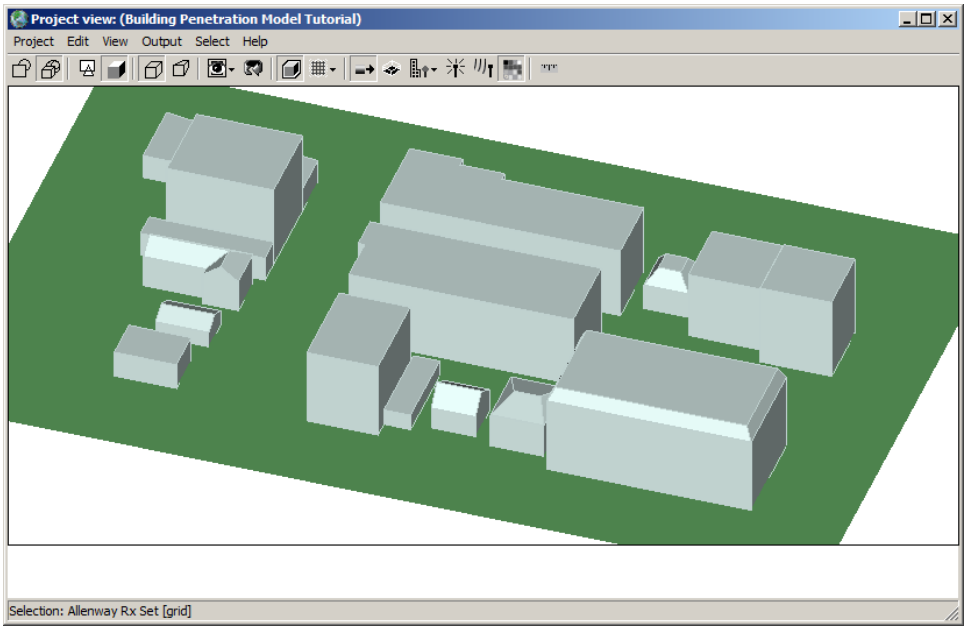


Figure 9.2: Building Penetration Model Project Geometry

Creating a Waveform

Create a narrowband sinusoidal 2400 MHz waveform with a bandwidth of 1 MHz.

1. Click *File→New→Waveform*.
2. Select a waveform type of *Sinusoid*. Click OK.
3. The WAVEFORM PROPERTIES Window will appear. Enter a *Short description* of "Narrowband (2.4 GHz Sine)".
4. Enter a *Carrier frequency* of 2400 MHz and an *Effective Bandwidth* of 1 MHz.
5. Click OK.

Creating Antennas

Create a vertically polarized, short dipole antenna (for the TRANSMITTERS).

1. Click *File→New→Antenna* to bring up the CREATE NEW ANTENNA Window.
2. Choose an *Antenna type* of "Short dipole" and click OK.
3. Enter a *Short description* of "Short dipole".
4. Leave the *Maximum gain* automatic option checked.
5. Set the *Polarization* to "Vertical".

6. Leave the other fields at their default settings, and click OK.

Now create a vertical isotropic antenna (for the RECEIVERS).

1. Click *File→New→Antenna* to bring up the CREATE NEW ANTENNA Window.
2. Choose an *Antenna type* of "Isotropic" and click OK.
3. The ANTENNA PROPERTIES Window for the isotropic antenna appears. Enter a *Short description* of "Vertical isotropic".
4. Leave the *Maximum gain* automatic option checked.
5. Set the *Polarization* to "Vertical".
6. Leave the other fields at their default settings, and click OK.

Creating Transmitter Sets

For this project, two TRANSMITTER points will be placed next to one of the central buildings. Each transmitter site will be placed on a separate *Points* set.

1. Click *File→New→Transmitter Set→Points*.
2. In the GEOMETRY VIEW, right-click in the area where the coordinates are displayed (in the bottom-right corner) and choose *Cartesian*.
3. In the GEOMETRY VIEW, click anywhere close to X =0 m, Y =8.0 m. Right-click to finish.
4. In the TX/RX PROPERTIES Window, enter a *Short description* of "Street Location 1".
5. Click the TRANSMITTER PROPERTIES button.
6. The "Short dipole" antenna should be selected.
7. Enter 25 dBm as the *Input power*.
8. Click OK to exit all open windows.

Repeat the process for the other transmitter:

- Place the transmitter point near X =30.0 m, Y =0.0 m.
- *Short description*: "Street Location 2", *Input power* =25 dBm.

Creating a Building Penetration Rx Grid

Building Penetration receiver grid sets are tied to a building within the project geometry, so they are created differently than other receiver sets. To create an *Building Penetration Rx Grid*,

1. Go to the GEOMETRY VIEW and choose *Select→Structure*
2. In the GEOMETRY VIEW, click on the centrally located building as shown in 9.3. It will highlight when it has been successfully selected.

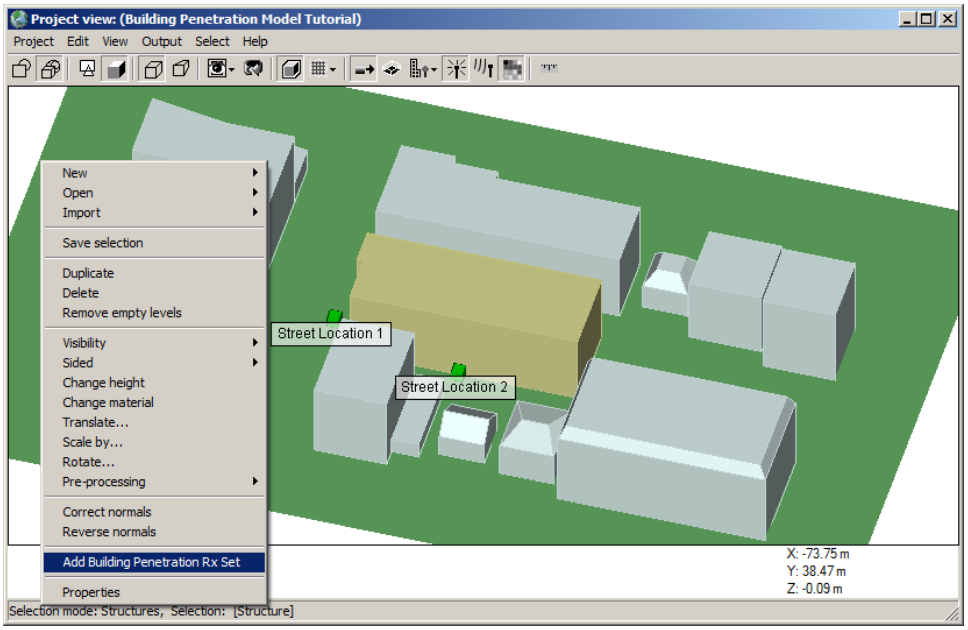


Figure 9.3: Building selection for creating a building penetration model receiver set.

3. Right-click anywhere in the GEOMETRY VIEW to access the context menu for the face. Choose *Add Building Penetration Rx Set*.

4. Enter “Allenway Rx Set” as the *Short description*.
5. Define the *Beta* as 2.0 dB/m.
6. Click the LAYOUT PROPERTIES button.
7. Define the *Spacing* as 0.5 m.
8. Click the EDIT CONTROL POINTS button. Verify that the Z-value of the control point is 2 m.
9. Click OK to close all open windown and finish the creation of the *Building Penetration Rx Grid*.

The resulting Receiver *Building Penetration* is displayed in the GEOMETRY VIEW as purple squares, as seen in Figure 9.4. To view the receiver set,

1. Go to the GEOMETRY VIEW and choose *Select→Face*
2. In the GEOMETRY VIEW , click on the roof of building containing the *Building Penetration Rx Grid* set. It will highlight when it has been successfully selected.
3. Right-click anywhere in the GEOMETRY VIEW to access the context menu for the face. Choose *Visible* to make the roof face invisible.

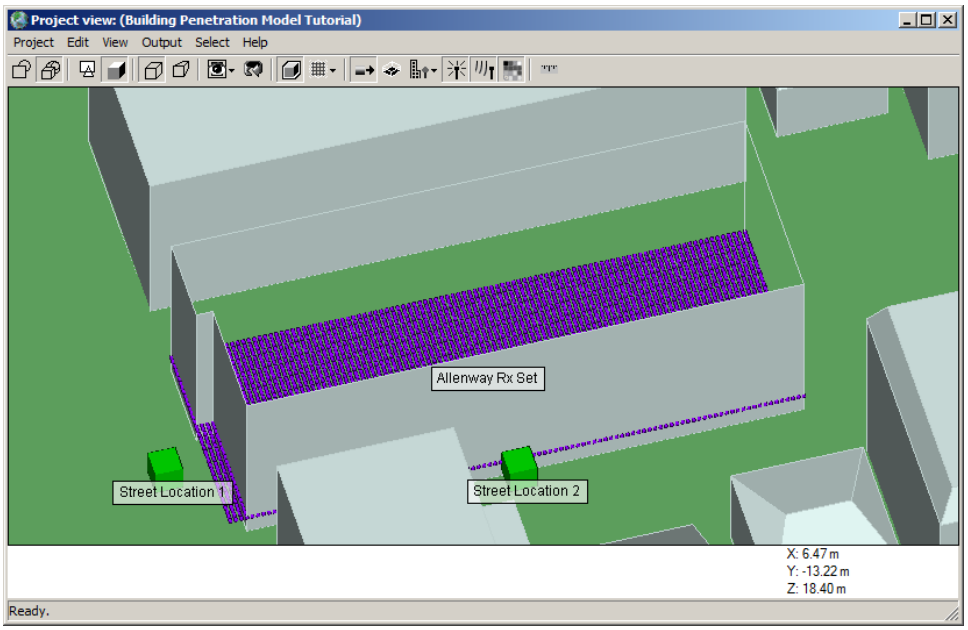


Figure 9.4: Building penetration receiver set within a building.

Creating a Study Area and Selecting a Propagation Model

To create a STUDY AREA:

1. Click *File→New→Study area*.
2. Select *Fit to geometry* and click BEGIN.

3. In the STUDY AREA PROPERTIES Window, enter "X3d_area1" as the *Short description*.
4. Select "X3D" as the *Propagation model*.
5. Deselect the *Automatic* checkbox on *Ray spacing* and set the spacing to "0.25°".
6. Deselect the *Automatic* checkbox on *Number of reflections* and set this to "2".
7. Deselect the *Automatic* checkbox on *Number of transmissions* and set this to "0".
8. Deselect the *Automatic* checkbox on *Number of diffractions* and set this to "1".
9. Click OK to finish the creation of the STUDY AREA.

Running a New Calculation

To start a calculation, click on RUN→*New* in the MAIN WINDOW. If you have not saved all changes made to this point in your Wireless InSite project, you will be prompted to do so.

Viewing Output

Viewing Received Power

1. In the PROJECT TREE, expand *Study areas and Output*→*Area: X3d_area1*→*Point to multipoint*→*Received Power*→*Street Location 1*→*Allenway Rx Set*.
2. Select *View* from the context menu for *Allenway Rx Set* to see a color display of the power coverage from the Street Location 1 transmitter.
3. To turn off receivers in the GEOMETRY VIEW, click on *View*→*Receivers*. This should disable the display of RECEIVERS and show the received power colors more clearly.
4. To see the coverage map more clearly, change the color bar scale. In the GEOMETRY VIEW, right-click on the color bar scale and select *Range Options*→*Manual Scaling* from the context menu. Set the *Minimum value* to "-90" dBm and the *Maximum value* to "-20" dBm.
5. Repeat steps 1 and 2 to view the power coverage from the Street Location 2 transmitter. The coverage maps for both transmitters are seen in Figures 9.5 and 9.6.

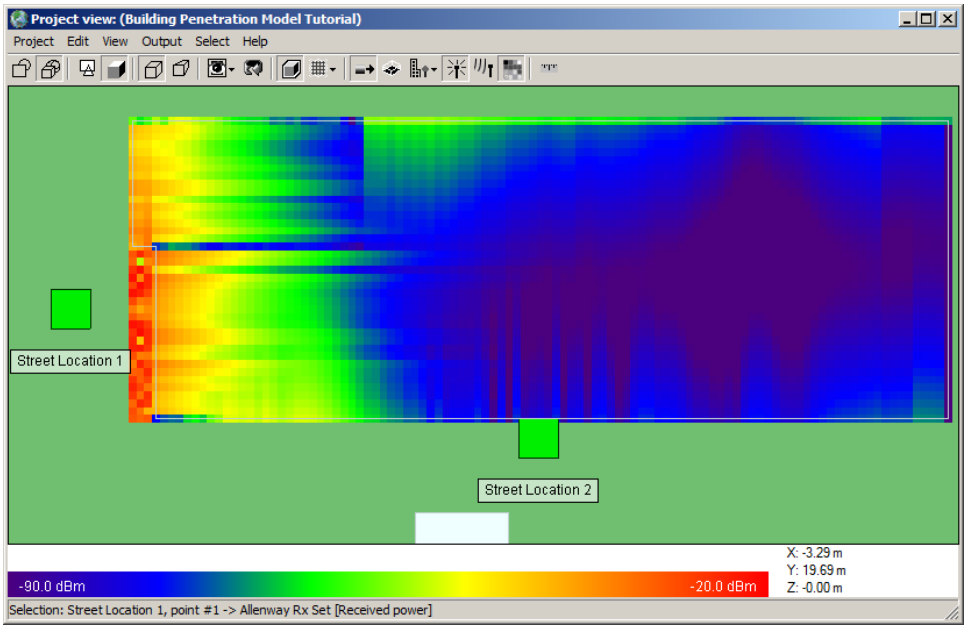


Figure 9.5: The power coverage maps for Street Location 1 transmitter.

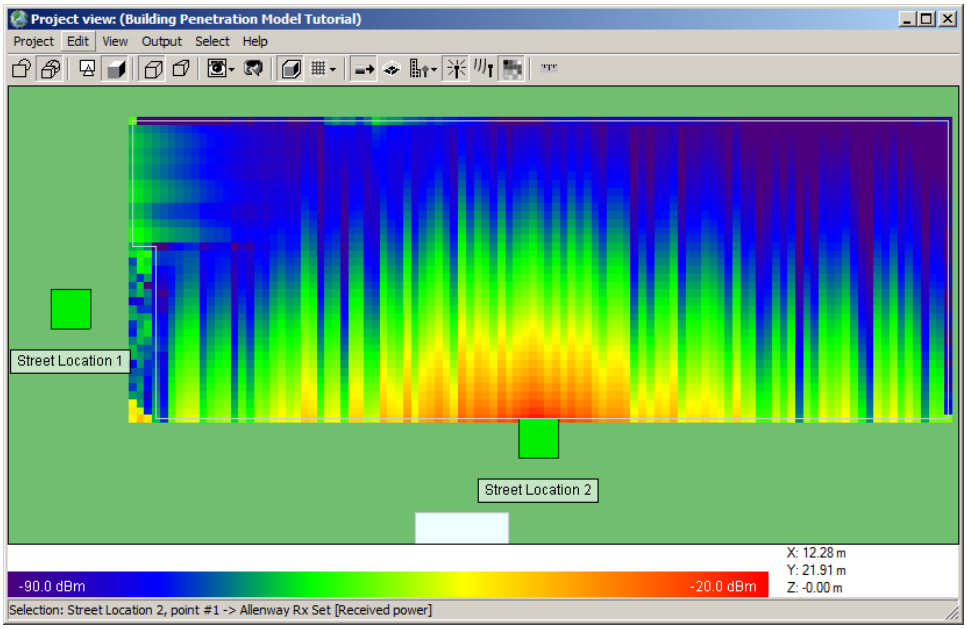


Figure 9.6: The power coverage maps for Street Location 2 transmitter.

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